AvareX User Manual

This manual documents all features in the current AvareX app, including:

- **What features exist**
- **How to reach them in the UI**
- **What each feature does**
- **Platform-specific availability**
- **Annotated screenshots of every major screen**
- **Step-by-step walkthroughs for common workflows**

> Tip: Screenshots in this manual were captured from the macOS desktop build. Mobile builds (iOS/Android) reflow some controls into bottom sheets, but the labels and workflow remain the same.

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1) Before You Start

1.1 Safety and intended use

AvareX displays an in-app Terms of Use during onboarding that states this is **not an FAA-certified GPS** and must not be relied on as a sole safety-critical system.

1.2 Internet and GPS requirements

- Internet is required for:
  - Chart/database downloads
  - Weather/doc products
  - Flight Intelligence (AI)
  - FAA plan services / cloud backup
- GPS can come from:
  - Internal device GPS
  - External ADS-B / GPS streams over **UDP ports 4000, 43211, 49002**
  - External Bluetooth stream (Android IO screen)

1.3 First-run onboarding checklist

On first run, complete onboarding pages:

1. **Sign Terms of Use** (required to continue).
2. Select **Day/Night theme**.
3. Select units:
   - Maritime (NM/knots/feet)
   - Imperial (SM/MPH/feet)
4. Confirm GPS permissions/settings.
5. When you reach the **Databases and Maps** page, AvareX **automatically** downloads what is missing for your current GPS region — the **DatabasesX** package plus the region's **Sectional**, **Plates**, and **CSUP** — in one blocking step. Nothing to tap. (Use **Choose Manually** to open the full Download screen if you want to pick specific data.)
6. Optional: open **Download** later to add more chart regions (IFR, TAC, Helicopter, etc.).
7. Optional: register your 1800wxbrief.com email for FAA flight-plan workflows.

You can reopen onboarding later from the drawer header icon.

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2) Platform Feature Availability

Feature	Availability
Core app (Map/Plate/Plan/Find, downloads, documents, aircraft & performance, checklist, logbook)	All supported platforms
Track Viewer with 2D/3D views (KML files)	All supported platforms
Vector map layer with airspace	All supported platforms
CAP Grid layer	All supported platforms
Bluetooth IO screen	**Android only**
Plan Transfer to/from Avidyne IFD (Wi-Fi)	All supported platforms

Feature	Availability
Avidyne IFD ADS-B traffic & weather (Capstone GDL90 over Wi-Fi)	All supported platforms
Pro Services (Flight Intelligence + Backup/Sync + Community + Aircraft Scheduler)	**iOS and Android only**
Airport Businesses & Reviews (free, sign-in required)	**iOS and Android only**
PDF viewing in Documents/Help	Not available on Linux
File sharing from Documents/Logbook export	Not available on Linux

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3) Main Navigation

After onboarding, the app opens to `MainScreen` with:

- **Bottom tabs**: `MAP`, `PLATE`, `PLAN`, `FIND`
- **Left drawer menu** (opened by `Menu` button on map)
- **Warnings drawer** (right side, opened by red warning icon when issues exist)

3.1 Bottom tabs

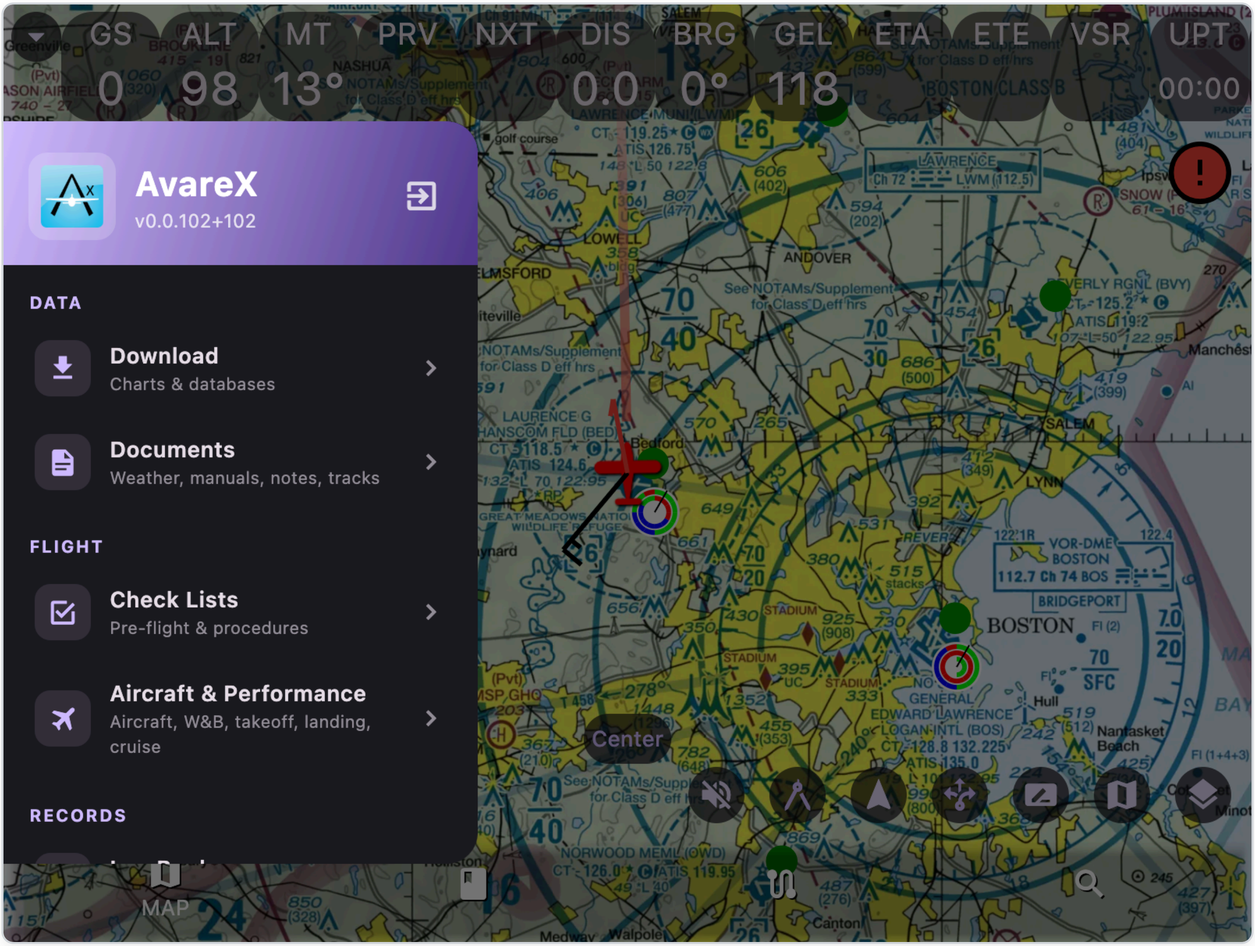
1. **MAP** — moving map, overlays, weather, traffic, route visualization, tools.
2. **PLATE** — approach plates/airport diagrams/CSUP with ownship overlay.
3. **PLAN** — build, edit, brief/file, and manage flight plans.
4. **FIND** — search destinations, recent list, nearest-airport shortcuts.

3.2 Drawer menu entries

Open from MAP with **Menu** button (bottom-left):

- Download
- Documents
- Aircraft & Performance

- Check Lists
- Log Book
- IO (Android only)
- Help (opens User Manual PDF; not available on Linux)



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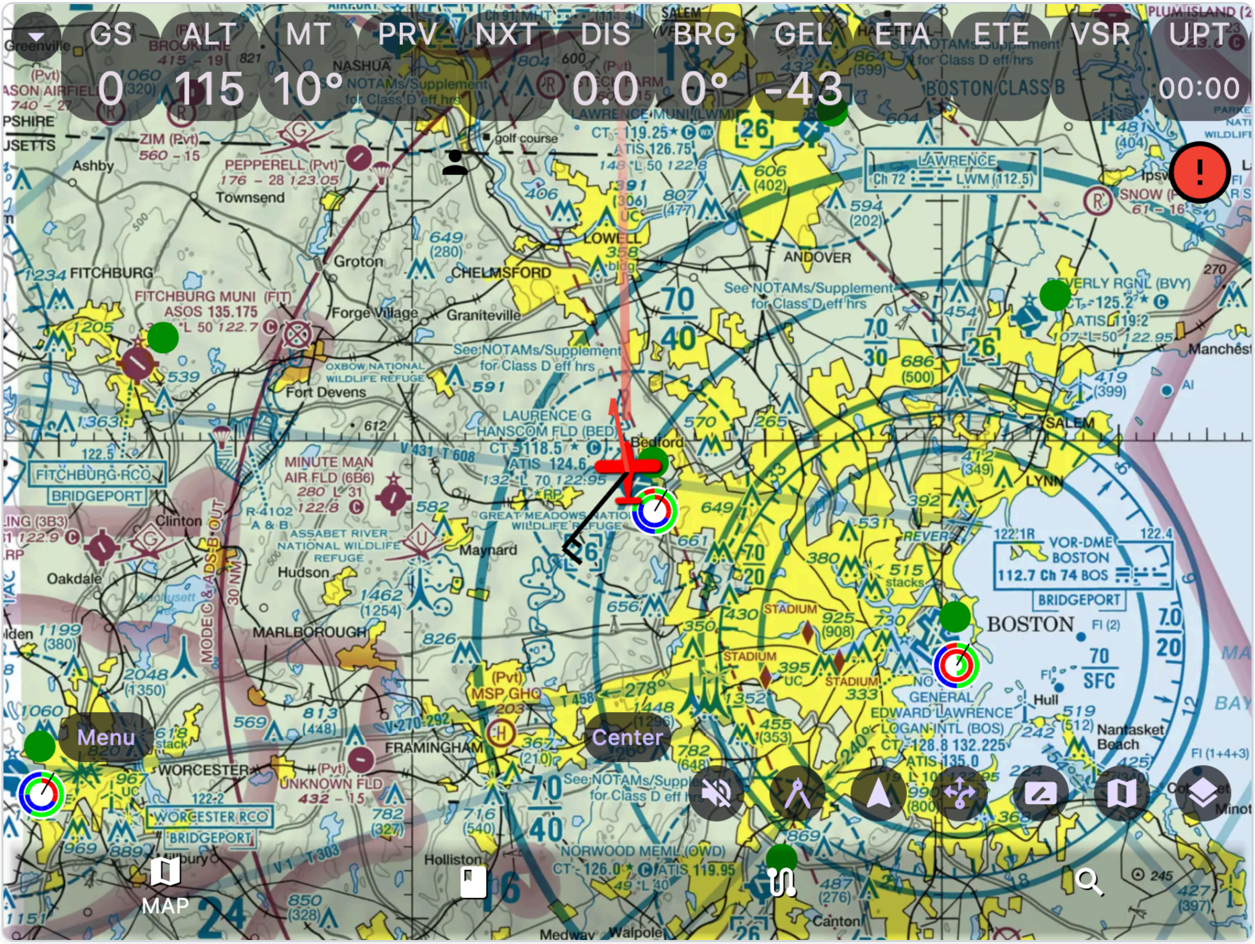
4) MAP Tab (Primary Flight Display + Map Tools)

4.1 How to access

Tap bottom tab **MAP**.



The **instrument tiles** float as a movable overlay over the map (GS, ALT, MT, PRV, NXT, DIS, BRG, GEL, ETA, ETE, VSR, UPT...); each tile can be dragged anywhere and its position is remembered. Bottom-left is **Menu** (drawer); bottom-center is **Center**; bottom-right has **Mute, Ruler, Track-up toggle, Rubber-banding, Notes, Chart Type, Layers**, plus the warning icon top-right.

4.2 Core interactions

- Pan/zoom map
- Track-up or North-up orientation toggle
- Long-press behavior:
  - If ruler mode OFF: opens destination popup near pressed location
  - If ruler mode ON: drops ruler points for distance/bearing measurement
- Single tap dismisses pop-up toasts

4.3 Map controls (buttons and menus)

Top-right

- **Pro icon** (iOS/Android): opens Pro Services login/paywall flow.

- **Red warning icon** (if active): opens troubleshooting drawer.

Bottom center

- **Center** button:
  - Tap: recenter on ownship (current zoom)
  - Long press: recenter and zoom to chart max zoom

Bottom-left

- **Menu** button: opens drawer list (Download, Documents, etc.)

Bottom-right control row (scrollable)

- **Mute** (volume icon): toggle audible alerts (traffic, GPWS, runway awareness) on/off.
- **Ruler** (compass icon): toggle measure mode; long-press map to add points. Red when active.
- **North-up / Track-up toggle**: switches orientation mode.
- **Rubber banding toggle**: enables dragging route waypoints directly on map. Red when active.
- **Notes icon**: opens handwriting Notes screen.
- **Chart type popup**: selects chart source type.
- **Layers popup**: per-layer on/off via opacity slider.

Top-left

- **Instrument tiles menu** (arrow dropdown): tile sizing, lock/reset layout, and show/hide individual instrument tiles (including the `ADSB` tile).

Traffic-only controls (show when Traffic layer > 0)

- **Traffic puck size** cycles: `S`, `M`, `L`
  - S: 20 aircraft, 3000 ft, 10 NM
  - M: 200 aircraft, 6000 ft, 50 NM
  - L: 1000 aircraft, 30000 ft, 500 NM

Altitude slider (right side)

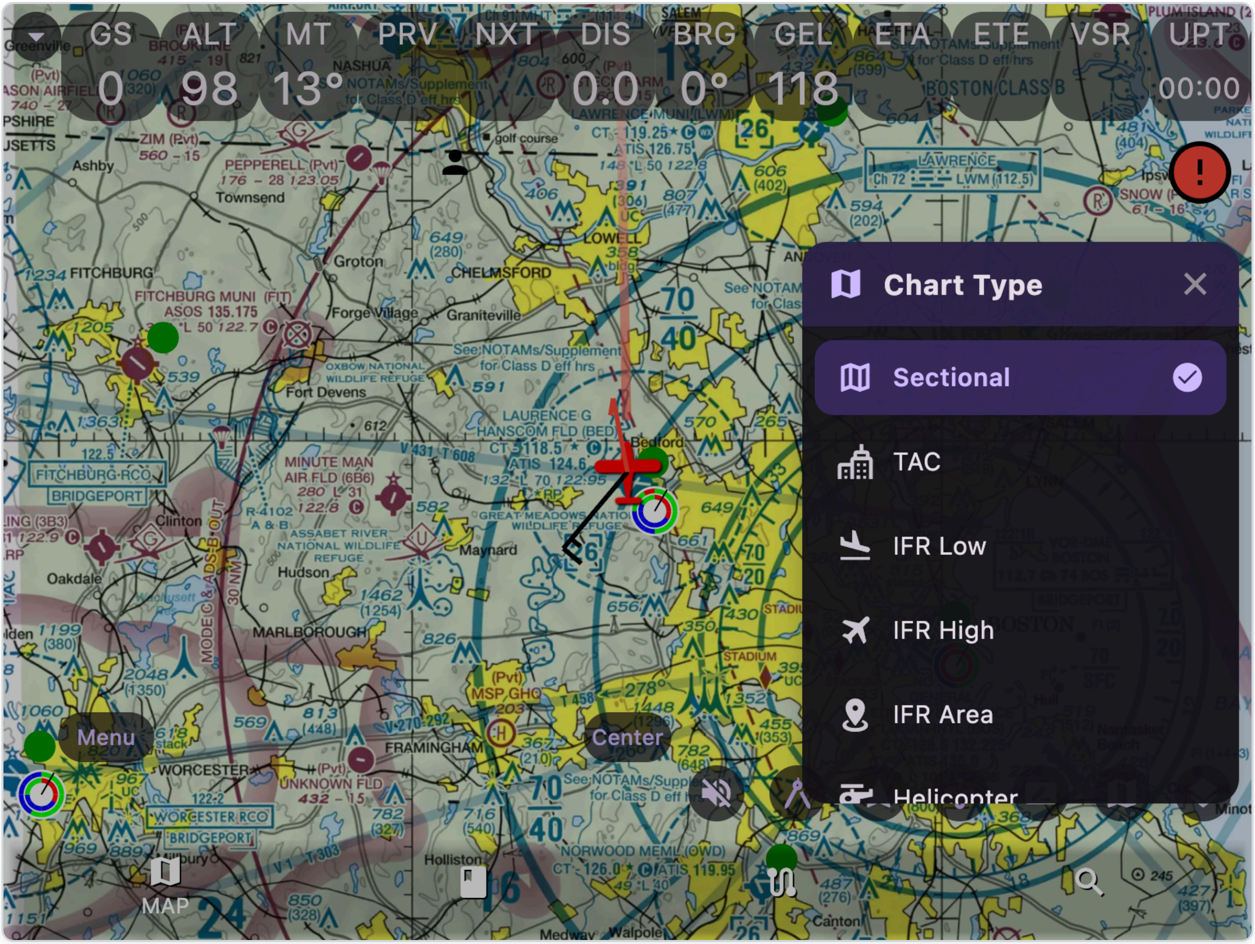
- Appears with the **Weather** layer when the product menu is set to Cloud tops, Icing, Turbulence, Ceiling, or Wind vectors.
- Range: 0 to 30,000 ft (1,000 ft increments).

- Product menu: with the **Weather** layer on, tap the cloud button on the right (above the traffic **S/M/L** control, below the altitude slider when shown) to open per-product opacity controls (same pattern as map layers). Enable any combination of **ADS-B Radar**, **ADS-B Cloud Tops**, **ADS-B Icing**, **ADS-B Turbulence**, **ADS-B Lightning**, **Radar**, **Ceiling**, and **Wind Vectors**. ADS-B items show an antenna icon; Radar, Ceiling, and Wind Vectors show an internet/radar icon. **Radar** is internet Mesonet radar animation.

4.4 Map chart types

Selectable map chart types with max zoom levels:

Chart Type	Max Zoom	Description
Sectional	10	VFR sectional charts
TAC	11	Terminal Area Charts (metro areas)
IFR Low	10	IFR Enroute Low Altitude charts
IFR High	9	IFR Enroute High Altitude charts
IFR Area	11	IFR Area charts
Helicopter	12	Helicopter route charts
Flyway	11	VFR Flyway planning charts



4.5 Map layers and what they do

Layer list from settings (with per-layer opacity 0-100%):

Layer	Description
Nav	Route lines (cyan=passed, purple=current, gray=next), runway depiction, waypoint markers/labels, ownship symbol, wind barb, north indicator.
Circles	Range rings (10/5/2 NM black rings), speed ring (blue, 1-minute travel distance), glide circle (purple, power-off glide range at the selected aircraft's best glide speed) + labels. The 10 NM ring carries a magnetic compass rose with ticks every 5° and labels at `N`, `NE`, `E`, `SE`, `S`, `SW`, `W`, `NW`.
Chart	Downloaded FAA chart tiles (offline chart base).
Vector Map	NASR vector tiles from MBTiles files, with Class B/C/D airspace and SUA (MOA, Restricted, Warning, Alert, Prohibited, NSA) rendered with standard aviation colors.

Layer	Description
CAP Grid	Civil Air Patrol grid overlay with grid identifiers (e.g., BOS42, SEA123). Only visible at zoom level 9+.
Topo	USGS topo online base map (max zoom 16).
Elevation	Downloaded elevation tiles with color-coded terrain.
Weather	METAR/TAF/PIREP/AIRMET/SIGMET symbols plus selectable weather products (multiple at once) from the right-side **Weather** menu: ADS-B Radar, ADS-B Cloud Tops, ADS-B Icing, ADS-B Turbulence, ADS-B Lightning, Radar (internet), Ceiling, and Wind Vectors — each with its own opacity slider like map layers. Use the altitude slider when tops/icing/turbulence/ceiling/winds are on.
TFR	Temporary flight restriction shapes/markers with time validity. Red=active, orange=future.
Plate	Georeferenced plate overlay on map (when loaded).
Traffic	Avare-style simple traffic dots with a 1-minute projection line drawn in the direction of travel. Cyan circle (with black outline) for normal/proximate traffic, red circle (with black outline) for threat traffic (advisory or resolution alert). Each dot is annotated with the relative flight-level offset, vertical trend arrow, and callsign. Integrates with audible alerts.
Obstacles	Obstacle markers (red squares) in your vicinity.
Tape	Distance tape labels from ownship upward in NM.
GeoJSON	Imported user GeoJSON polygons/markers.
PFD	Inset Primary Flight Display panel (artificial horizon, speed/altitude tapes, VSI, compass, CDI/VDI, AOA indicator, turn coordinator). Requires AHRS data.
Tracks	Ownship breadcrumb/polyline track recording (green line).

Notes:

- Turning **Tracks** layer OFF saves a KML track file into Documents/tracks and clears active track recording.
- **PFD** requires AHRS data from GDL90-compatible devices to show attitude.

4.6 Instrument overlay (movable tiles)

Instrument tiles float as a full-screen overlay on top of the map. Each tile can be **dragged independently** to any position on the screen, and its position is remembered across app restarts (separate layouts are kept for portrait and landscape orientation). By default only the first few tiles are shown; every tile has a row in the menu with a **+/–** toggle to show or hide it. Available tiles include:

Tile	Description	Tap Action
GS	Ground speed (knots/mph)	—
ALT	GPS altitude (feet)	—
MT	Magnetic track (degrees)	—

Tile	Description	Tap Action
PRV	Previous waypoint identifier	Jump to previous waypoint
NXT	Next waypoint identifier	Jump to next waypoint
DIS	Distance to next waypoint (NM/mi)	—
BRG	Bearing to next waypoint (magnetic)	—
GEL	Ground elevation (requires Elevation charts)	—
ETA	Estimated time of arrival (HH:MM)	—
ETE	Estimated time en-route (HH:MM)	—
VSR	VSI required to arrive 1000ft above destination	—
UPT	Up timer (count up, green when running)	Start/stop timer
DNT	Down timer (count down from 30min, red when expired)	Start/stop timer
UTC	Current UTC time (HH:MM)	—
SRC	GPS source mode. Shows `Internal`, `External`, `Internal-A`, or `External-A` (A=Auto mode)	Tap to cycle modes: Auto, Internal, External. Green=Internal, Blue=External
FLT	Total flight time in hours	Reset timer
ADSB	ADS-B receiver status (from GDL90 heartbeat + FIS-B uplinks). Shows your own tail number when the receiver reports it; otherwise a status circle — green filled circle ● = connected with GPS, yellow filled circle ● = connected without GPS (partial), empty circle ○ = disconnected. Only the text/circle is colored; the tile background is unchanged. Enable from the tiles menu.	Open the **ADS-B Status** screen

Interactions:

- **Drag** any tile to move it anywhere on the screen; its position is saved automatically.
- **Dropdown arrow** (top-left menu): **Expand/Contract** tile sizes, **Lock Tiles/Unlock Tiles** to prevent or allow moving tiles (unlocked by default), a row for every tile with a **+** (show) / **–** (hide) toggle, **Reset Layout** to restore the default tiles and positions, and a **Help** entry with tile descriptions.
- **Tap the `ADSB` tile** to open the **ADS-B Status** screen (back button in the app bar). It shows live receiver status (connected, GPS position valid, UTC timing OK, UAT initialized, ground stations received) and a scrolling log of the **last 50 received messages**, newest first. Use the **filter** button (funnel icon) to choose which message types appear in the log — toggle Heartbeat, Uplink, Ownship, Traffic, Basic/Long report, AHRS, Device, and more. **AHRS is off by default** because it arrives many times per second and would crowd out other messages. For FIS-B uplinks, the collapsed row also shows a short summary of the products carried (e.g. `uplink (0x07) — METAR, NEXRAD, TFR`). **Tap any message** to expand its decoded fields along with the **raw bytes** (selectable hex). Decoded detail per message type includes:
    - **Heartbeat**: GPS position valid, maintenance required, GPS battery low, UTC timing OK, UAT initialized, receiver UTC time, and uplink/traffic message counts.
    - **Traffic**: alert status, address type (ADS-B/TIS-B/surface/beacon), NIC/NACp integrity and accuracy, emergency/priority code, callsign, ICAO, position, altitude, ground speed, track/heading, vertical speed, emitter category, and airborne state.
    - **Basic/Long report**: target type, NIC, address qualifier, plus the common traffic fields above.
    - **Ownship**: ICAO, position, altitude, ground speed, track/heading with track type, vertical speed, NIC/NACp, report (updated/extrapolated), and airborne state.
    - **FIS-B uplink**: ground station position, position-valid flag, slot ID, TIS-B site ID, and each weather product carried in the frame, decoded individually:
        - **METAR/SPECI/TAF/PIREP/Winds Aloft**: the full textual report.
        - **NOTAM / NOTAM-TFR**: NOTAM text, or for TFRs the report number, polygon vertex count, altitude band, and effective/expiry times.
        - **D-ATIS / TWIP**: airport digital ATIS and terminal weather text (also stored for the airport **NOTAM** tab).
        - **NEXRAD radar** (regional or CONUS): the bin block number and whether the block carries precipitation or is empty.
        - **Icing / Turbulence / Cloud tops / Lightning**: block number, scale, altitude band (icing and turbulence), and whether the block carries data or is empty.
        - **SIGMET / AIRMET / SUA**: location, polygon vertex count, effective times, and any embedded text.

- **Device report**: battery charge percentage and charging state.

  Each message type has its own **color**, so you can tell types apart at a glance. The same colors are used in the **filter** list (funnel icon) next to each message type, so the log and the filter selector stay in sync. When a traffic target is suppressed, the row also carries a text tag — `filtered: ownship` (its ID matched yours, so it was treated as "us") or `filtered: out of range` (outside the current S/M/L traffic distance/altitude window). This makes it easy to confirm whether a nearby aircraft (for example one received via **TIS-B**) is being received but suppressed rather than never received at all.

  Use **Pause/Resume** to freeze or continue the live log. Leaving the screen automatically pauses the log (it resumes when you reopen the screen).

- **Ground stations**: under the **Ground stations received** count, each currently-heard FIS-B ground station (tower) is listed, nearest first, showing its identity (**TIS-B site** number when it provides TIS-B, plus its UAT **slot**), its **distance and bearing** from your position, its latitude/longitude, and how many seconds ago it was last heard. Distance/bearing are shown only when your position is known.

- **Stratus Open ADS-B Mode**: a one-shot command via the antenna icon (top-right of the ADS-B Status screen) that broadcasts the command to put an Appareo **Stratus 3 / 3i** into Open ADS-B (standard GDL90) mode so AvareX can receive its traffic and weather. Join the Stratus Wi-Fi network first (and grant **Local Network** on iOS). Tap the icon once; a toast confirms the command was sent. Official configuration is also available in Appareo's **Stratus Horizon Pro** app.

- **Diagnostics** (expandable card under the receiver status): reception-quality counters to help debug ADS-B problems. It shows the total decoded messages and average message rate, per-type counts (Heartbeat, Ownship, Traffic, Uplink), how long ago the last heartbeat/ownship/traffic message was seen, the uplink/traffic counts reported by the latest heartbeat, how many aircraft are currently tracked, how many traffic targets were filtered (ownship-match vs out-of-range), and error counts (**CRC**, **frame**, and **parse** errors — highlighted red when non-zero). Tap **Reset** to zero the counters. Use these to tell, for example, whether traffic is simply not being received versus being received but filtered.

4.7 GPWS (Ground Proximity Warning System)

- Monitors terrain ahead in direction of flight
- Predicts collisions within 3-minute lookahead
- Audible "PULL UP" alerts

- Requires: 30+ knot ground speed, Elevation data downloaded

4.7.1 Runway awareness (crossing alert)

Announces "approaching runway" when your ground track is about to meet a runway centerline, so you get a callout before taxiing onto an active runway.

Runways come from the **closest airport** to ownship (updated with the normal area refresh).

The callout fires when all of these hold:

- Ownship flight phase is **Taxi** (not Airborne) and moving at least 3 knots.
- Within about 250 ft of the runway centerline, or already on the pavement.
- Tracking across the runway rather than along it (heading more than 25° off the runway heading).
- The projected path meets the centerline within about 15 seconds, between the two runway ends.

A takeoff or landing roll — aligned on a runway above 15 knots — suppresses the alert, so rolling through an intersecting runway stays quiet. Once the flight phase is Airborne there is no crossing callout. Each runway announces once per approach and re-arms after you are clear of it.

Alerts share the global **Mute** control with traffic and GPWS. This is an advisory aid only, not a substitute for ATC clearances or looking outside.

4.8 Weather marker interactions

- **METAR tap**: Shows full METAR text with flight category icon
- **TAF tap**: Shows TAF forecast
- **PIREP tap**: Shows pilot report
- **AIRMET/SIGMET tap**: Shows advisory text. **Long-press**: toggles shape visibility on map
- **TFR tap**: Shows TFR details (altitudes, times)

4.9 Ruler/measurement tool

- Activate via compass icon (turns red when active)
- Long-press map to add measurement points
- Shows distance (NM) and bearing (degrees) between points
- Multiple segments supported

4.10 Rubber banding

- Activate via toggle button (turns red when active)
- Long-press and drag waypoint icons to reposition
- On release, snaps to nearest navaid/fix from database
- Must be explicitly enabled to prevent accidental edits

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5) Destination Popup (Map Long-Press or Find Tap)

5.1 How to access

- Long press on map (when ruler mode is off), or
- Tap an item in FIND list

5.2 Top action buttons

- **→D**: set Direct-To destination, center map there, return to map
- **+Plan**: insert destination into current plan at the current waypoint position
- **↓Plan**: append destination to the end of the plan
- **Plates** (airport only): jump to PLATE tab at that airport

5.3 Information pages (shown when data exists)

Possible tabs:

- **Main**: destination summary (airport info, runway diagram, or VOR/navaid info with nearby VORs)
- **AD**: airport diagram/runway depiction (interactive viewer with zoom)
- **METAR**: METAR + TAF text with flight category color indicator
- **NOTAM**: fetched NOTAM list (downloaded async)
- **SUA**: special-use airspace data for the area
- **Wind**: nearest winds-aloft station data at multiple altitudes
- **ST**: sounding chart/image for the area (Skew-T diagram)
- **Business**: for airports, lists the crowd-sourced businesses/FBOs for that airport inline (free; sign-in required) with details and reviews shown in place — see Section 11.7

If nearby alternatives exist, a horizontal "Nearby" selector appears.

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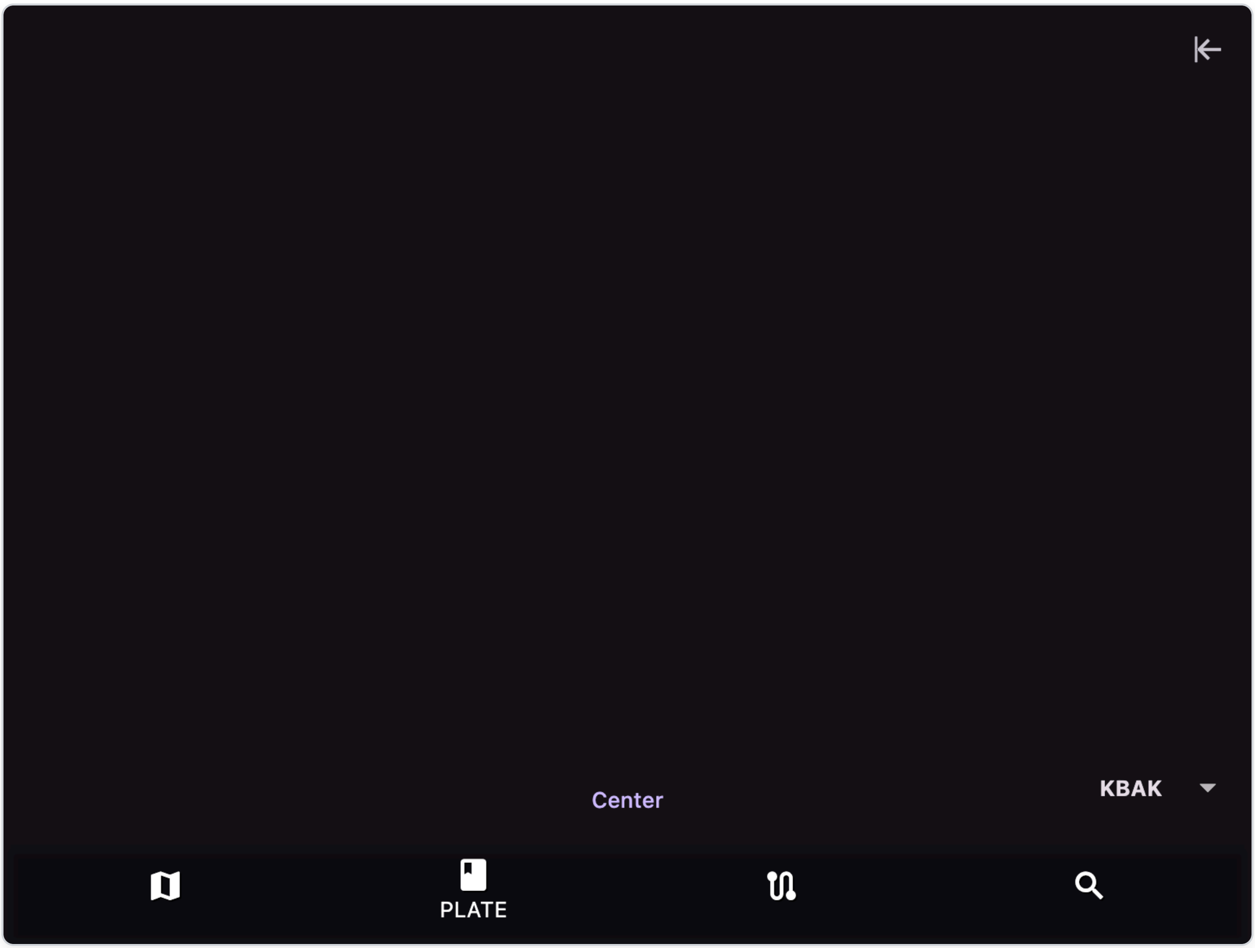
6) PLATE Tab

6.1 How to access

- Bottom tab **PLATE**, or
- From destination popup using **Plates**



6.2 What it does

- Displays downloaded plates/diagrams/CSUP
- Draws ownship and heading on georeferenced plates
- Can overlay a selected business marker on airport diagrams (from the crowd-sourced Airport Businesses directory; iOS/Android, sign-in required)
- Supports zoom/pan via InteractiveViewer (up to 8x)

6.3 Controls

- **Airport selector** (bottom-right): choose airport from recent airports list
- **Plate selector** (bottom-left, always shown): choose plate within airport (color-coded by type). Always contains an `APS-AERIAL VIEW` entry — once downloaded it loads the

georeferenced Google Maps satellite picture like any other plate; until then it appears as an `APS-AERIAL VIEW (Get)` placeholder that downloads the image when selected. When downloaded, a trash icon on that entry **deletes** the stored picture immediately; you can download it again later. The picture is stored in the airport's plates folder as `APS-AERIAL VIEW.png` with ownship drawn on it
- **Procedure menu** (plus icon near bottom-right):
  - Tap a procedure to show its profile, draw the CIFP route on georeferenced plates (progress-colored), and add its waypoint sequence to the plan
- **Instrument overlay show/hide** toggle (top-right arrow icon)
- **Profile close (X)** closes procedure profile card and clears the route overlay
- **IAF chips** (black with white text, above profile when multiple transitions exist): switch IAF without re-adding to the plan

6.3a Procedure route on plate

When you select a procedure from the plate procedure menu (`+` / road icon):

- The vertical profile card appears
- On **georeferenced** plates, the CIFP route is drawn with colors counted back from the final fix (same on the vertical profile):
  - **Green** — final leg
  - **Yellow** — 1 before final
  - **Brown** — 2 before final
  - **Blue** — 3 before final
  - **Red** — 4 before final
  - **Orange** — 5+ before final
- Fix names are not drawn on the plate (keeps chart symbology clear)
- When the chart has multiple IAFs, **black chips** (white text) above the profile switch transitions (without re-adding to the plan)
- Closing the profile (X), or changing airport, removes the overlay
- Non-georeferenced plates show the profile only (no route overlay)

Route geometry follows the CIFP fix sequence in the databases (great-circle chords between fixes). Path-terminator arcs/holds are not drawn until those ARINC fields ship in a future DatabasesX cycle.

6.4 Plate type color coding

Color	Plate Type
Green	Airport Diagram (APD)
Blue	CSUP
Pink	Departure Procedure (DP)
Purple	Instrument Approach (IAP)
Cyan	STAR
Brown	Minimums (MIN)
Red	Hot Spots (HOT), Land and Hold Short (LAH)
Teal	Satellite View (`APS-AERIAL VIEW`)

6.5 Terrain caution overlay on plates

When map **Elevation** layer has opacity > 0, plate rendering can include colored terrain risk cells:

- **Yellow**: terrain within 500-1000 ft below aircraft
- **Red**: terrain within 500 ft or above aircraft

6.6 Business selector

On airport diagrams, when businesses with a known location exist for the airport, a right-side selector (business icon) lets you:

- Select a business from the crowd-sourced Airport Businesses directory
- See its location marked on the airport diagram, with its name labeled next to the marker

The businesses come from the shared Firestore directory (see Section 11.7), so this selector appears only on iOS/Android and when you are signed in.

6.7 Automatic behavior on landing

Flight-state logic can auto-switch plate context to nearest airport diagram after landing (if available).

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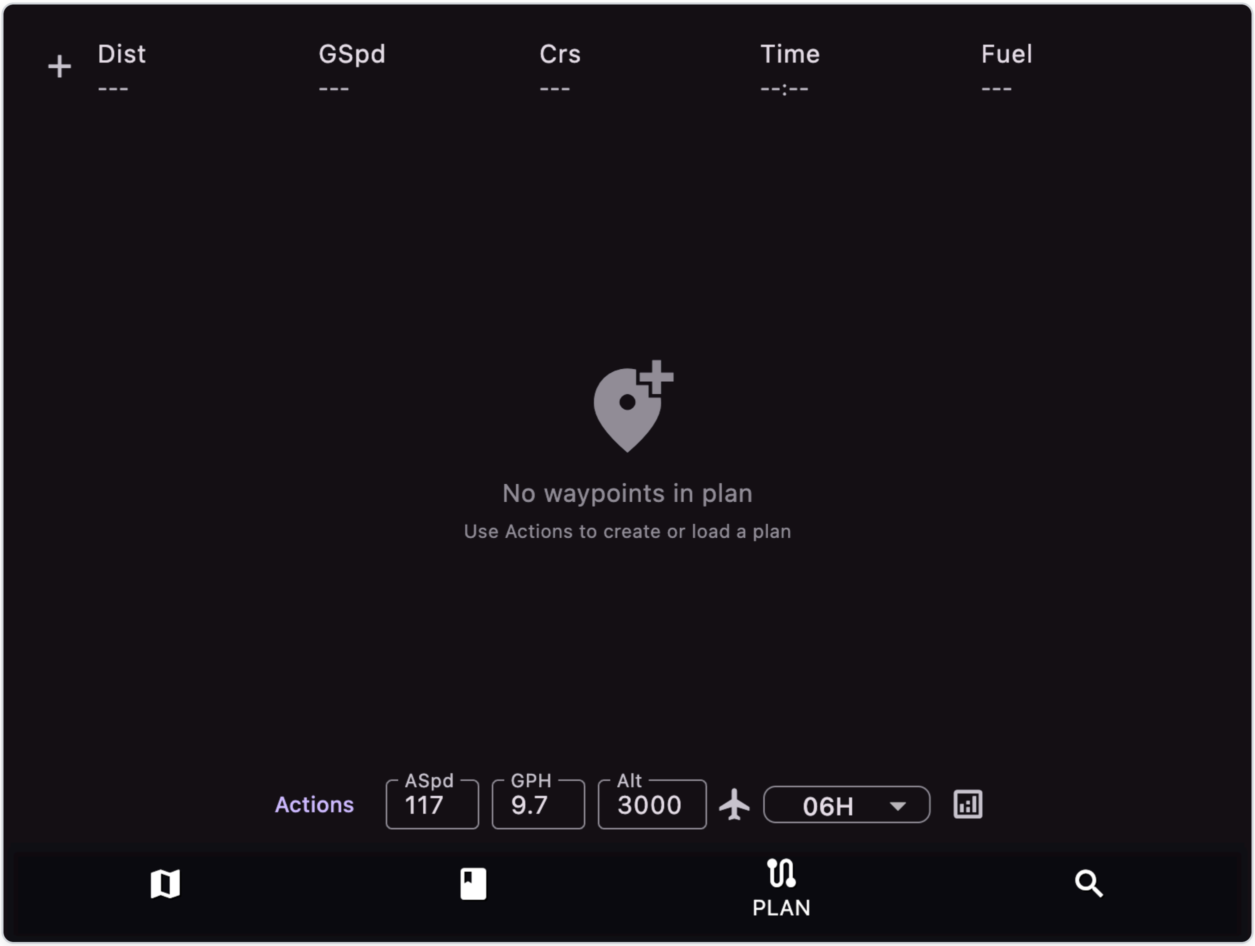
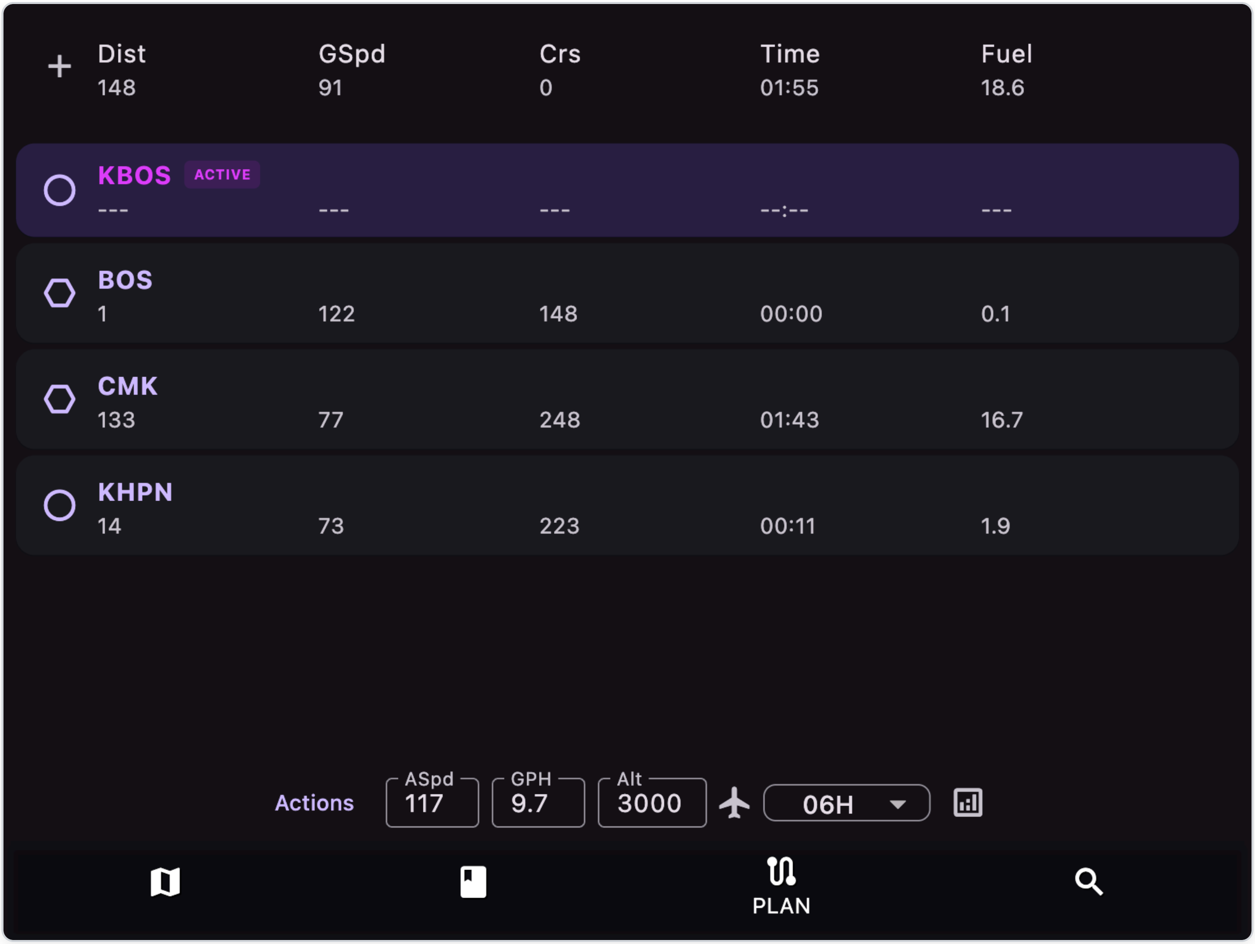
7) PLAN Tab

7.1 How to access

Tap bottom tab **PLAN**.



After waypoints are added (e.g. `KBOS BOS CMK KHPN`), each row shows distance/ground speed/course/time/fuel calculations:



7.2 Main plan editor

- Reorderable waypoint list (long-press and drag to reorder)
- Swipe row right-to-left to delete waypoint
- Tap row to set current waypoint (highlighted in purple with "ACTIVE" badge)
- **Insert waypoint**: tap a waypoint to select it, then press the **+** button to insert a new waypoint after that position (opens FIND tab; next selection inserts after the selected waypoint)
- Long press (grab) a waypoint to move it up and down
- Header row shows plan-wide calculated totals (Distance, Ground Speed, Course, Time, Fuel)
- **Delete entire plan** (trash icon, top-right of header): confirms, then removes all waypoints from the active plan (hidden when the plan is empty)
- Airways and procedures show as expandable items with nested waypoints

Bottom controls are arranged in two rows. The **settings row** (on top) is only shown on the **Plan** tab and holds the plan calculation controls:

- **ASpd** field: true airspeed used in calculations (knots)
- **GPH** field: fuel burn (gallons per hour)

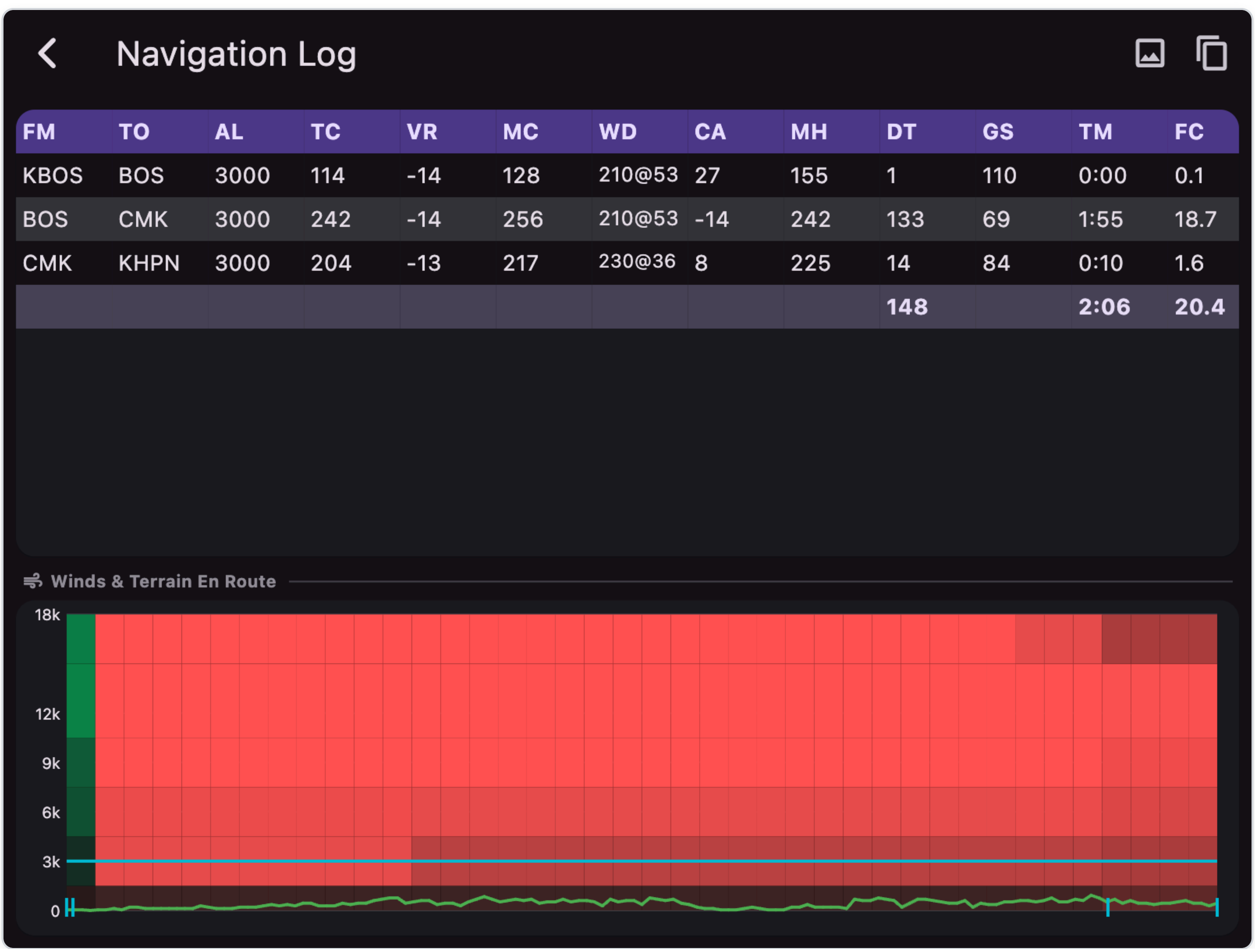
- **Alt** field: plan altitude (feet, default 3000)
- **Aircraft icon** (next to Alt): loads **ASpd** and **GPH** using the same aircraft as the **Performance** screen app-bar **dropdown** (latest selection), interpolated at the plan **Alt** using the **saved cruise power %** from the Performance **Cruise** tab (ISA deviation 0; custom *My Aircraft* cruise table when that tail is selected, otherwise built-in POH when a template is selected). If no power has been saved yet, **65%** is used. If you have never opened Performance, aircraft selection falls back to the map aircraft setting when set.
- **Forecast horizon** selector: 06H / 12H / 24H (winds aloft forecast period)
- **Analytics icon** (chart): opens full-screen Navigation Log

The **tab row** (below the settings row) switches what the PLAN tab shows; the active tab is highlighted:

- **Plan** (first tab): the waypoint list / plan editor (default)
- **Load & Save**, **Create**, **Brief & File**, **Manage**, **Transfer**: each shows the matching action sub-page inline; tap **Plan** to return to the waypoint list

7.3 Navigation Log dialog

Open from the PLAN tab by tapping the **analytics** icon (graph, tooltip *"Navigation log and terrain"*) at the far right of the bottom row.

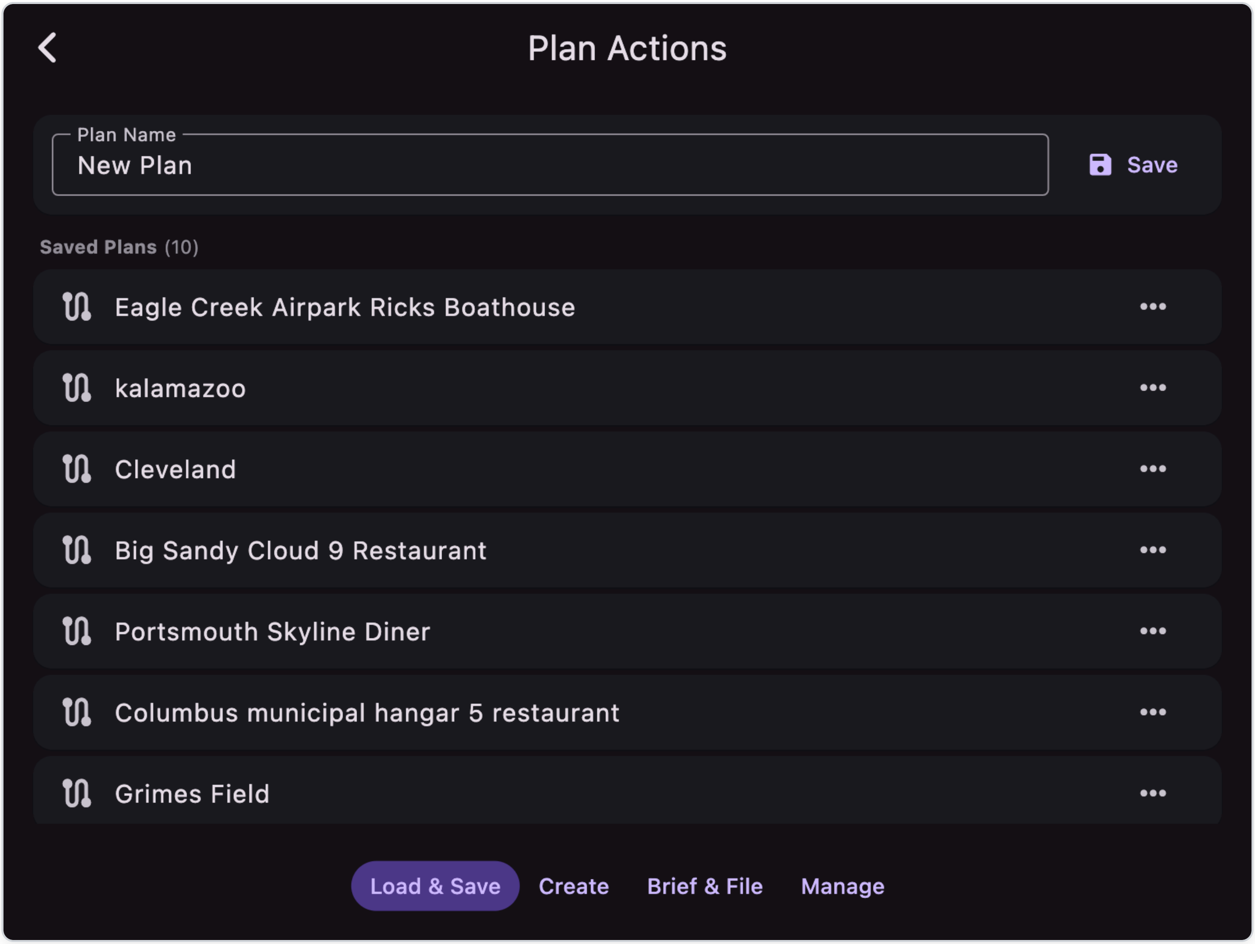


Includes:

- **Per-leg log grid** with columns: FM (from), TO (to), AL (altitude), TC (true course), VR (variation), MC (magnetic course), WD (wind direction@speed), CA (wind correction angle), MH (magnetic heading), DT (distance), GS (ground speed), TM (time), FC (fuel consumption). The bottom row of the grid totals DT/TM/FC for the whole plan.
    - Grid is zoomable/pannable; **double-tap** to reset zoom
- **Winds & Terrain En Route diagram**: combined visualization showing wind components and terrain profile:
    - **Wind heat map**: color-coded cells at altitudes 0-18000 ft along route (green = tailwind, red = headwind, darker = lighter winds, brighter = stronger winds up to 50 kt scale)
    - **Plan altitude line**: cyan horizontal line showing your planned altitude
    - **Terrain profile overlay**: terrain elevation curve with color coding (green = safe below plan altitude, red = at or above plan altitude)
    - **Waypoint markers**: cyan tick marks at the bottom showing waypoint positions along the route
    - **Interactive tap**: tap anywhere on the diagram to display a tooltip with:
        - Nearest waypoint (if within 5% of route position)
        - Altitude at tap position (snaps to altitude bands: 0, 3000, 6000, 9000, 12000, 18000 ft)
        - Terrain elevation at that point
        - Course direction
        - Wind direction and speed
        - Tailwind/headwind component in knots
- **Copy plan to clipboard** button (app bar)

7.4 Plan actions (inline sub-pages)

The action sub-pages are shown inline in the PLAN tab by tapping one of the tab buttons at the bottom. The first tab, **Plan**, is the waypoint list / editor; the remaining tabs are **Load & Save**, **Create**, **Brief & File**, **Manage**, and **Transfer**. The tapped button is highlighted while its sub-page is displayed; tap **Plan** (or complete an action such as loading/creating a plan) to return to the waypoint list.

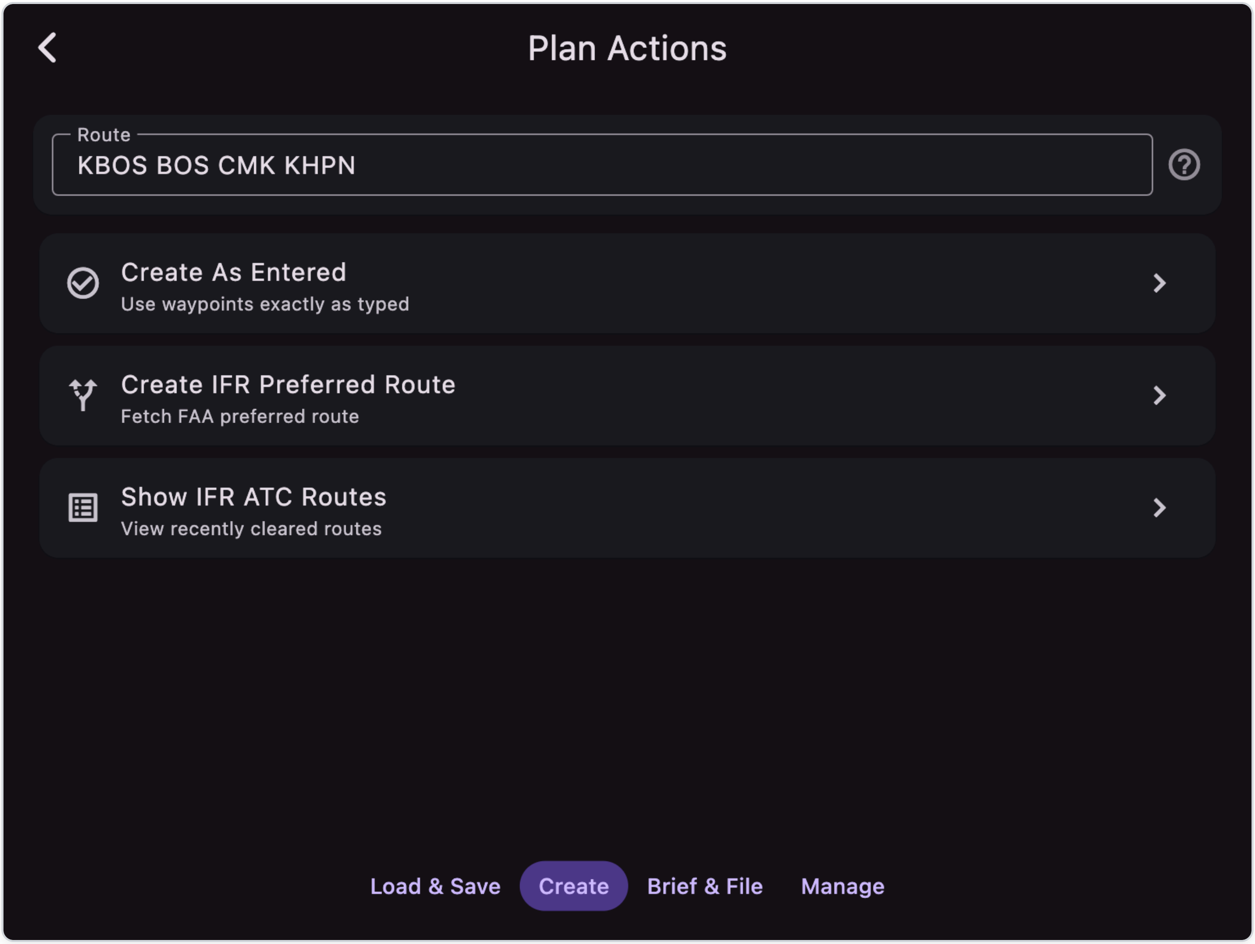


A) Load & Save

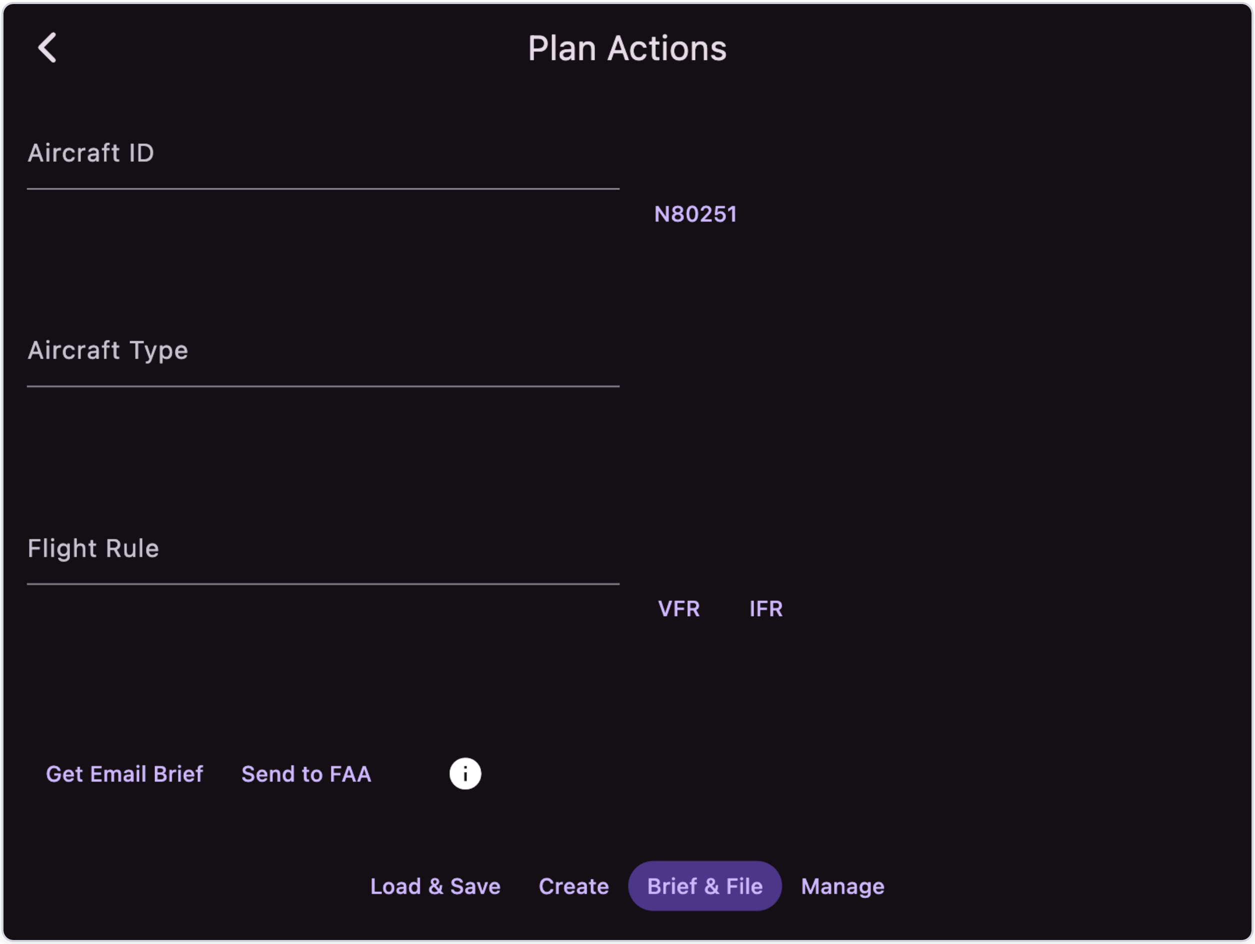
- Save current plan by name
- Load saved plan
- Load reversed (loads plan in reverse order)
- Delete saved plan
- Tap any plan to load immediately

B) Create

- **Create As Entered**: parse typed route string (space-separated waypoints/airways, e.g., `KBOS V123 KHPN`)
- **Create IFR Route**: build a plan between two airports (enter `DEPART DEST`) using the built-in offline route generator, which produces a waypoint route over all airways — joining near the departure, following the airway network, and leaving near the destination
- **Create IFR LA Route**: same as above but restricted to low altitude airways
- **Show IFR ATC Routes**: show recent ATC routes between departure/destination from 1800wxbrief with last departure times



C) Brief & File (FAA)

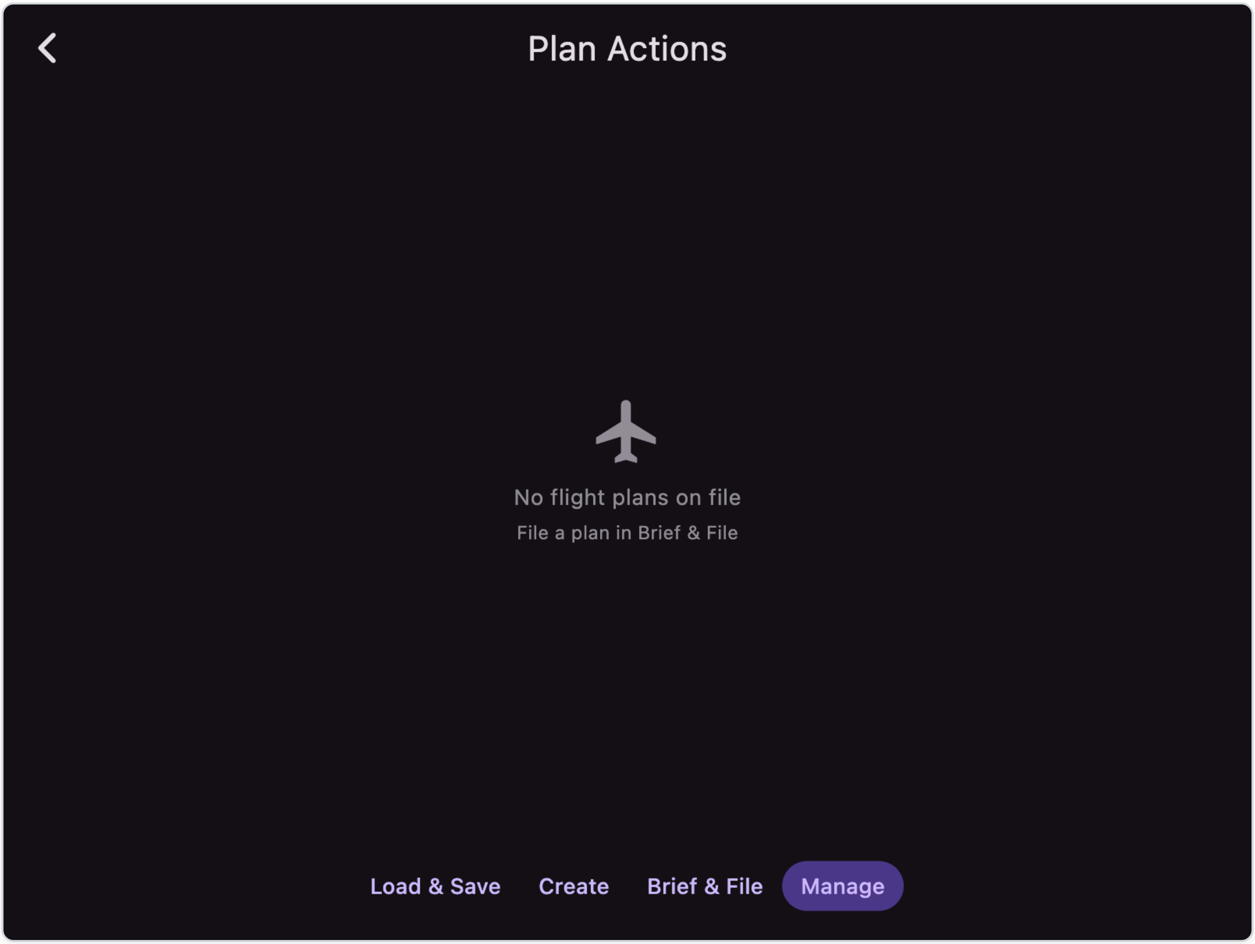


- Detailed ICAO form fields:
    - Aircraft ID, Type, Flight Rule (VFR/IFR), Flight Type, Number of Aircraft
    - Wake Turbulence (LIGHT/MEDIUM/HEAVY)
    - Aircraft Equipment, Surveillance Equipment
    - Departure, Departure Date/Time Zulu
    - Cruising Speed, Altitude, Route
    - Destination, Total Elapsed Time
    - Alternate Airports (1 & 2)
    - Fuel Endurance, People On Board
    - Aircraft Color, Remarks
    - Pilot in Command, Pilot Information
- Quick-fill buttons from current plan and stored aircraft
- Buttons:
    - **Get Email Brief**: sends briefing request to 1800wxbrief (50nm corridor)
    - **Send to FAA**: files flight plan via LMFS API
- Status indicator (check mark=success, info icon with tooltip=error)

Requires configured 1800wxbrief-compatible account email.

D) Manage (FAA)

- Retrieve filed plans
- State-aware actions with color coding:
  - **PROPOSED** (blue): **Cancel** or **Depart** (activate with chosen time - opens time picker with sunrise/sunset)
  - **ACTIVE** (green): **Close**
  - **CLOSED** (grey): view only



E) Transfer

The `Transfer` sub-page exchanges flight plans with a panel-mounted **Avidyne IFD440/540/550** over Wi-Fi, in both directions.

- Connect your device to the **same Wi-Fi network** as the IFD (use the IFD's Maintenance-mode network page to configure it).
- AvareX automatically searches for IFDs; tap the **search icon** (top-right of the Avidyne section) to re-broadcast discovery. Discovered units appear in a list showing their IP address and software version. Each has two buttons:
  - **Send** — uploads the active AvareX plan to the IFD as a stored route. Requires a route with at least 2 waypoints, and the IFD must have flight-plan input enabled (the list

notes when it is not). The uploaded route reproduces the AvareX waypoints on the IFD ("stick route"); review and activate it on the IFD.
    - **Get** — downloads the IFD's flight plan and loads it into the active AvareX plan (replacing the current one). Each downloaded waypoint is matched to the AvareX database by identifier (closest to the IFD's coordinate); waypoints that cannot be matched are added as GPS coordinate points. Airway/procedure legs are reproduced as their coordinate waypoints.
- Below the device list, a scrolling **Messages** log shows the last 50 WiFiCrChannel packets exchanged with the IFD during Send/Get — **TX** (AvareX → IFD) and **RX** (IFD → AvareX), newest first. Tap a row to expand the **raw bytes** as selectable hex. Use **Pause** / **Resume** to freeze the log, and **Clear** to empty it.
- **iOS only**: the **Local Network** permission must be granted for discovery and transfer to work.

Avidyne ADS-B (Capstone): While AvareX is running on the same Wi-Fi network as an IFD equipped with an ADS-B receiver, it also receives the IFD's **Capstone** ADS-B stream automatically. Capstone is GDL90-format data, so traffic, ADS-B weather (NEXRAD/METAR/TAF/D-ATIS/TWIP/icing/turbulence/cloud tops/lightning/AIRMET/SIGMET via FIS-B), and receiver status appear the same way as with any other GDL90 device — no separate setup is needed beyond being on the IFD's network (and, on iOS, the **Local Network** permission).

7.5 Route building details

Waypoint types supported:

Type	Description
GPS	User-entered lat/lon coordinates
Airport	Airport from database
Navaid	VOR/NDB
Fix	Named fixes/intersections
Airway	Victor (V) and Jet (J) routes
Procedure	SID/STAR/Approach (format: `AIRPORT.PROCEDURE.TRANSITION`)

Airway processing:

- Automatically finds entry and exit points when airway is between two waypoints
- Max segment length: 500nm (separates AK/HI from lower 48)
- Populates actual fix names asynchronously

7.6 Automatic waypoint passage

When within 2nm of a waypoint and moving away from it, the plan automatically advances to the next waypoint.

7.7 Automatic plan resume after restart or crash

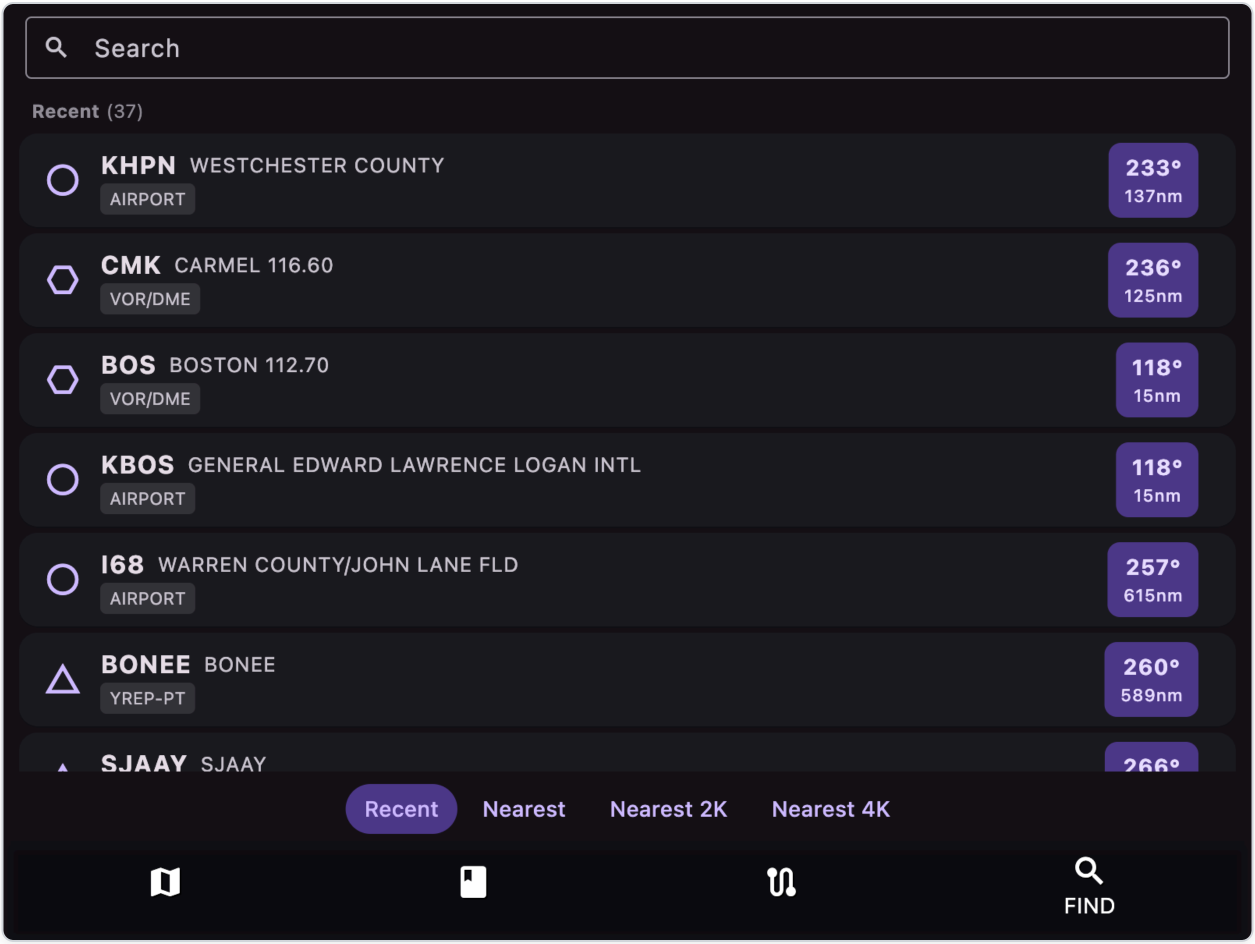
The active plan and the waypoint you are currently flying to are saved automatically as they change. If the app is closed or crashes, the next launch **reloads the same plan and resumes on the waypoint** (and airway/procedure leg) you were on. No action is required; there is no button to press. Clearing the plan clears the saved copy as well.

---

8) FIND Tab

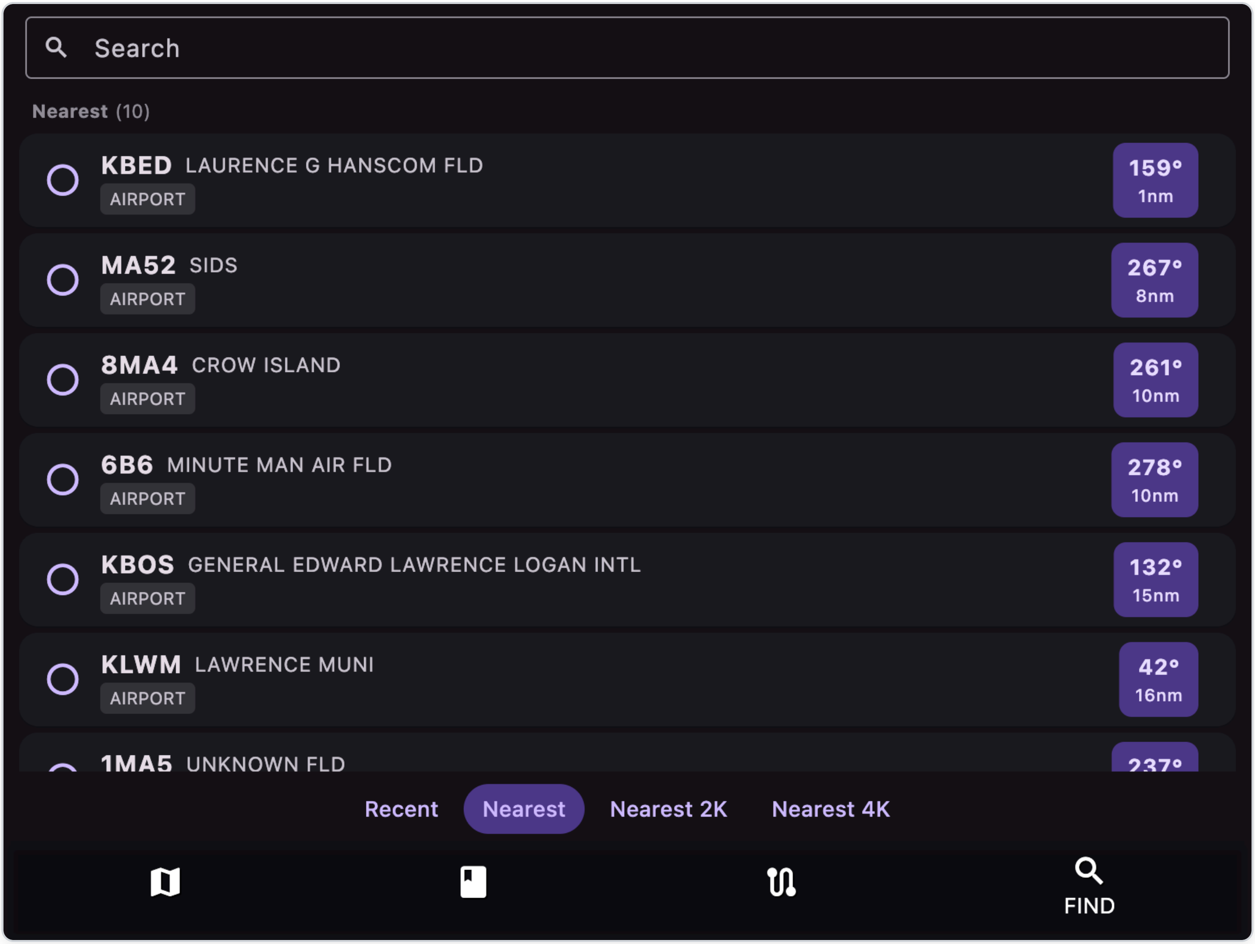
8.1 How to access

Tap bottom tab **FIND**.

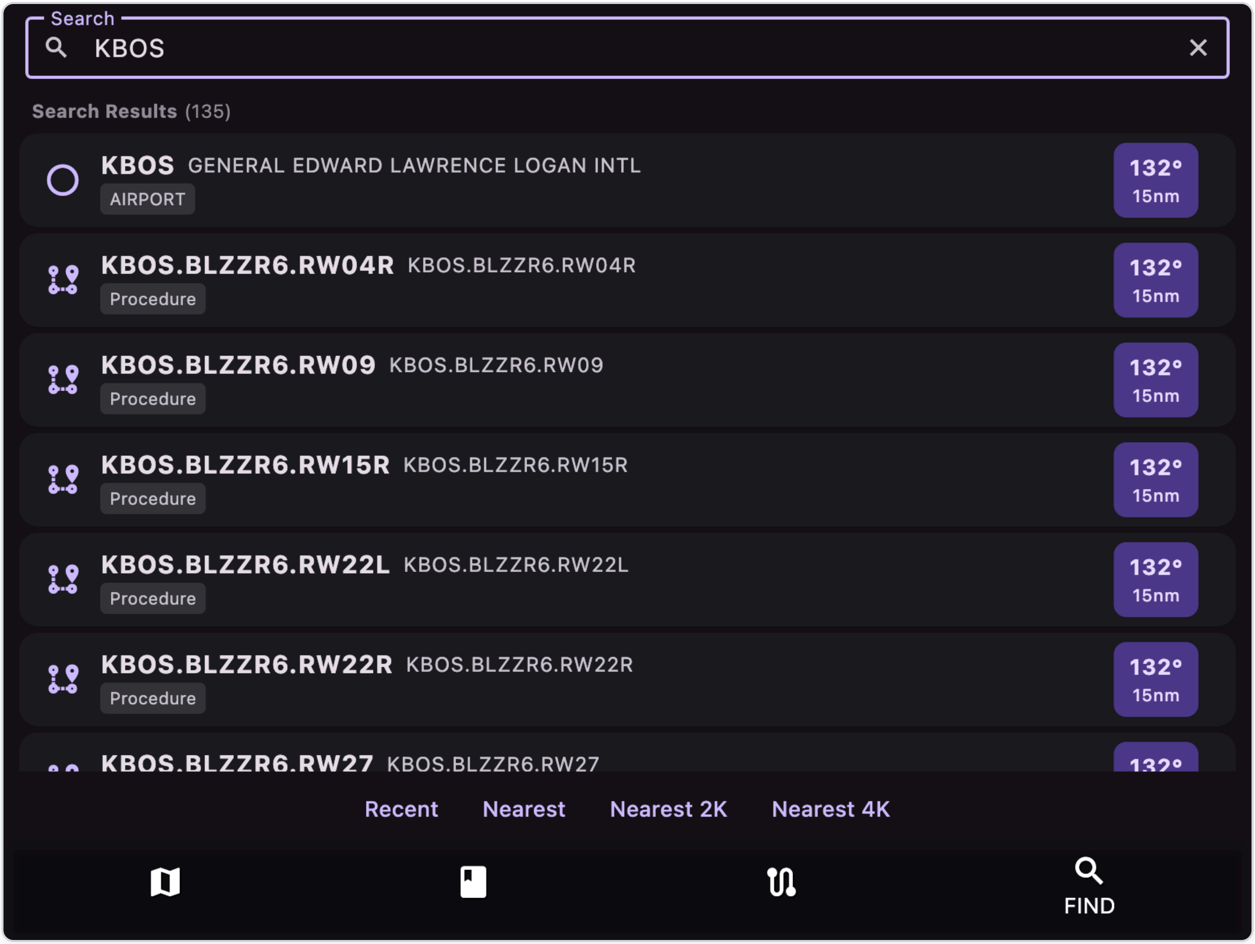


8.2 What it does

- Search destinations by text (airports, navaids, fixes, GPS coordinates)
- Browse:
  - Recent (previously viewed destinations)
  - Nearest (nearest airports)
  - Nearest 2K (runway >= 2000 ft)
  - Nearest 4K (runway >= 4000 ft)



Type into the **Search** box to find airports, navaids, fixes, or procedures by identifier:



8.3 Row actions

- Tap row: opens destination popup/details
- Tap left icon (marked `→D`): **Direct-To** the destination (same as the `→D` button on the destination popup) — sets it as your active target, shows it on the map, and switches to MAP
- Right-side bearing@distance button: centers map on item and switches to MAP
- Swipe right-to-left to delete recent entry
- For GPS-type recents, tap edit icon to modify custom facility name

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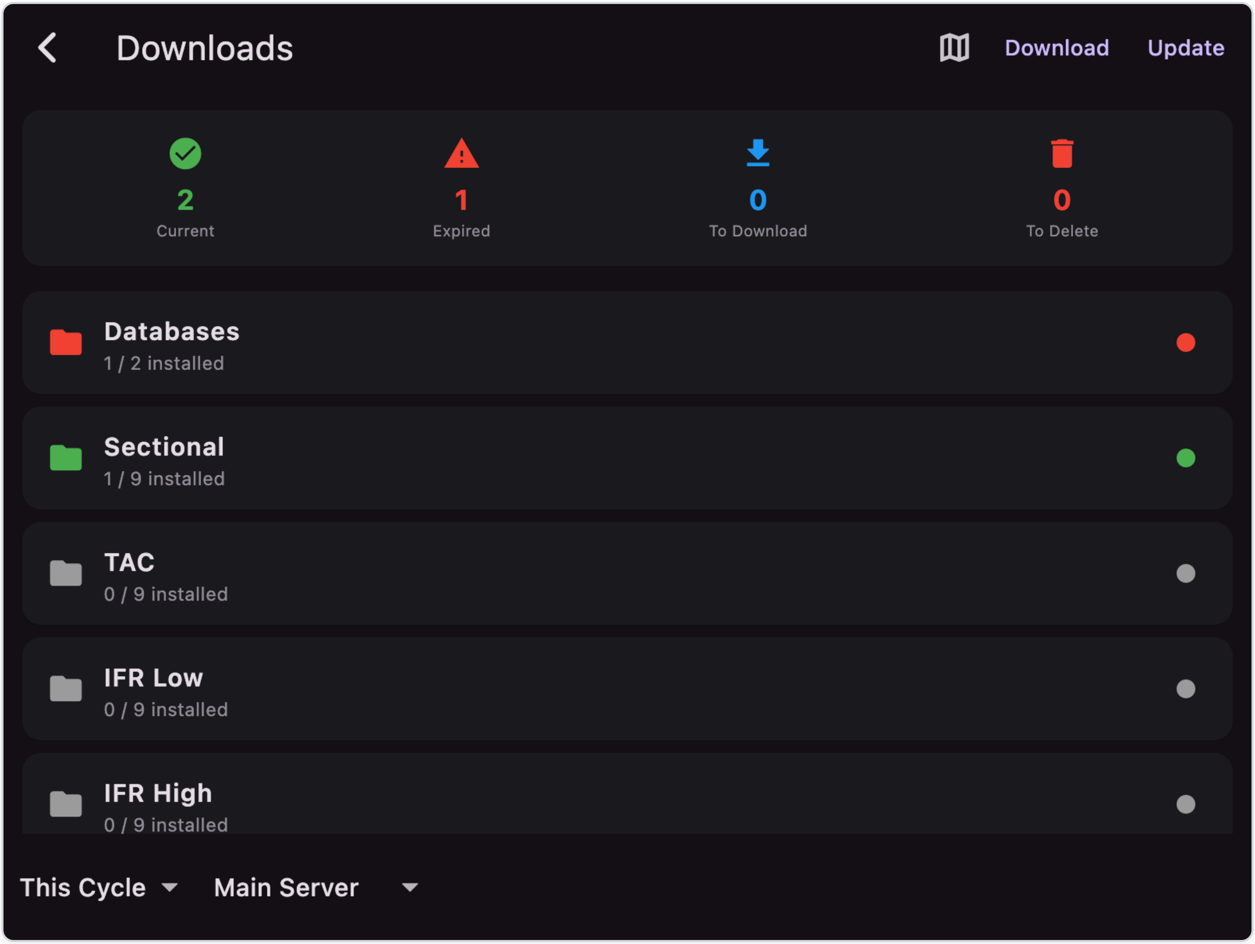
9) Drawer Features

9.1 Download

Open: **Menu → Download**



Purpose:

- Install/update/delete downloadable data sets by chart category and region
- Monitor progress with per-item progress ring and stop button

Status Summary Card:

- Shows counts of Current, Expired, To Download, and To Delete items

State Indicators:

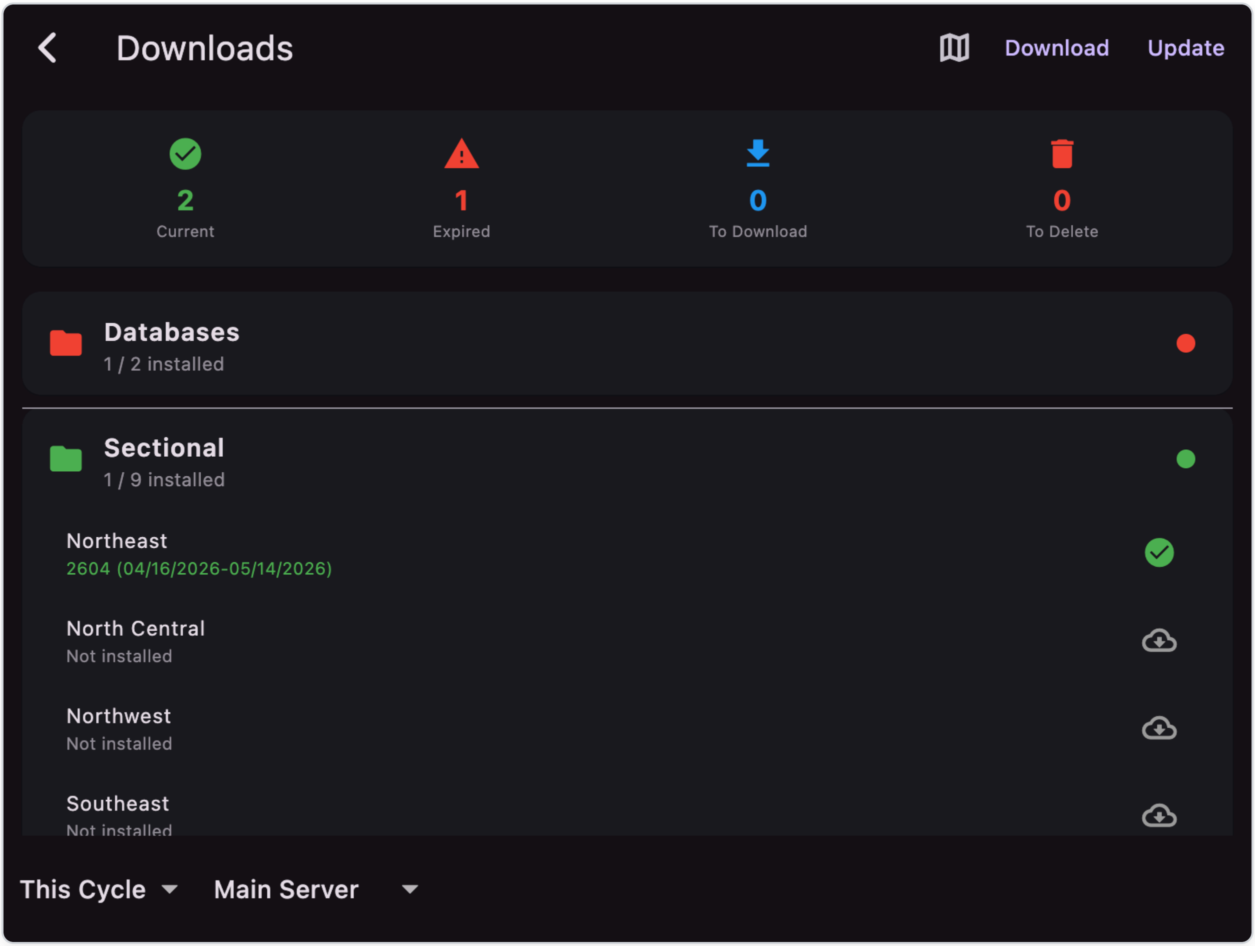
- Green = Current (up-to-date)
- Orange = Expired (needs update)
- Gray = Absent (not downloaded)

Controls:

- Tap chart item to queue for download/update/delete
- **Download** button processes all queued items
- **Update** button automatically updates all expired charts
- Toggle **This Cycle / Next Cycle**
- Toggle **Main Server / Backup Server**

- Info map icon shows regional coverage image

Tap a category (e.g. **Sectional**) to expand and see regional charts that can be selected individually:



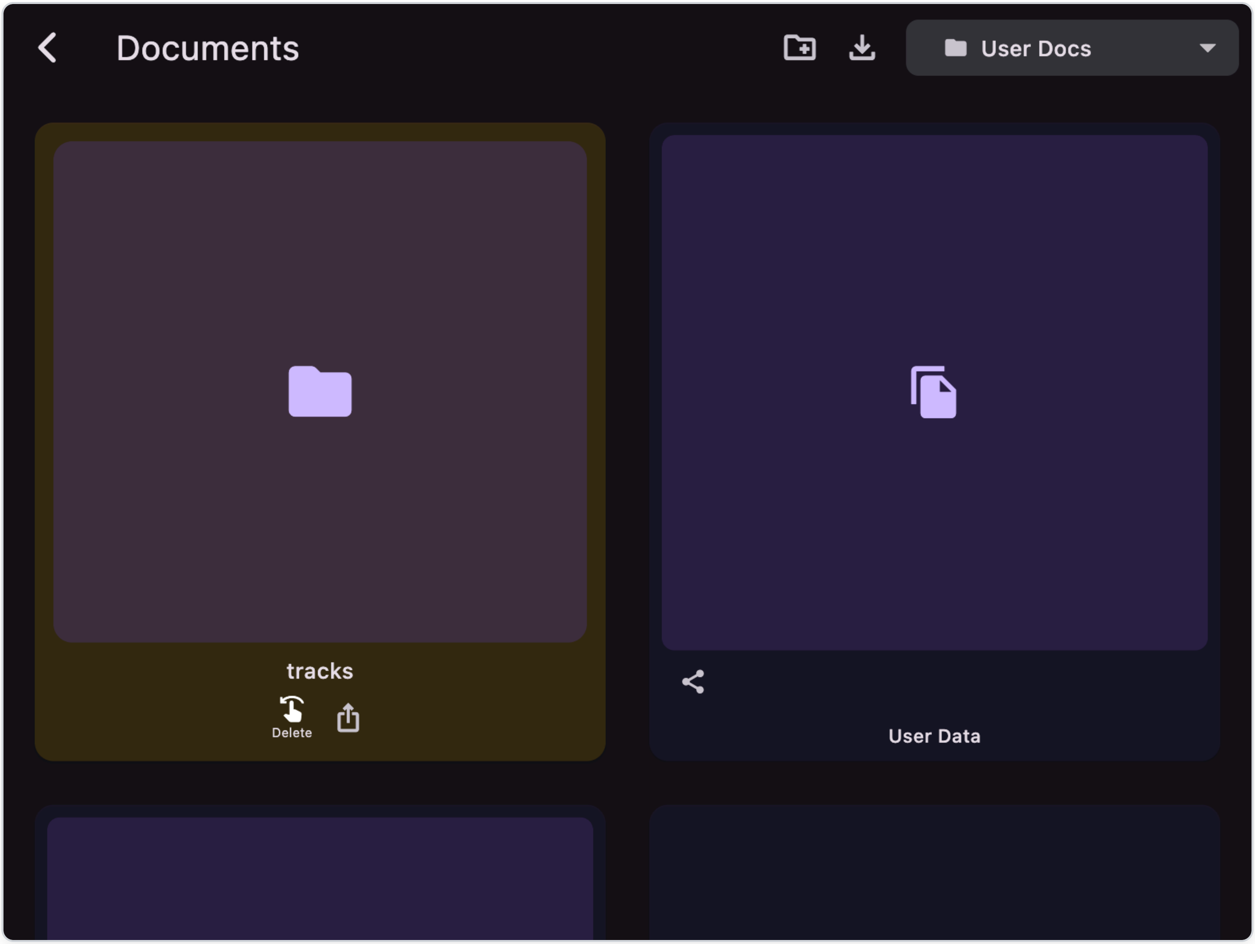
Major download categories include:

- **Databases**: DatabasesX (required for core nav data; fetched automatically during onboarding for your GPS region)
- **VFR Charts**: Sectional, TAC, Flyway, Helicopter
- **IFR Charts**: IFR Low, IFR High, IFR Area
- **Procedures**: Plates, CSUP
- **Terrain**: Elevation

Regional coverage (9 regions): Northeast, North Central, Northwest, Southeast, South Central, Southwest, East Central, Alaska, Pacific

9.2 Documents

Open: **Menu → Documents**



Includes:

- Built-in weather/prog chart products (fetched from aviationweather.gov):
  - WPC Surface Analysis & Prognostic charts (6/12/18/24/30/36/48/60/72HR)
  - SigWx Low Level & Mid Level charts (00/06/12/18HR)
  - Radar & SIGMETs
  - AIRMET Tango (turbulence), Sierra (mountains/IFR), Zulu (icing) - 00/03/06/09/12HR
  - Surface Forecast charts (03-18HR)
  - Clouds Forecast charts (03-18HR)
  - Winds/Temperature at altitudes 5000-30000ft (06/12HR)
- User documents (imported or generated)
- User database file (`user.db`) entry
- Subfolders for organization:
  - `tracks/` — saved flight tracks (KML files)
  - `notes/` — saved notes screenshots

Folder management:

- **Create folder**: tap folder icon in app bar to create a new subfolder
- **Delete folder**: use delete icon below folder to remove folder and contents
- **Move files**: use move icon on user files to relocate to a folder or back to root

- **Export folder**: use share icon below folder to export all folder contents (not Linux)
- **Navigate**: tap folder to enter, use back arrow to go up

Filter dropdown: Filter by document category (All Documents, User Docs, WPC, SigWx, etc.)

Import button supports:

- `.txt` — text files
- `.adsb` — ADS-B capture logs (daily files written by external ADS-B/GDL90 receivers)
- `.geojson` — GeoJSON geographic data
- `.kml` — KML track files
- `.pdf` (except Linux)
- `user.db` — user database backup

Document behavior:

- Text: in-app text reader
- PDF: in-app PDF viewer (if supported)
- GeoJSON: parsed into map shapes (visible when GeoJSON layer is on)
- KML: opens Track Viewer with 2D map, altitude profile, and 3D terrain view
- Images: zoomable preview
- Share button available where supported (not Linux)
- Swipe delete for user files (preserves User Data entry)

Track Viewer (KML files)

When you tap a KML file in Documents, the Track Viewer opens with three view modes (toggle via app bar icons):

- **2D Map** (map icon): displays the track on a USGS topo background with:
  - Flight track polyline (blue)
  - Start marker (green takeoff icon)
  - End marker (red landing icon)
  - **Log Flight** FAB button to create logbook entry
- **Altitude Profile** (chart icon): shows altitude vs. distance graph:
  - Line chart of altitude (ft) over distance (NM)
  - Min/max altitude display
  - Total distance in nautical miles
  - Shaded area under curve

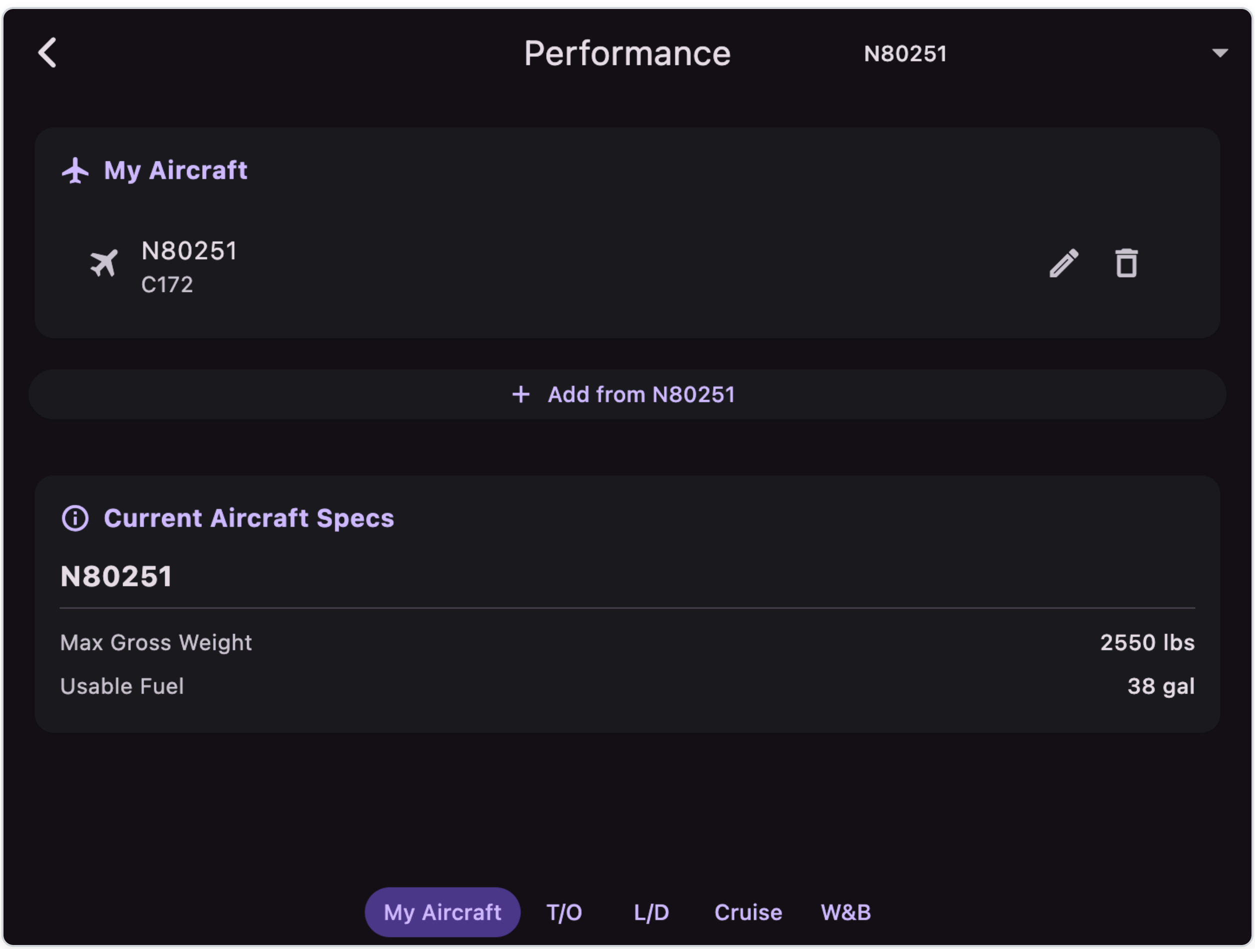
- **3D View** (AR icon): interactive 3D visualization:
    - Terrain grid with elevation data (fetched for 20nm buffer)
    - Topo map texture projected onto terrain mesh
    - Flight path with color gradient (green→yellow→orange→red)
    - Drop lines connecting track to terrain surface
    - Vertical scale with altitude tick marks (1000ft increments)
    - Drag to pan, two-finger rotate, pinch to zoom
    - Reset View button

Auto Logbook Entry Creation (Log Flight button):

- Uses nearest airports for route (takeoff and landing)
- Calculates flight time from track timestamps or from distance/cruise TAS
- Creates entry with aircraft make/model and tail from selected aircraft profile

9.3 Aircraft & Performance

Open: **Menu → Aircraft & Performance**



A unified screen with tabs for managing aircraft profiles, performance calculations, and weight & balance:

Tabs Overview

Tab	Purpose
My Aircraft	Create, edit, and save your own aircraft profiles (POH-style tables and flight-plan fields)
T/O	Calculate takeoff ground roll and 50ft obstacle clearance distances
L/D	Calculate landing ground roll and 50ft obstacle clearance distances
Cruise	Calculate cruise performance (TAS, fuel flow) at altitude/power settings
W&B	Weight and balance calculations with CG envelope

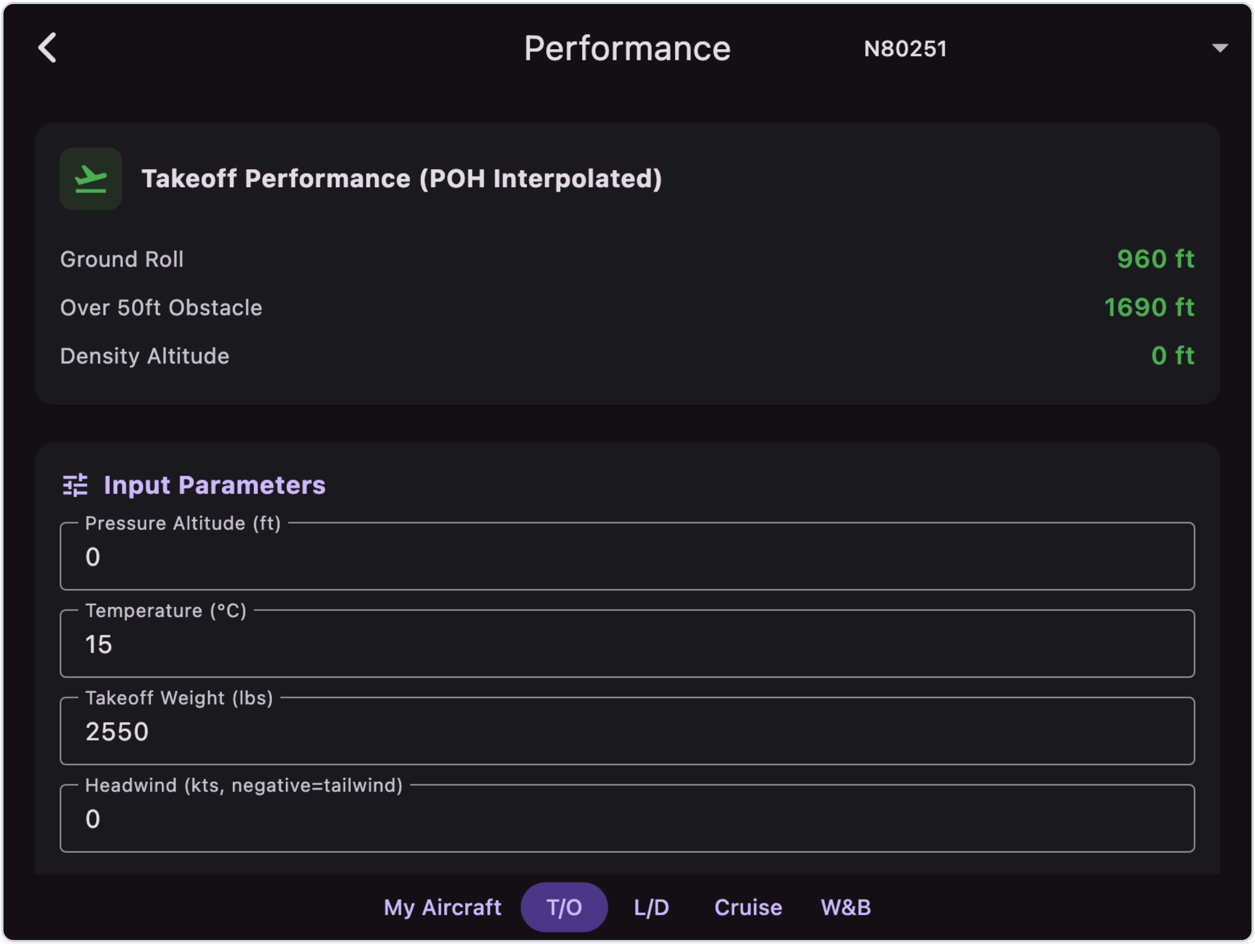
Helicopter (map icon): When the **selected** aircraft uses the helicopter icon (per-aircraft in **My Aircraft**, or the app-wide default in settings if that aircraft has no saved profile row), the **T/O**, **L/D**, and **Cruise** tabs are hidden—only **My Aircraft** and **W&B** remain. The same applies to takeoff, landing, and cruise **POH-style performance tables** when **adding or editing** a profile whose **Aircraft Icon** is **Helicopter**.

Aircraft Selection

Use the dropdown at the top to select from:

- **Built-in aircraft**: C152, C172S, C182T, PA-28-181, PA-28-161, PA-44, Beech A36, Cirrus SR22, Diamond DA40
- **My Aircraft profiles**: User-created profiles saved in your database (same list as in **My Aircraft** tab)

Takeoff Tab



Enter conditions to calculate takeoff performance:

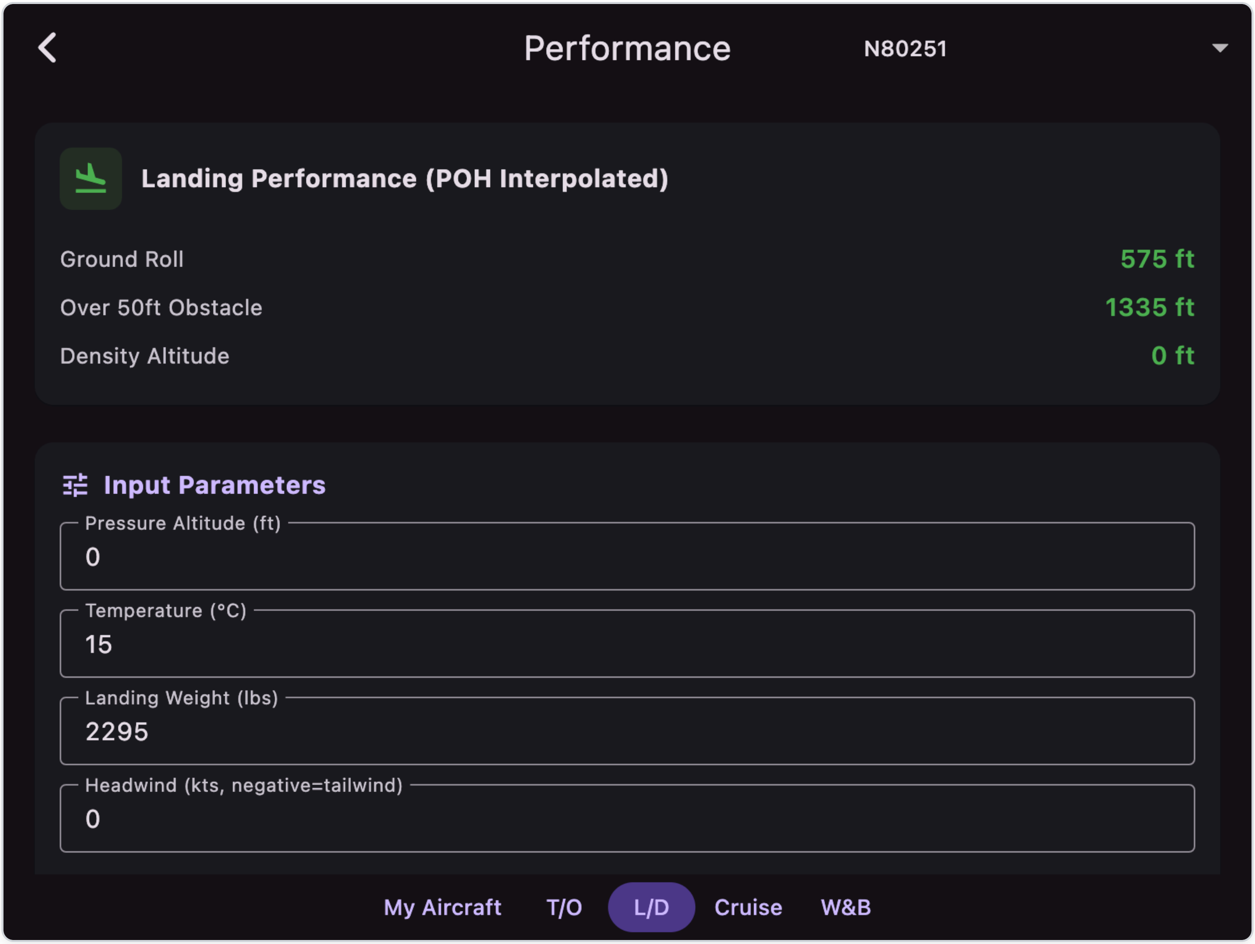
- **Pressure Altitude** (ft)
- **Temperature** (°C)
- **Weight** (lbs)
- **Headwind** (negative for tailwind)

Results show:

- Ground Roll
- Distance to clear 50ft obstacle
- Density altitude (ft) — computed as pressure altitude plus 120 × (OAT − ISA temperature), with ISA 15°C at sea level and −2°C per 1000 ft pressure altitude
- Soft field adjustments (if configured)

Speed and distance values follow the app unit setting (onboarding) and are shown without unit suffixes.

Landing Tab



Enter conditions to calculate landing performance:

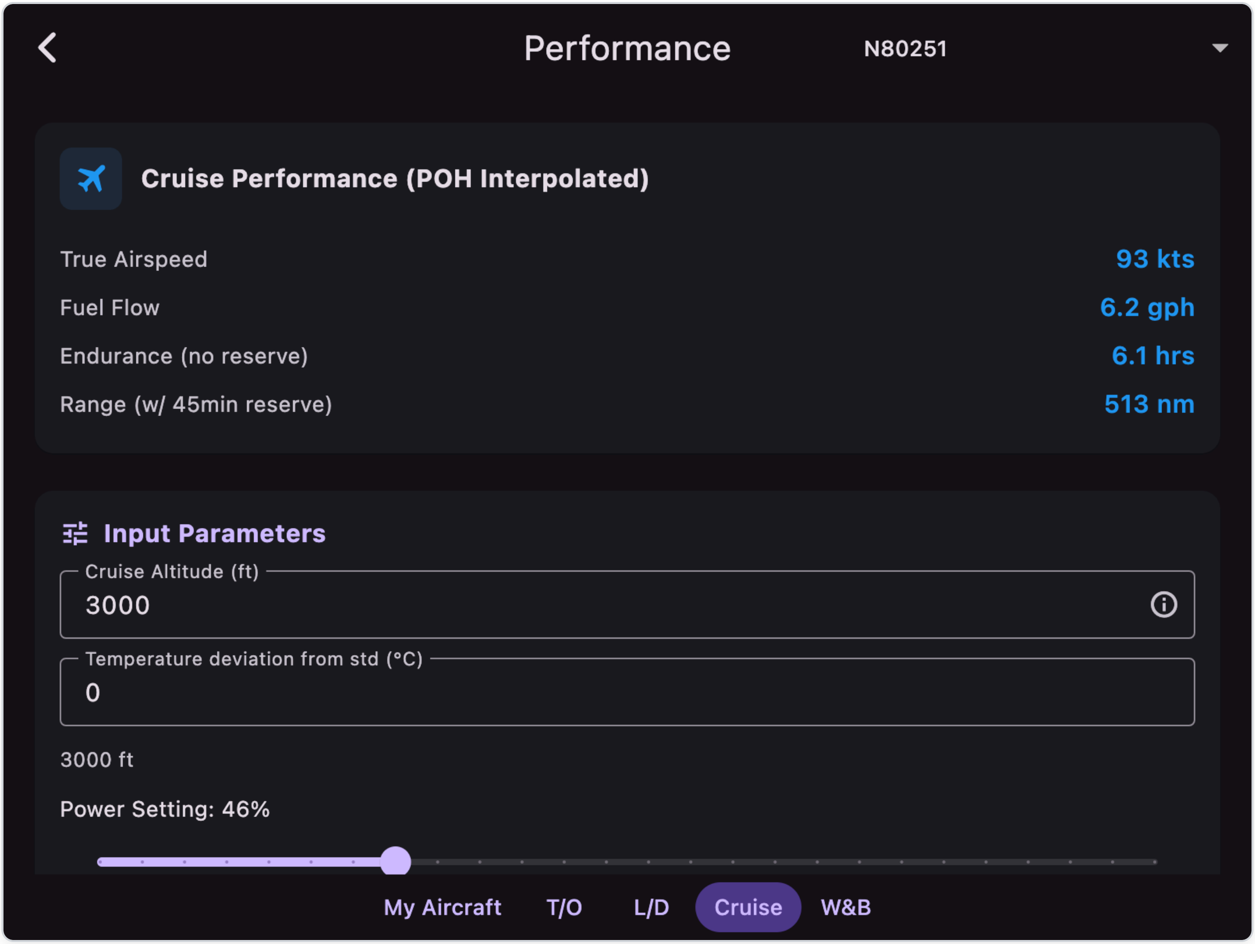
- **Pressure Altitude** (ft)
- **Temperature** (°C)
- **Weight** (lbs)
- **Headwind** (negative for tailwind)

Results show:

- Ground Roll
- Distance to clear 50ft obstacle
- Density altitude (ft) — same formula as takeoff (OAT vs ISA at that pressure altitude)
- Soft field adjustments (if configured)

Speed and distance values follow the app unit setting and are shown without unit suffixes.

Cruise Tab



Enter cruise conditions:

- **Altitude** (ft) — prefilled from the **PLAN** tab **Alt** when you open the Cruise tab (or open Aircraft & Performance); you can still edit it for what-if
- **Temperature deviation** (°C from standard)
- **Power setting** (%) — remembered between app sessions via **app settings** (same preference store as **ASpd** / **GPH** on the Plan tab), not the aircraft database table

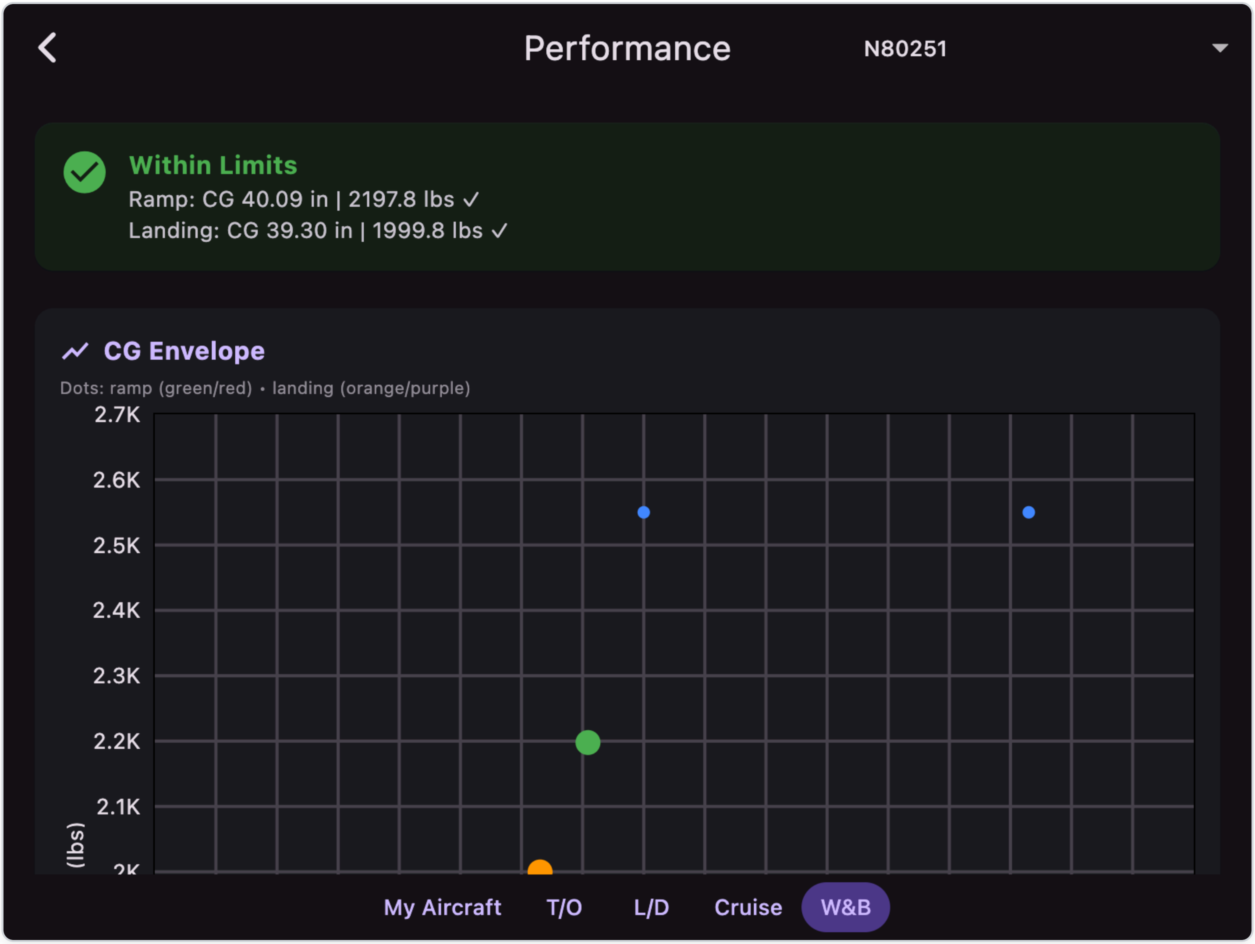
Below altitude and temperature deviation, the screen shows **density altitude** (ft) for those inputs (read-only).

Results show:

- True Airspeed
- Fuel Flow (GPH)
- Range (with 45 min reserve)

Speed and range follow the app unit setting and are shown without unit suffixes.

W&B Tab



Status Card: Green "Within Limits" only when **both** ramp and landing CG are inside the envelope; otherwise red "Outside Limits". Shows ramp and landing lines (CG in inches, weight in lbs, each with ✓ or ✕).

Helicopter map icon: If the aircraft's **map icon** is set to helicopter (per-aircraft in **My Aircraft**, or the app default in settings when no saved profile applies), the W&B tab adds **lateral** CG alongside **longitudinal**: a second status line for ramp and landing, a **Lateral CG** scatter chart (weight vs inches from centerline), and extra table columns (**Lateral** arm, **Longitudinal moment**, **Lateral moment**). Longitudinal behavior is unchanged. "Within limits" requires **all four** checks (ramp longitudinal, ramp lateral, landing longitudinal, landing lateral). Re-open the **W&B** tab after changing the icon so the layout refreshes.

Interactive CG Envelope Chart:

- Scatter plot showing weight (Y-axis) vs arm (X-axis); helicopter mode labels this chart **Longitudinal CG**
- Blue dots: envelope boundary points
- Large **green/red** dot: ramp / takeoff CG (uses all stations except **Landing fuel**)
- Large **orange/purple** dot: landing CG (uses **Landing fuel** instead of **Fuel (lbs)**)
- For custom aircraft: tap chart in edit mode to add/remove envelope points (CG dots are not removable)

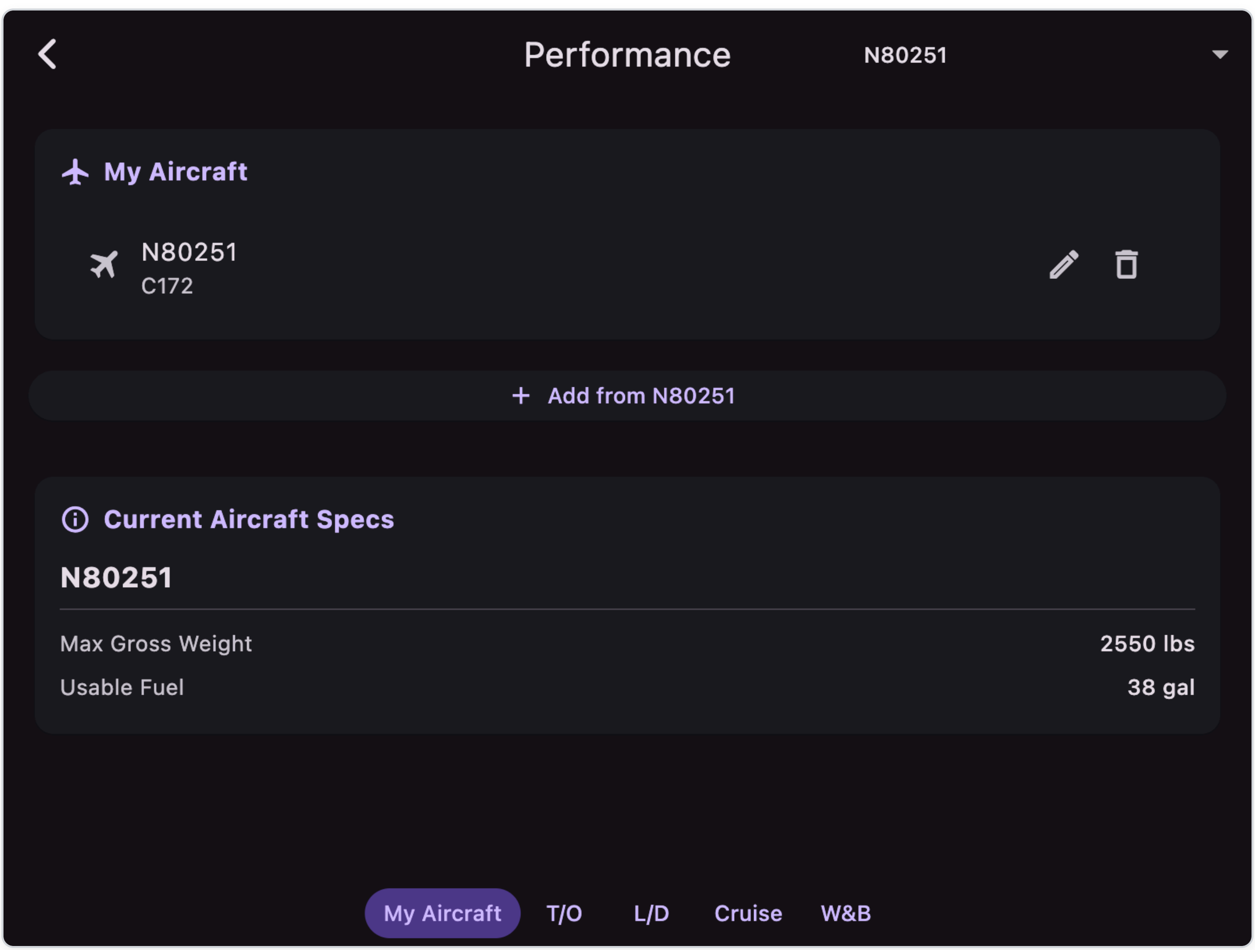
Weight Stations Table:

- Editable weight entries for each station
- Name, weight (lbs), arm (inches); helicopter mode also **Lateral** (lateral arm, inches from centerline) and a **Longitudinal** column for longitudinal arm
- **Landing fuel (lbs)**: expected fuel remaining at landing (same fuel arms as **Fuel (lbs)** by default); excluded from ramp totals
- Auto-calculated **longitudinal** moment (weight × longitudinal arm); helicopter mode also **lateral** moment (weight × lateral arm)
- **TOTAL (ramp)** and **TOTAL (landing)** rows with weight, CG arm(s), and summed moment(s)

For **custom aircraft**: Use **Edit/Save** button to modify envelope points and station definitions.

For **built-in aircraft**: W&B data is from POH (read-only envelope, editable weights).

My Aircraft tab



Create and manage your saved aircraft profiles:

Aircraft list: Tap a row to select that profile for the other tabs. Use the **edit** (pencil) icon to change an existing profile and **Save changes**; use **delete** to remove it from the database.

Add from template: Starts a new profile using the currently selected built-in or saved aircraft as a starting point, then **Save aircraft** to store it.

Aircraft Identification:

- Tail Number, Type, Wake Turbulence
- Color & Markings, Mode S Code
- Map icon selection (plane, helicopter, canard)

Pilot Information:

- PIC name and contact info
- Home Base

Flight Plan Fields:

- Equipment codes, Surveillance codes
- Cruise Speed, Fuel Endurance
- Other Information

Glide circle (MAP `Circles` layer):

- **Best Glide Speed** — POH best-glide speed of the **selected** aircraft (Performance dropdown). Used instead of current GPS ground speed. Entered in the same speed units as the rest of the app (onboarding unit setting).
- **Sink Rate** (fpm) — power-off sink rate at best glide
- If Best Glide Speed is empty, Cruise Speed is used; if that is also empty, 65 is assumed.

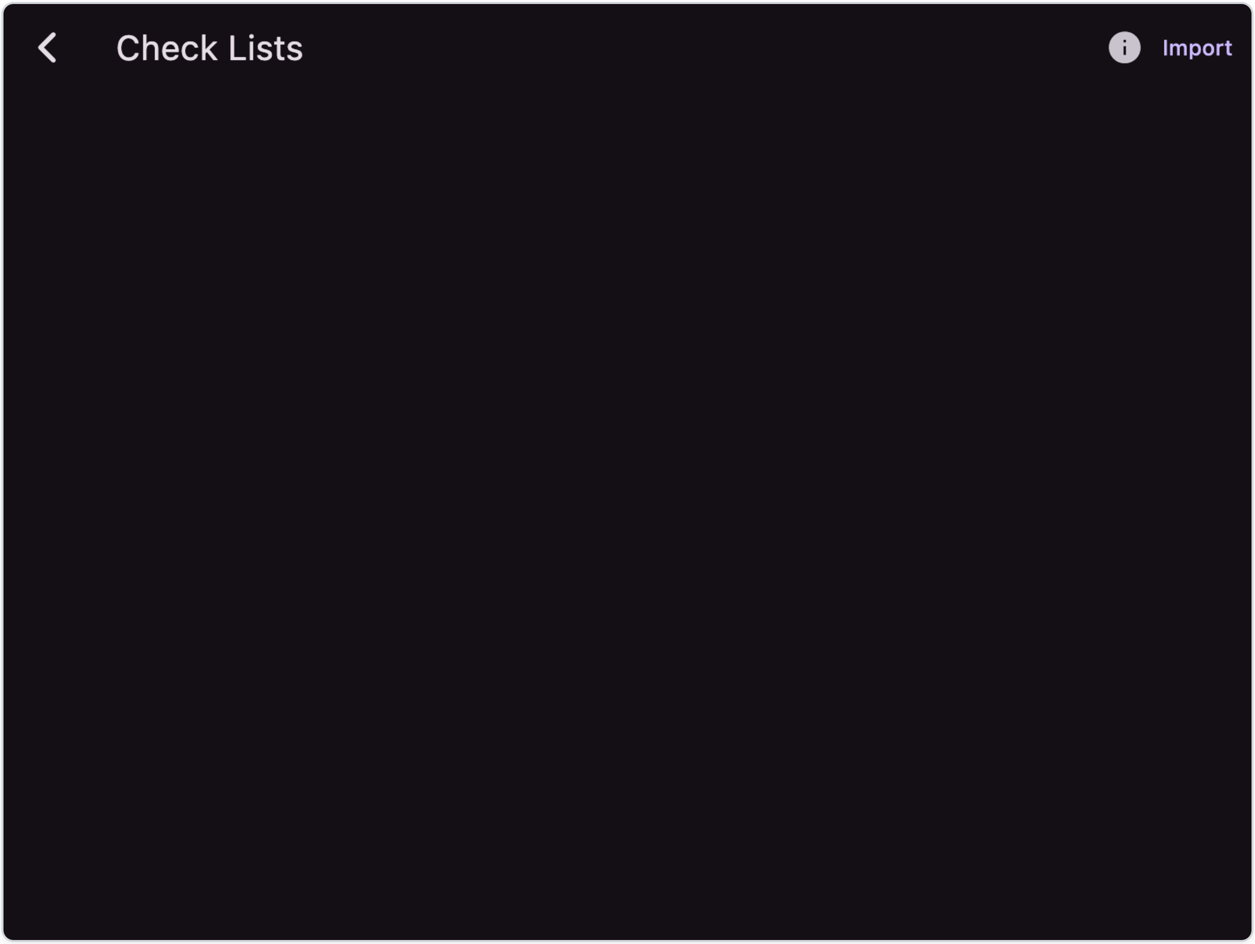
Performance Data:

- Max Gross Weight, Usable Fuel, Empty Weight
- Takeoff/Landing correction factors (headwind %, tailwind %, soft field %)
- Custom performance tables (altitude, temperature, weight entries)

Performance Tables: Add rows for takeoff, landing, and cruise performance at various conditions (omitted when **Aircraft Icon** is **Helicopter**). The app interpolates between entries for accurate calculations. Under each row, a read-only **density altitude** (ft) is shown from that row's altitude and temperature (takeoff/landing use °C OAT; cruise uses the same ISA temperature deviation convention as the Cruise tab).

9.4 Check Lists

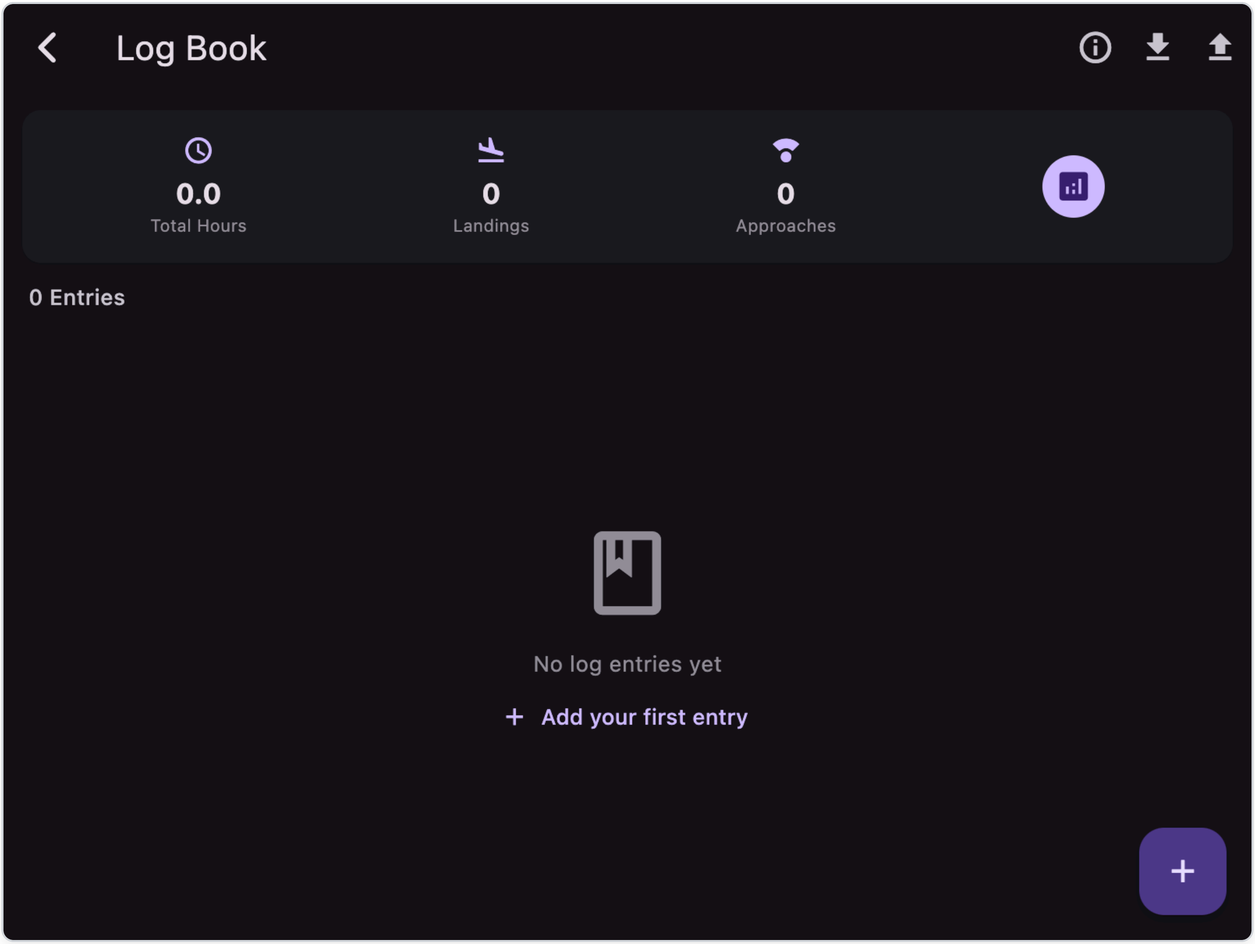
Open: **Menu → Check Lists**



- **Progress bar**: shows completion percentage with completed/total count
- **Checkbox items**: tap to check off steps
- **Visual feedback**:
  - Checked items: green background, strikethrough text
  - All complete: green progress bar and background
- **Reset button**: reset all items to unchecked
- **Multiple checklists**: dropdown to switch between saved checklists
- Import checklist from `.txt` (first line is title, following lines are steps)
- Swipe delete to remove checklist
- Info tooltip explaining import format
- Persistent state during session

9.5 Log Book

Open: **Menu → Log Book**



Features:

- **Summary Card**:
    - Total Hours, Total Landings, Total Approaches
    - Analytics button to view dashboard
- **Flight List**:
    - Flight time (hours), aircraft make/model (tail)
    - Date, route
    - Day/Night landings, instrument approaches
    - Tap entry to edit, or use FAB to add new entry
- **Log Entry Form** (sections):
    - Flight Info: Date, Aircraft Tail, Type, Route
    - Flight Time: Total, Day, Night, Cross Country, Solo
    - Pilot Function: PIC, SIC, Dual Received, Instructor, Examiner
    - Instrument: Actual IMC, Simulated, Approaches, Holds
    - Landings: Day, Night
    - Training & Simulation: Ground Time, Simulator, Instructor Name/Certificate

    - Remarks: Free-form notes

- **CSV Import/Export**:

    - Import from CSV file (header row required)
    - Export to CSV (share via system share sheet, not Linux)
    - Info tooltip explaining format

Logbook Dashboard (Details)

Open by tapping **Details** button:

- **Currency Status Card**: Day currency (3 landings/90 days), Night currency (3 landings/90 days), IFR currency (6 approaches/6 months)
- **Summary Card**: Total hours, PIC, Night, XC, Landings, Approaches
- **Filters**: Year, Aircraft Type, Tail Number, Remarks keywords (Check Ride, IPC, Flight Review, Favorite)
- **Bar Charts**:
    - Flight Hours by Year
    - Hours by Aircraft Type
    - Hours by Tail Number
    - Hours by Category (Solo, Dual, PIC, SIC, Day, Night, IMC, XC, etc.)
    - Landings & Approaches
- **Filtered flight list**: scrollable list of filtered entries

9.6 IO (Bluetooth) — Android only

Open: **Menu → IO**

Functions:

- **Device Discovery**: auto-discovery starts when screen opens
- **Device List**: shows device name, address, signal strength (RSSI in dBm)
    - RSSI color-coded: green = strong, red = weak
    - Icons indicate connection status: connected (↔), paired/bonded (link icon)
- **Device Operations** (tap device for dialog):
    - **Pair**: bond with unpaired device
    - **Unpair**: remove device bond
    - **Connect**: establish Bluetooth SPP connection (for paired devices)
- **Connection Status Bar**: shows currently connected device with disconnect button

Connected stream feeds external data input into app parser for GPS, traffic, weather, AHRS. Also used for autopilot output of navigation data.

9.7 Help

Open: **Menu → Help**

- Opens the User Manual PDF for in-app reference.
- Available on platforms that support PDF viewing (not Linux).

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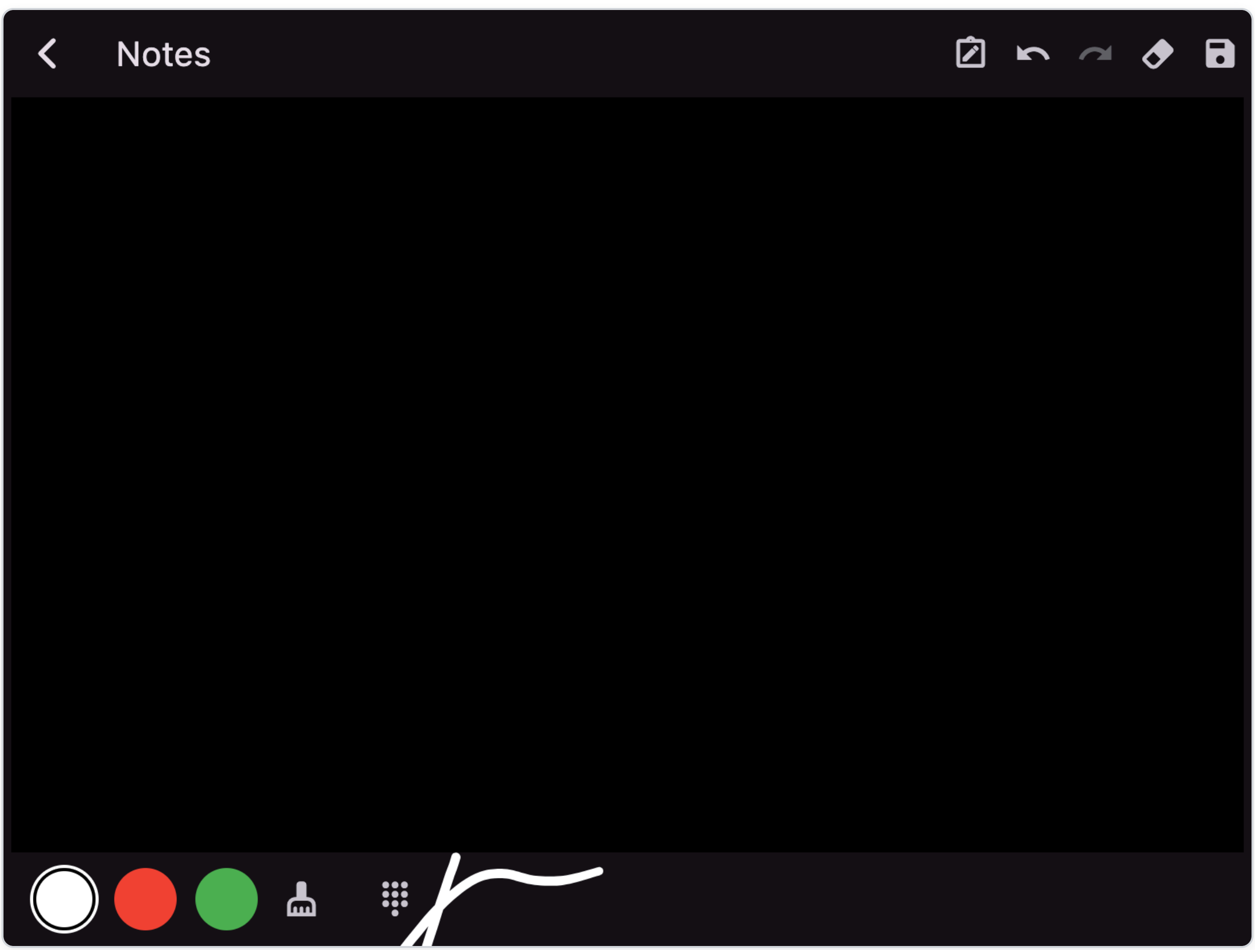
10) Notes Screen

10.1 How to access

On MAP tab, tap the notes/pen (transcribe) icon in bottom-right controls.

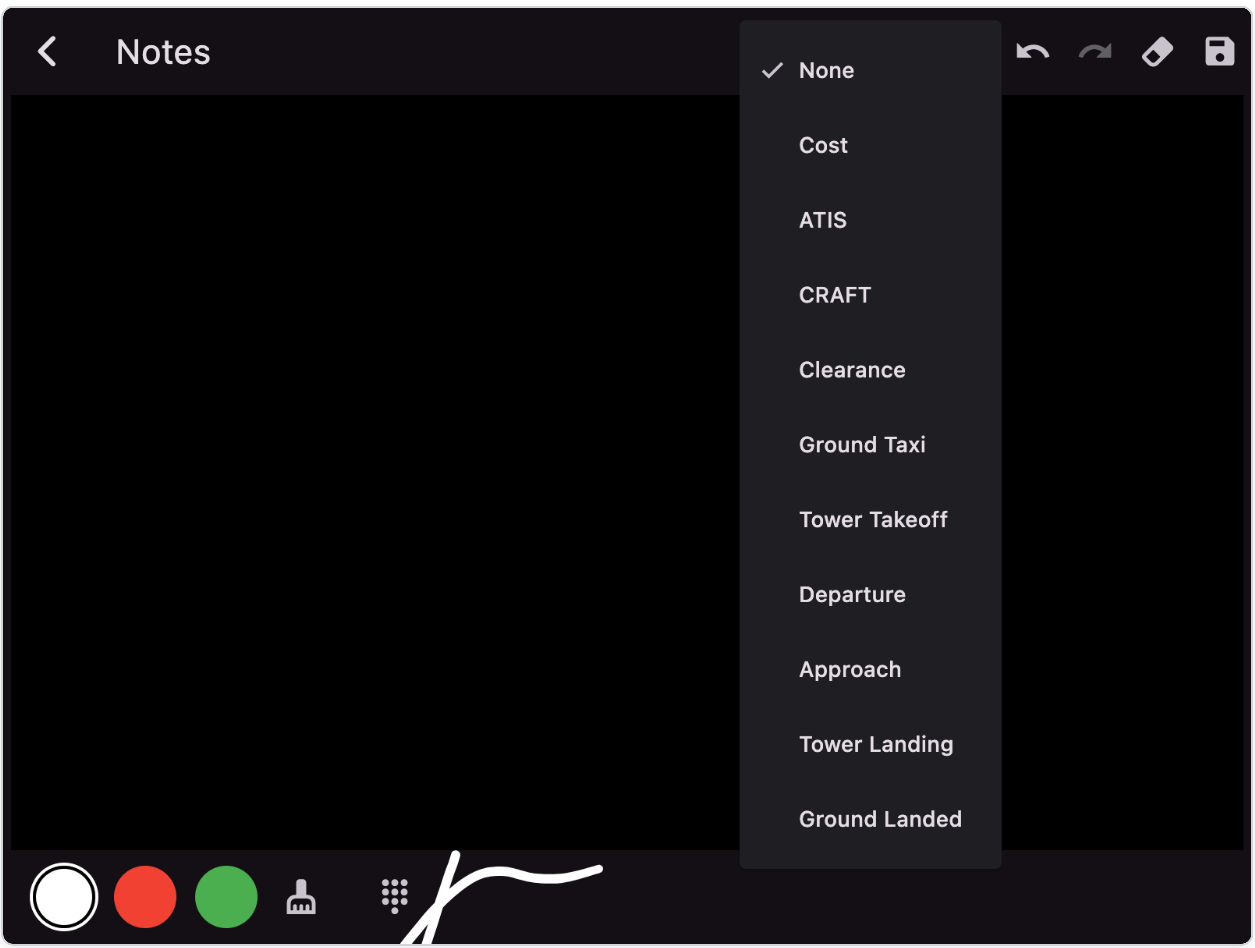


10.2 Features

- **Freehand drawing** with finger or stylus
- **Color choices**: Black/White (theme-adaptive), Red, Green
- **Eraser mode**: tap cleaning services icon in toolbar (wide stroke eraser)
- **Undo/Redo**: full stroke history
- **Clear canvas**: eraser icon in app bar
- **Save snapshot**: saves to Documents/notes as `notes_<timestamp>.png`
- **Auto-save**: current sketch autosaves on exit and reloads next time

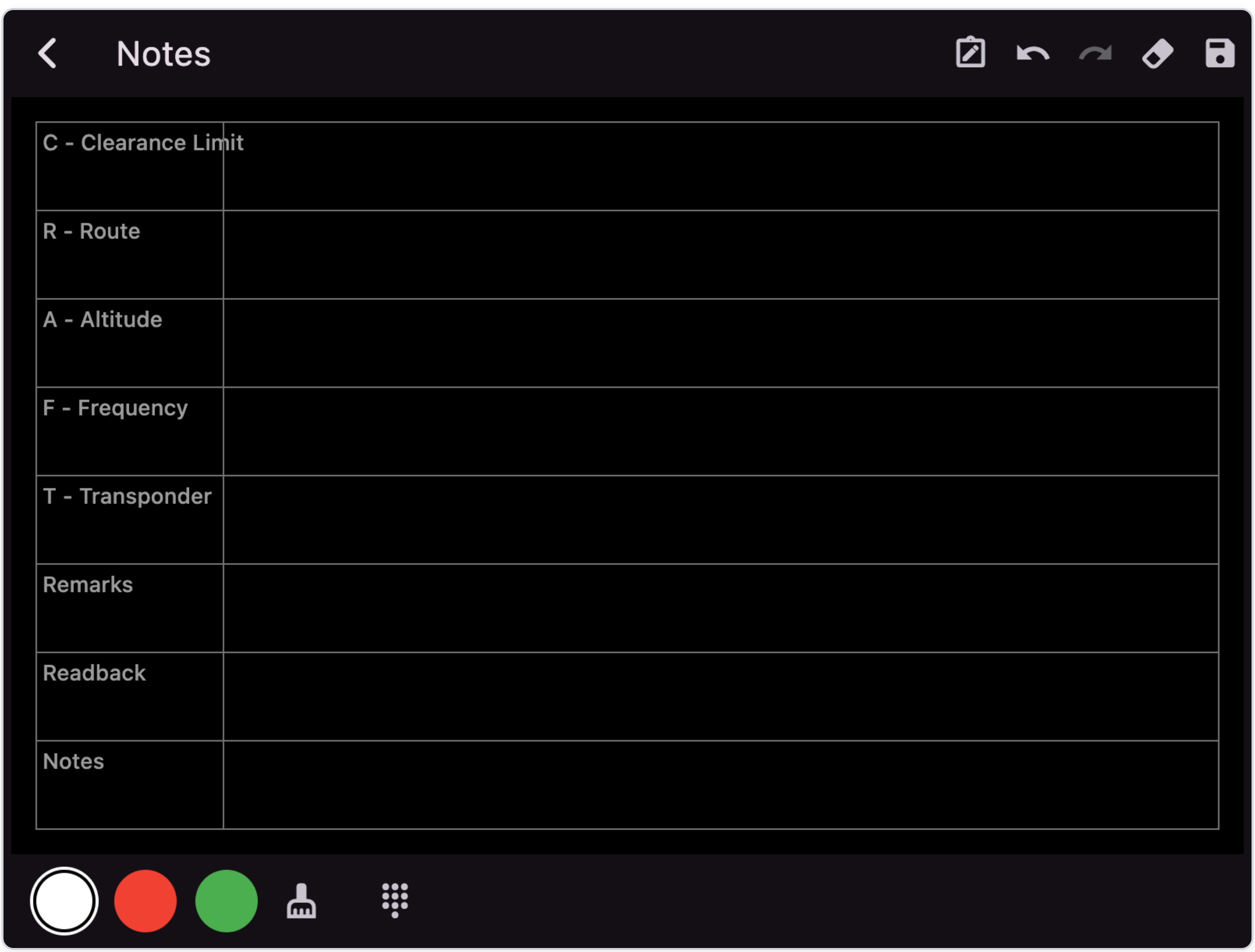
10.3 Aviation background sheets

Use the sheet icon in the app bar to select a background template for common aviation communications:



Sheet	Purpose
None	Blank canvas
Cost	Flight times (Hobbs, Tach, Fuel, Oil)
ATIS	VFR ATIS copydown fields (Airport, ATIS letter, weather, runways, NOTAMs)

Sheet	Purpose
CRAFT	IFR clearance (Clearance limit, Route, Altitude, Frequency, Transponder)
Clearance	VFR flight following clearance
Ground Taxi	Ground control taxi instructions
Tower Takeoff	Tower takeoff clearance
Departure	Departure control contact
Approach	Approach control contact
Tower Landing	Tower arrival/landing
Ground Landed	Ground control after landing



Each sheet maintains its own saved sketch state. Switching sheets saves the current sheet and loads the new one.

10.4 Number keypad

Tap the dialpad icon in the toolbar to display an on-screen keypad:

Number pad (default):

- Digits 0-9, decimal point
- Space, newline, backspace
- **C** (Clear): removes the last text entry; press repeatedly to remove more
- **ABC**: switches to QWERTY letter keyboard

QWERTY keyboard:

- Letters A-Z in standard QWERTY layout
- Space, backspace, newline
- **C** (Clear): removes the last text entry
- **123**: switches back to number pad

General usage:

- **Tap anywhere on canvas** while keypad is active to create a new text entry at that location
- Type to add text at the current location, then tap elsewhere to add text at another spot
- Supports multiple text entries at different positions on the same sheet
- All text entries are saved with the sheet and persist across sessions

---

11) Pro Services (iOS/Android only)

11.1 Access

From MAP top-right account icon, or by routed requests from some features (e.g., translate actions).

11.2 Login and subscription flow

- Sign in/register with email authentication (Firebase)
- Paywall handled through RevenueCat entitlement (`Pro`)
- After login, bottom sheet shows available Pro features

11.3 Flight Intelligence (AI)

Screen title: **Flight Intelligence**

Powered by Gemini 2.5 Pro with Google Search integration.

Capabilities:

- Ask free-form aviation/trip questions (internet required)
- Question must be 256 characters or less
- Context limit: 10,000 tokens

Context toggles (tap to highlight, highlighted context sent with question):

- **Logbook icon**: include your 50 most recent logbook entries
- **Aircraft icon**: include tail number and type from first aircraft
- **Route icon**: include current flight plan
- **Cloud icon**: include winds aloft from departure and destination

Query/answer workflow:

- Type question in text box
- Tap **Ask** to submit
- Response appears in text box with disclaimer
- History drawer (clock icon): view previous questions/answers, choose question, show answer, delete history item
- **Clear (X)** button clears text box

Disclaimer: Responses are generated by an AI model and may not be accurate or reliable. Do not use when life, health, property are at stake.

11.4 Pilot Community

Screen title: **Pilot Community**

A pilot-only social network backed by Firebase. Pilots can build a public profile, create groups, and join existing ones. **Private** groups require owner approval; **public** groups are open to all signed-in Pro users.

Disclaimer (shown as a strip at the top of the screen and via the info icon in the app bar):

- Apps4Av is **not responsible for data loss** in the Pilot Community. Keep your own copies of anything you want to retain.
- **Do not share sensitive information** (passwords, government IDs, financial details, medical records, etc.). Anything you post may be visible to other pilots.

- Apps4Av **does not moderate** community activity. Group owners are responsible for their own groups.
- Apps4Av **will not share community data with third parties**. Data is stored only in the project's Firebase backend to operate this feature.

Tabs:

- **My Groups**: groups you own or have been approved to join.
- **Discover**: searchable list of public groups, sorted by member count.
- **Profile**: your shared pilot profile (display name, home airport, ratings, aircraft, bio).

Top action:

- **New Group** (floating action button): create a new group with name, description, optional home airport (ICAO), and visibility (Public or Private).

Group detail screen tabs:

- **Feed**: scrollable list of topics (top-level posts). Active members can **Post** (floating action button) up to 1000 characters and optionally attach:

    - **Photos** (up to 4 per post, JPEG/PNG, automatically downscaled to 1920px wide and re-encoded at quality 80, which also strips EXIF). Tap a photo to zoom.
    - **A flight plan** — tap **Attach plan** to share your current PLAN. Other members see the route inline with a **Load** button that imports the plan into their PLAN tab (after a confirmation dialog).
    - **An airport ICAO** quick-tag.

    Threaded discussions: each topic keeps its own thread of replies. Tap **Reply** on a topic to add a reply (text and/or photos), or **tap the topic** to open its discussion thread — the topic at the top followed by all replies, oldest first, with a **Reply** button. The reply count is shown on each topic in the feed. Replies stay attached to their topic, so a conversation never gets mixed up with unrelated posts.

    Unread topics: topics posted since you last opened a group's feed are shown in **bold**. Opening the feed marks it read, so they return to normal weight on your next visit (your own posts are never shown as unread).

    Authors can delete their own posts; the group owner can delete any post. Deleting a topic also deletes all of its replies. Deleting a post removes its attached photos from cloud storage on a best-effort basis.

- **Members**: list of active members with their home airport. The owner sees a separate **Pending requests** section at the top with **Approve** / **Reject** controls and can **Remove** any non-owner member.

- **About**: owner, visibility, home airport, member/post counts, creation date.

Membership controls (shown as a banner inside each group):

- **Join** (public group): immediate active membership.
- **Request to Join** (private group): puts you in `pending`; owner must approve before you see the feed or can post.
- **Requested**: you have a pending request; tap to cancel.
- **Leave**: removes you from the group. Owners cannot leave — they must **Delete group** from the app bar (deletes all posts and memberships).

Visibility rules:

- Public groups are discoverable and any signed-in Pro user can read the feed.
- Private groups are discoverable by name but the feed is hidden from non-members. Owners get a red badge over the **Members** tab when pending requests exist.

Notifications: in-app only (no push). A **bell icon** appears both in the `Pilot Community` app bar and above the `Community` button on the Pro Services screen; it shows an unread count and opens the **Notifications** screen. You are notified when:

- someone **replies to a topic you started**, or
- someone **replies to any post in a group you own**.

On the Notifications screen, unread items are shown in **bold** with a dot. You can **tap** a notification to jump straight to that discussion thread (which marks it read), **swipe** a notification away to dismiss it, **mark all read** (`✓✓` icon), or open **notification settings** (gear icon). Settings let you turn reply notifications **off globally**, or **mute individual groups** — muted groups and a global "off" stop counting toward the unread badge and are hidden from the list. Pending join requests still appear as a red badge over the **Members** tab when you open a group you own.

11.5 Aircraft Scheduler

Screen title: **Aircraft Scheduler**

A shared booking calendar and club dispatch board for flying clubs, partnerships, and flight schools, backed by Firebase. It works like the Pilot Community: an owner creates a scheduler, members request to join, and members reserve the resources the owner adds. **All schedulers**

are private — they are discoverable by name, but the owner approves every member, and the schedule and roster are hidden from non-members.

Tabs:

- **My Schedulers**: schedulers you own or have been approved to join.
- **Discover**: type a name to find a scheduler to request to join. There is no public browse list because all schedulers are private.

Top action:

- **New Scheduler** (floating action button): create a scheduler with name, description, and optional home airport (ICAO). The creator becomes the owner. Every scheduler is private; set booking limits afterwards from the **Booking Rules** screen.

Scheduler detail screen tabs:

- **Schedule**: the booking timeline. Resources are stacked **vertically** (one row each) and time runs **horizontally** across the day. Colors indicate slot state:
  - **Green** = available (free) slot on an in-service resource.
  - **Blue** = booked. A solid block is the main reservation; a thin lighter-blue bar at the bottom of the row is a backup. Your own reservations have a white border.
  - **Red** = unavailable (the resource is marked out of service by the owner/dispatcher).

  Use the date bar (**‹ / ›**, the date button to pick a day, and **Today**) to change days. Tap a **green** slot to open the **Book** dialog, which pre-fills the tapped day/time as the start; choose a **start date + time** and **end date + time**. Bookings may span **multiple days** (up to 14 days). A multi-day reservation appears as a booked (blue) block on every day it covers. If the resource is already booked for any of that time, you are added to the **backup queue** instead. When booking an aircraft you can optionally assign an **instructor** and/or link a **lesson pack**. Tap any reservation block to see details (including training assignment) and cancel it. Aircraft with an open **grounding squawk** or overdue **100-hour** hobbs cannot be booked.

- **Dispatch**: club ops board for the shared fleet (members only):
  - **Fleet**: status cards for each aircraft (READY / ATTENTION / MX DUE / GROUNDED / OUT OF SERVICE) with hobbs, tach, annual / 100-hour / transponder / ELT due, open squawk counts, and MX notes. Tap a card to view details. Owners and **dispatchers** can **Update meters / MX** and toggle availability; any member can **File squawk**.
  - **Squawks**: open and resolved discrepancy reports (Grounding / Caution / Info). An open **Grounding** squawk blocks booking and shows the aircraft as GROUNDED on

    the fleet board. The reporter, owner, or dispatcher can **Resolve** a squawk.
  - **Lessons**: prepaid **lesson packs** (hour blocks) assigned to students. Owners create packs and can cancel/delete them. The student, assigned instructor, owner, or dispatcher can **Log hours**; when used hours reach the total, the pack completes automatically.

- **Mine**: your own reservations in this scheduler, split into **Upcoming** and **Past**. Upcoming entries can be cancelled here directly. When the owner has set booking limits, chips at the top show your current usage (e.g. `2/3 active`, `1/1 weekend`). Weekend reservations and backups are tagged.

- **Members**: list of active members with **club roles** (Pilot / Student / Instructor / Dispatcher). The owner sees a **Pending requests** section with **Approve** / **Reject** controls, can **Remove** any non-owner member, and can set club role via the badge icon. Students can be linked to an **assigned instructor**; that instructor is pre-selected when the student books.

- **About**: owner, visibility, home airport, member/resource counts, booking limits, creation date, and a short explanation of how booking and dispatch work.

Owner app-bar actions:

- **Booking rules** (tune icon): open the **Booking Rules** screen to set limits.
- **Delete scheduler** (trash icon): remove the scheduler and all resources/reservations/memberships/squawks/lesson packs.

Resources (owner only): tap **Add resource** on the Schedule tab to add an **Aircraft** or **Instructor** (name plus optional tail number/identifier). Tap a resource's name in the left column to **toggle availability** (mark it out of service for maintenance — shown in red) or **Delete** it (which also removes its reservations). Hobbs/tach and MX due dates are edited from the **Dispatch → Fleet** board (owner or dispatcher).

Booking Rules screen (owner only): set limits that apply to members (the owner is exempt). A value of **0 means unlimited**.

- **Reservations per member**: the most current/upcoming reservations a member may hold at once.
- **Weekend reservations**: the most Saturday/Sunday reservations a member may hold at once.

Limits are enforced when a member books; exceeding a limit shows an explanatory message and the booking is blocked.

Booking and backups:

- Any active member can book an available resource that is not already taken, subject to the owner's booking limits and dispatch holds (OOS, grounding squawk, overdue 100-hour).
- If a resource is already booked for an overlapping time, the new request joins the **backup queue** (Backup #1, #2, ...).
- When a **main** reservation is cancelled, the next backup in line is automatically **promoted** to the main reservation.
- **Only the member who made a reservation or the scheduler owner can cancel it.** The owner can cancel any member's reservation.

Club roles:

Role	Typical use
Pilot	Default member; can book and file squawks
Student	Linked to an assigned instructor; bookings can carry that instructor and a lesson pack
Instructor	Appears in booking and lesson-pack instructor pickers
Dispatcher	Same fleet/MX/squawk resolve powers as the owner (without owning the scheduler)

11.6 Backup/Sync

Screen title: **Backup/Sync**

Cloud operations for `user.db` using Firebase Storage:

- **Backup** (upload icon): upload local database to cloud storage
  - Warning: overwrites existing cloud backup
- **Restore** (download icon): download cloud backup over local database
  - Warning: overwrites existing local data
- Progress indicator shows transfer percentage
- Both actions show confirmation warnings before proceeding
- Data stored under user's Firebase UID

11.7 Airport Businesses & Reviews

A crowd-sourced directory of airport businesses/FBOs that pilots build together. Business names are pre-seeded from the app's built-in business database; everything else is contributed by signed-in pilots.

This is a free feature — it is **not** part of a Pro subscription. It only requires a signed-in account (the same account used for Pro Services) so that contributions are accountable rather than anonymous. It is available on iOS and Android, where cloud accounts are supported.

Access: destination popup → `Business` tab. When you are signed in, the airport's businesses are listed **directly inside the tab** — there is no separate screen to open. If you are not signed in, the tab shows a sign-in prompt; tapping it takes you to the sign-in screen (the only time this feature navigates away), and no subscription or paywall is involved. Once signed in the tab refreshes to show the businesses in place.

What you can do (all in place, without leaving the tab):

- **Add**: tap **Add** to create a new listing at the airport with its name, services, fuel types, operating hours, phone number, and radio frequency.
- **View details & reviews**: tap a listing to expand it inline and see its fuel, services, operating hours, phone number, radio frequency, and reviews.
- **Edit details**: on an expanded listing, tap **Edit details** to add/update its services, fuel available, operating hours, phone number, and radio frequency. The phone number and radio frequency fields use a numeric keypad. Details are community-owned, so any signed-in pilot can enrich a listing.
- **Set fuel prices**: on an expanded listing, tap **Set fuel prices** to enter a price per gallon for any fuel type. Each price is shown with the date it was last set; leave a field blank to clear that fuel's price.
- **Add Review**: on an expanded listing, tap **Add** next to *Reviews*, choose a **1–5 star** rating and optional text. You can post **one review per business** — reviews are permanent and can't be edited or deleted.

Ordering: listings that pilots have created, updated or reviewed are floated to the top of both the `Business` tab list and the `PLATE` business selector (most recently touched first), marked with a small verified-user icon. The remaining (seeded) listings follow **nearest-first** from the airport, so the most relevant businesses appear before distant ones. The “interacted” status is stored on the listing in the cloud, so that part of the ordering is the same for everyone.

Accountability: every listing and review is tagged with your profile display name — nothing is anonymous, and each display name is **unique** across pilots so it can't be used to impersonate someone else. To keep a durable, attributable record, **listings and reviews cannot be deleted or edited once posted** (only detail fields like services/fuel/hours can be updated), and each pilot may post **one review per business**. A listing's average rating and review count update automatically as reviews are added.

Offline: businesses are cached on your device, so airports you've already opened stay browsable without a connection. An **offline banner** appears when you're viewing cached data. While offline: airports you've never opened may show nothing until you reconnect, star ratings show **"Ratings offline"** (they're computed on the server), and anything you add or edit is **saved on your device and uploads automatically when you're back online** (you'll see a note instead of the usual confirmation).

On the airport diagram: listings that carry a location (e.g. those seeded from the built-in business database) can be pinned on the `PLATE` airport diagram using its business selector — see Section 6.6.

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12) Warnings and Troubleshooting Drawer

When warning state is active, a red alert icon appears on MAP top-right. Tap it to open issues drawer.

Possible issue items:

- GPS permission denied (opens app settings)
- GPS disabled (opens location settings)
- No GPS lock
- Critical data/charts missing (shortcut to Download)
- Data expired (shortcut to Download)
- Runtime exception notices

iOS: traffic / ADS-B weather / external GPS not appearing

If AvareX shows ownship moving but no traffic, no FIS-B weather, and no external GPS lock from your receiver — even though other apps on the same phone work — iOS has blocked AvareX's local network access.

Fix: `Settings → Privacy & Security → Local Network → AvareX → ON`, then force-quit and relaunch AvareX. See FAQ-16 for full details.

This is the most common cause of "AvareX traffic doesn't work" on iPhone/iPad and only affects iOS, not Android.

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13) Data Lifecycle and Auto-Update Behavior

- On reaching the onboarding **Databases and Maps** page, AvareX automatically downloads the **DatabasesX** package plus the **Sectional**, **Plates**, and **CSUP** for your current GPS region whenever any of them are missing/expired (skips items already current).
- Weather downloads refresh periodically (10-minute cycle in storage timer).
- GPS source modes (tap `SRC` tile to cycle):
  - **Auto**: Prefers external GPS when available, falls back to internal after 30s timeout. Shows `Internal-A` or `External-A`.
  - **Internal**: Uses only internal GPS, discards external GPS data. Shows `Internal` with green background.
  - **External**: Uses only external GPS, discards internal GPS data. Shows `External` with blue background.
- Flight status tracks taxi/airborne transitions and accumulates flight time.
- External autopilot/NMEA sentence output is generated continuously while app is running and IO connection exists.
- Track recording continues while Tracks layer is enabled; saves to KML when layer is turned off.

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14) Quick Feature Path Index

Feature	Path
Download charts/data	`MAP → Menu → Download`
Read weather docs / import files	`MAP → Menu → Documents`
View saved tracks (KML)	`MAP → Menu → Documents → tracks folder → tap KML file`
Create logbook from track	`Documents → tracks → tap KML → 2D Map → Log Flight`
Create folder for documents	`MAP → Menu → Documents → folder icon in app bar`
Build or modify plan	`PLAN tab`
Delete entire plan	`PLAN tab` → trash icon (top-right of header)
File FAA plan	`PLAN → Brief & File`

Feature	Path
Manage filed plans	`PLAN → Manage`
Load reversed plan	`PLAN → Load & Save → 3-dot menu → Load Reversed`
Create IFR preferred route	`PLAN → Create → Create IFR Preferred Route`
Show recent ATC routes	`PLAN → Create → Show IFR ATC Routes`
Send plan to / get plan from Avidyne IFD (Wi-Fi)	`PLAN → Transfer`
Destination details	Long-press on map or tap FIND result
Show plates for airport	Destination popup `Plates` or `PLATE tab`
Get airport satellite view	`PLATE tab → Plate selector → APS-AERIAL VIEW (Get)`
Delete airport satellite view	`PLATE tab → Plate selector → APS-AERIAL VIEW → trash icon`
Configure aircraft	`MAP → Menu → Aircraft & Performance → My Aircraft tab`
Set best glide / sink rate	`MAP → Menu → Aircraft & Performance → My Aircraft tab` (Best Glide Speed, Sink Rate)
Change aircraft map icon	`MAP → Menu → Aircraft & Performance → My Aircraft tab → icon dropdown`
Checklist operations	`MAP → Menu → Check Lists`
Weight and balance	`MAP → Menu → Aircraft & Performance → W&B tab`
Takeoff performance	`MAP → Menu → Aircraft & Performance → Takeoff tab` (fixed-wing icon only; hidden for helicopter)
Landing performance	`MAP → Menu → Aircraft & Performance → Landing tab` (fixed-wing icon only; hidden for helicopter)
Cruise performance	`MAP → Menu → Aircraft & Performance → Cruise tab` (fixed-wing icon only; hidden for helicopter)
Runway crossing alert	Automatic when Mute is off; uses closest-airport runways

Feature	Path
Fuel burn in plan	`PLAN tab` — set **GPH** manually or use the aircraft icon to load from performance data
Logbook + dashboard	`MAP → Menu → Log Book`
Logbook statistics	`MAP → Menu → Log Book → Details`
Currency status	`MAP → Menu → Log Book → Details` (currency card)
Bluetooth pairing/connection	`MAP → Menu → IO` (Android)
Check ADS-B receiver status	`MAP → ADSB instrument tile` (movable; tap opens ADS-B Status screen)
Enable Stratus 3 Open ADS-B Mode	`MAP → ADSB instrument tile → antenna icon (top-right)`
Notes/drawing	`MAP → Notes icon`
Notes with aviation sheet	`MAP → Notes icon → sheet icon → select template`
Notes number keypad	`MAP → Notes icon → dialpad icon`
Pro AI	`MAP top-right account icon → Flight Intelligence`
Pro Community	`MAP top-right account icon → Community`
Create a pilot group	`MAP top-right account icon → Community → New Group`
Join a public group	`MAP top-right account icon → Community → Discover → tap group → Join`
Approve pending members (owner)	`Community → group → Members tab → Approve / Reject`
Edit your pilot profile	`MAP top-right account icon → Community → Profile → edit icon`
Post a photo to a group	`Community → group → Post → Photo → choose source`
Share your current flight plan	`Community → group → Post → Attach plan`
Load a shared flight plan into PLAN	`Community → group → tap Load on a post with a route`

Feature	Path
Reply to a topic	`Community → group → tap topic → Reply`
View a topic's discussion thread	`Community → group → tap a topic in the Feed`
View notifications	`Community → bell icon` (also shown on the bell above the `Community` button on the Pro Services screen)
Turn reply notifications off / mute a group	`Community → bell icon → gear icon`
Aircraft Scheduler	`MAP top-right account icon → Scheduler`
Create a scheduler (private)	`MAP top-right account icon → Scheduler → New Scheduler`
Add an aircraft/instructor (owner)	`Scheduler → scheduler → Schedule tab → Add resource`
Book a resource	`Scheduler → scheduler → Schedule tab → tap a green slot → Book`
View / cancel my reservations	`Scheduler → scheduler → Mine tab`
Set booking limits (owner)	`Scheduler → scheduler → tune icon → Booking Rules`
Cancel a reservation	`Scheduler → scheduler → Schedule tab → tap reservation → Cancel`
Mark a resource out of service (owner)	`Scheduler → scheduler → Schedule tab → tap resource name → Available toggle`
Fleet / dispatch board	`Scheduler → scheduler → Dispatch tab`
Update hobbs/tach / MX due (owner/dispatcher)	`Scheduler → scheduler → Dispatch → Fleet → aircraft → Update meters / MX`
File a squawk	`Scheduler → scheduler → Dispatch → Squawks → File squawk`
Create a lesson pack (owner)	`Scheduler → scheduler → Dispatch → Lessons → New lesson pack`
Assign student → instructor (owner)	`Scheduler → scheduler → Members → badge icon`

Feature	Path
Airport Businesses & Reviews	`Destination popup → Business tab` (inline; free; sign-in required)
Add an airport business	`Destination popup → Business tab → Add`
Add services/fuel/hours to a business	`Business tab → tap listing → Edit details`
Set fuel prices for a business	`Business tab → tap listing → Set fuel prices`
Review an airport business	`Business tab → tap listing → Reviews → Add`
Cloud backup/restore	`MAP top-right account icon → Backup/Sync`
User Manual (Help)	`MAP → Menu → Help`
CAP Grid overlay	`MAP → Layers → CAP Grid slider > 0` (zoom to level 9+)
Enable wind vectors	`MAP → Layers → Weather > 0` → product menu **Wind** (use altitude slider)
Enable ceiling overlay	`MAP → Layers → Weather > 0` → product menu **Ceil** (use altitude slider)
Change GPS source mode	Tap `SRC` tile in instrument overlay
Measure distances	`MAP → Ruler icon → long-press map to add points`
Rubber band waypoints	`MAP → Rubber banding icon → long-press and drag waypoints`
Insert waypoint at position	`PLAN → long-press waypoint row → FIND → select destination`
View winds & terrain analysis	`PLAN → analytics icon → tap diagram for details`

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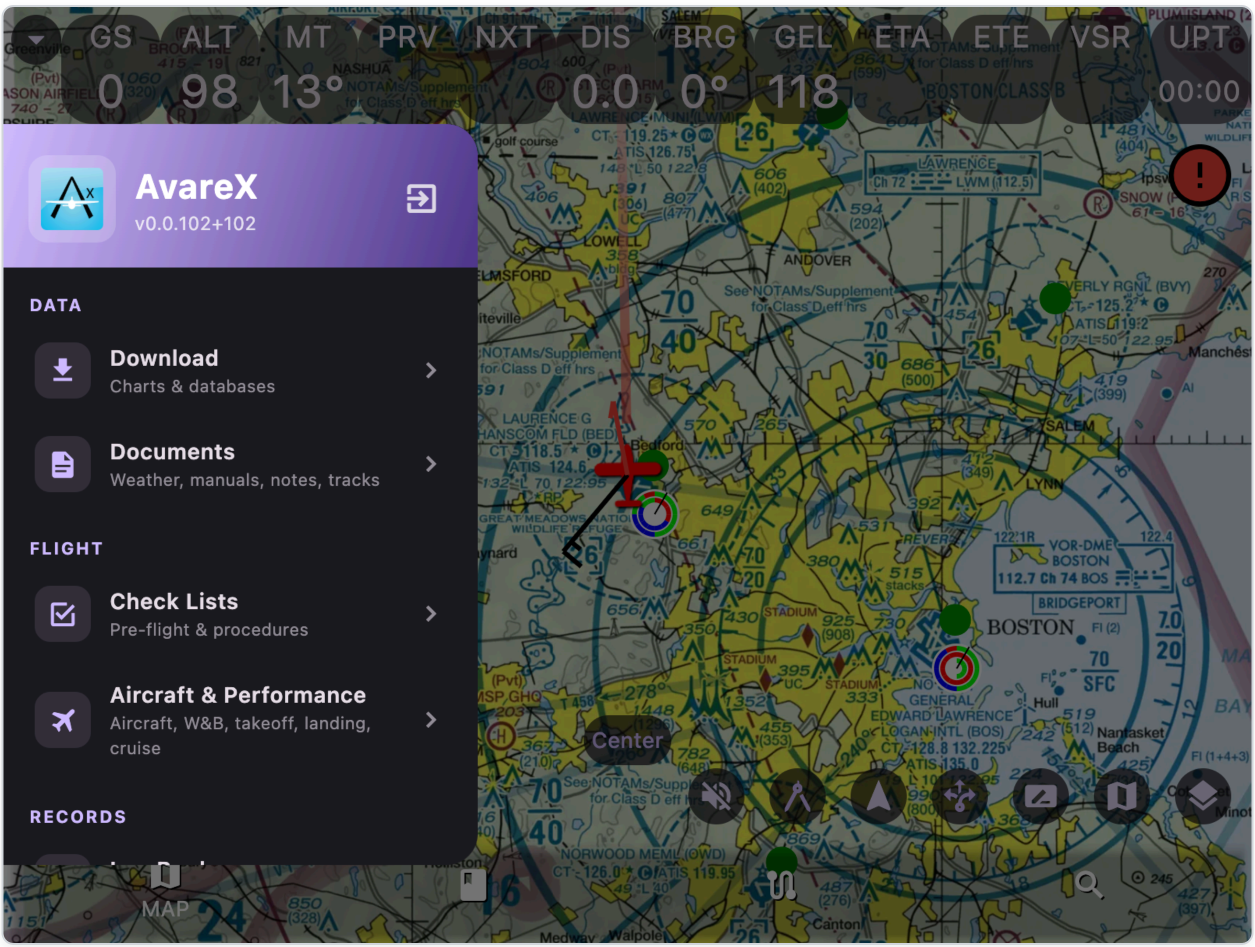
15) Step-by-Step Use Cases (Common Scenarios)

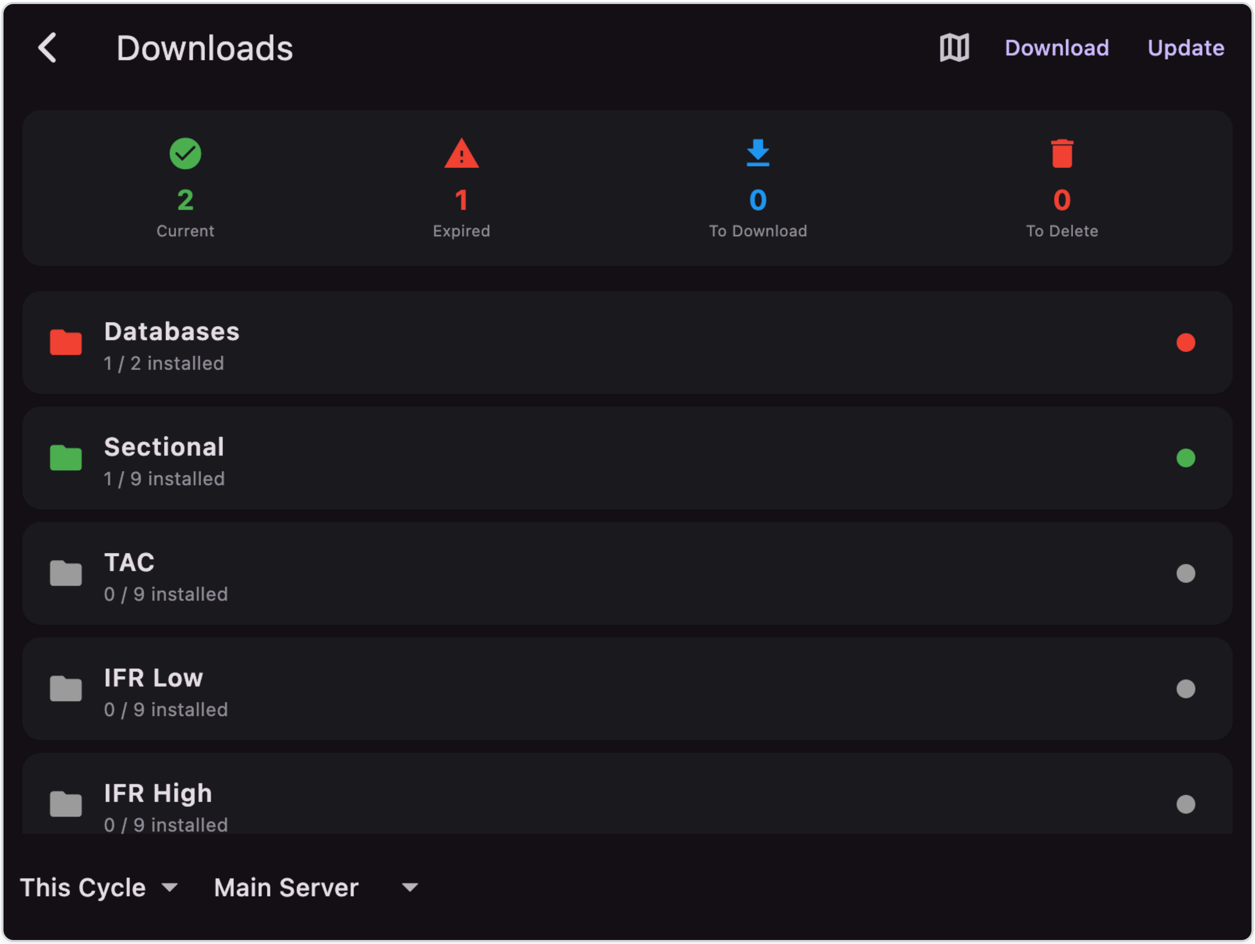
> Each use case is a numbered walkthrough you can follow end-to-end. Screen names match the labels in the app, and the screenshots above each step block show the exact place you should be.

UC-00: Download charts and databases

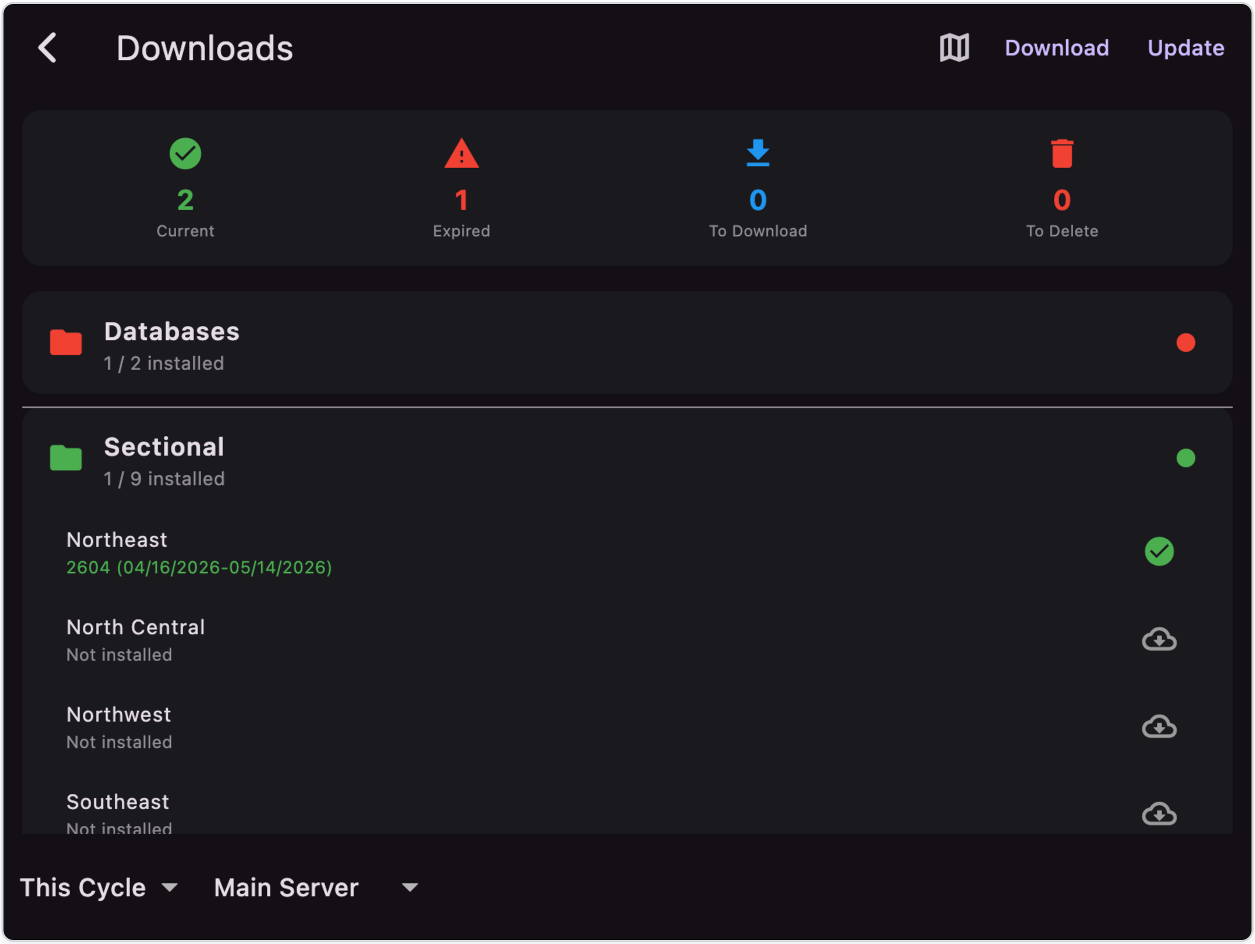
Without **DatabasesX** the airport database is empty and the FIND/PLAN tabs return no results. Onboarding already grabs the databases plus the Sectional, Plates, and CSUP for your GPS region **automatically** when they are missing. Use this walkthrough when you want to add or update **specific chart regions** manually.



1. On the **MAP** tab, tap **Menu** (bottom-left) to open the left drawer.
2. Tap **Download** under the **DATA** section.



3. Confirm the **cycle** dropdown is on **This Cycle** (use **Next Cycle** only if a new cycle is published and you want to pre-stage it). Use **Main Server** unless it is unreachable; then switch to **Backup Server**.
4. Tap **Databases** to expand it, then tap **DatabasesX** so it shows a download icon (queued).
5. (Optional) Tap **Sectional**, **TAC**, **IFR Low**, **IFR High**, **Plates**, **CSUP**, **Elevation**, etc. to queue the chart products you need.



6. Tap individual regions inside each category (e.g. *NE_SEC*, *SE_SEC*) — each tap toggles the queue state (download icon → trash icon → cleared).
7. Press **Download** in the app bar. A progress ring appears next to each item; the **Status Summary Card** updates the **Current / Expired / To Download / To Delete** counts.
8. Wait for downloads to complete before leaving the screen — exiting can abort an in-flight download.
9. To refresh expired data later, just press **Update** instead — it queues every expired item automatically.

> Tip: During **first-run onboarding**, the *Databases and Maps* page automatically downloads your GPS region's databases, Sectional, Plates, and CSUP. It also offers **Choose Manually** (opens this Download screen) if you prefer to pick data yourself. The manual flow is the same as above.

UC-01: Connect an external ADS-B/GPS receiver over Wi-Fi UDP

Best when your receiver broadcasts GDL90/NMEA over local network.

1. Power on your ADS-B/GPS receiver.
2. Connect your tablet/phone/computer to the receiver's Wi-Fi network (or same LAN).
3. Open AvareX and go to `MAP`.
4. **iOS only — first launch on a new device**: when iOS shows the *"AvareX would like to find and connect to devices on your local network"* prompt, tap **Allow**. Without this permission iOS will silently drop all ADS-B/GPS UDP packets and traffic/weather will never appear, even though ownship over internal GPS still works. If the prompt was previously dismissed or denied, enable it under `Settings → Privacy & Security → Local Network → AvareX`.
5. Wait for incoming data on supported UDP ports: `4000`, `43211`, or `49002` (automatic listener).
6. **Stratus 3 / 3i only**: if the receiver is not in Open ADS-B mode, open `MAP → ADSB` tile → **ADS-B Status**, then tap the antenna icon (top-right). Confirm the toast, then wait for the Receiver status to show Connected.
7. Turn on useful map layers:
   - `Traffic`
   - `Weather`
   - `Radar` (internet radar)
8. Verify data:
   - Ownship updates smoothly
   - Traffic symbols appear (if in range)
   - Weather products populate
9. If needed, open warning drawer from red icon and resolve GPS/data warnings.

UC-02: Connect an external ADS-B/GPS receiver over Bluetooth (Android)

1. Open `MAP → Menu → IO`.
2. Wait for device discovery to populate the list.
3. Select your receiver from list.
4. Pair (if needed), then tap **Connect**.
5. Confirm status line shows connected device.
6. Return to map and enable `Traffic` / `Weather` layers as needed.
7. To disconnect: `IO → Disconnect`.

UC-03: Rubber-band a route directly on the map

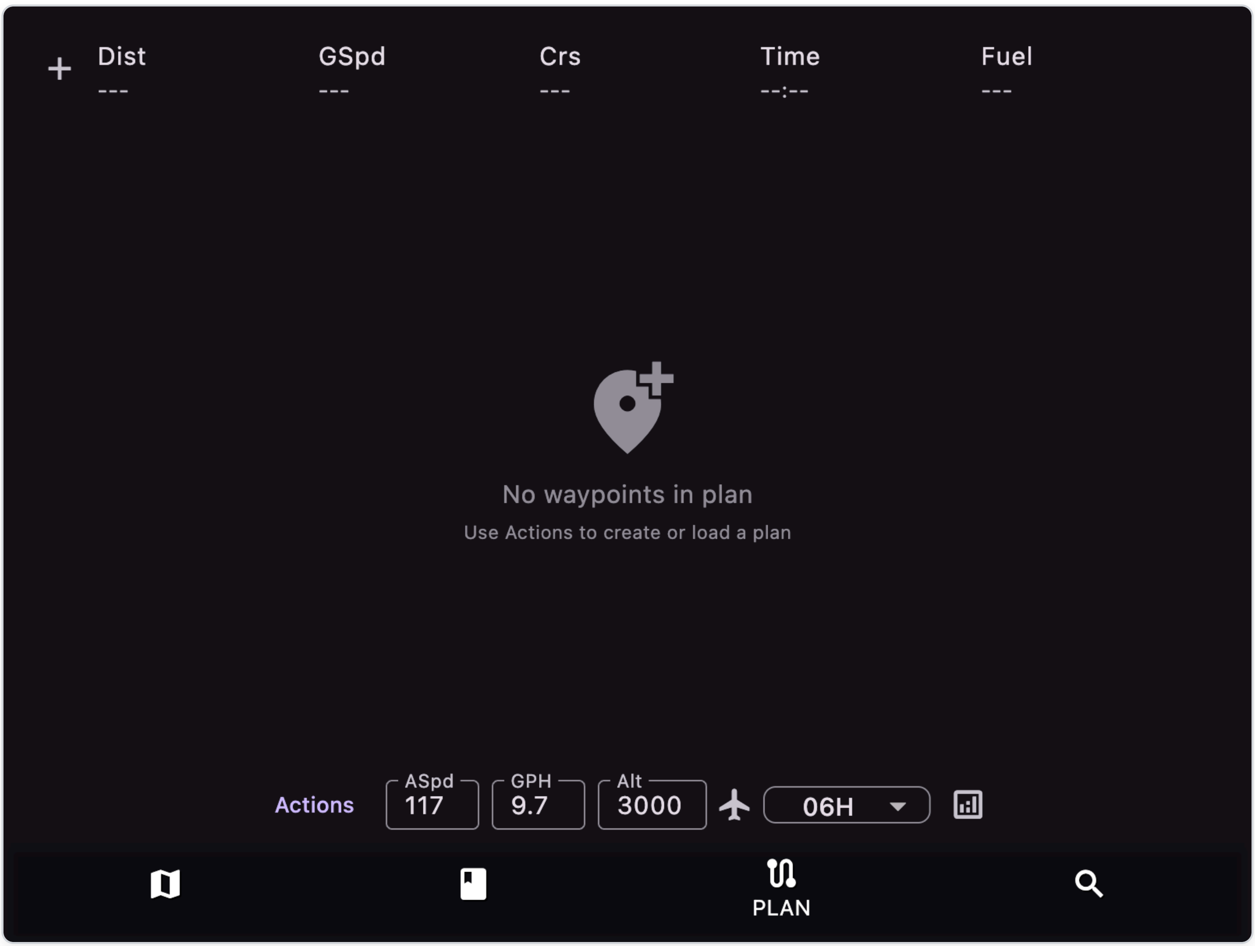
Use this to adjust waypoints graphically.

1. Ensure your route has waypoints (`PLAN` tab or destination popup `+Plan`).

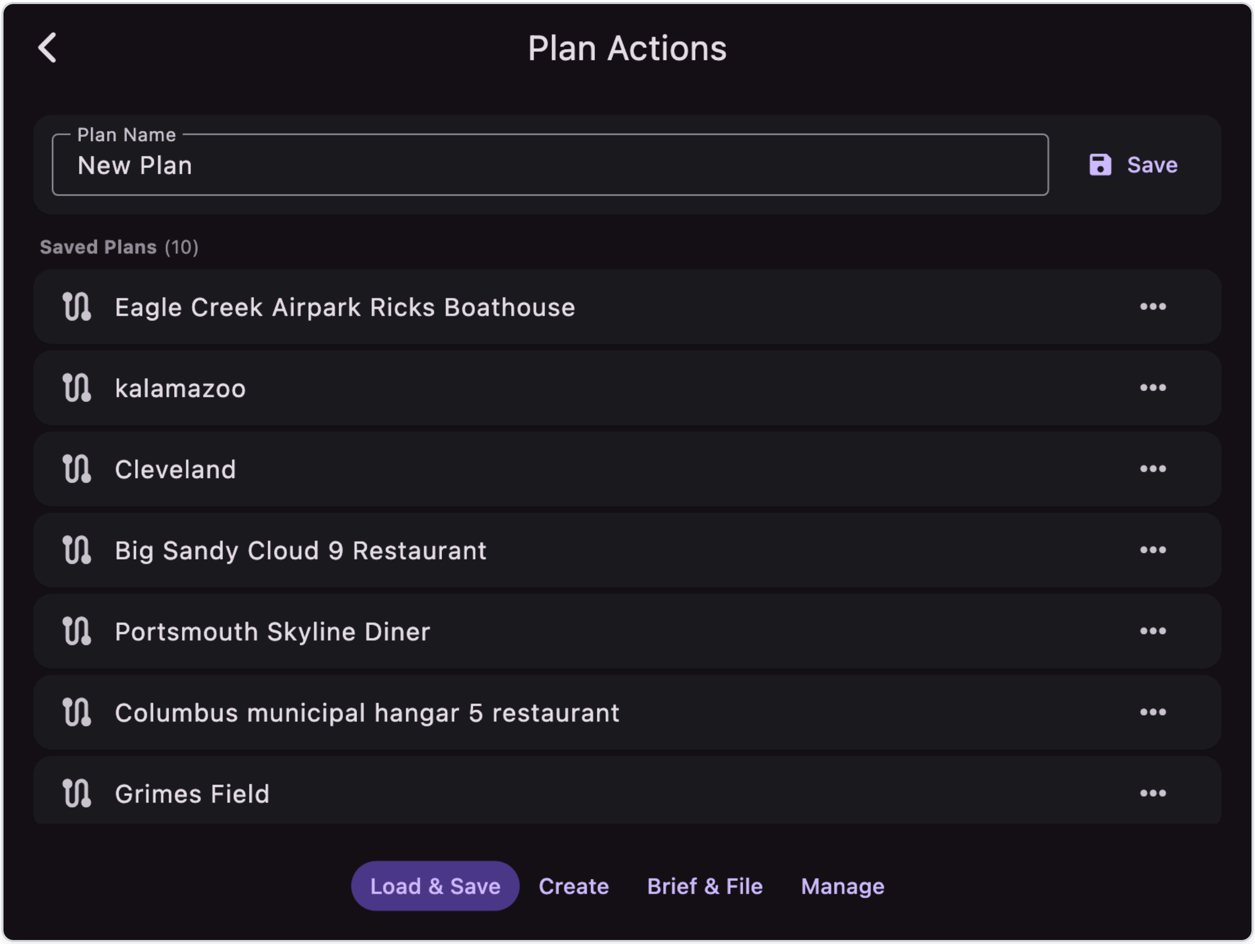
2. Go to `MAP`.
3. Tap the **rubber banding** icon (decision arrow) to enable it (icon turns red).
4. Long-press a waypoint marker or label and drag to a new position.
5. Release to snap/update from database lookup and rebuild route geometry.
6. Turn rubber banding off when done to avoid accidental edits.

UC-04: Build a flight plan from scratch (quick VFR workflow)

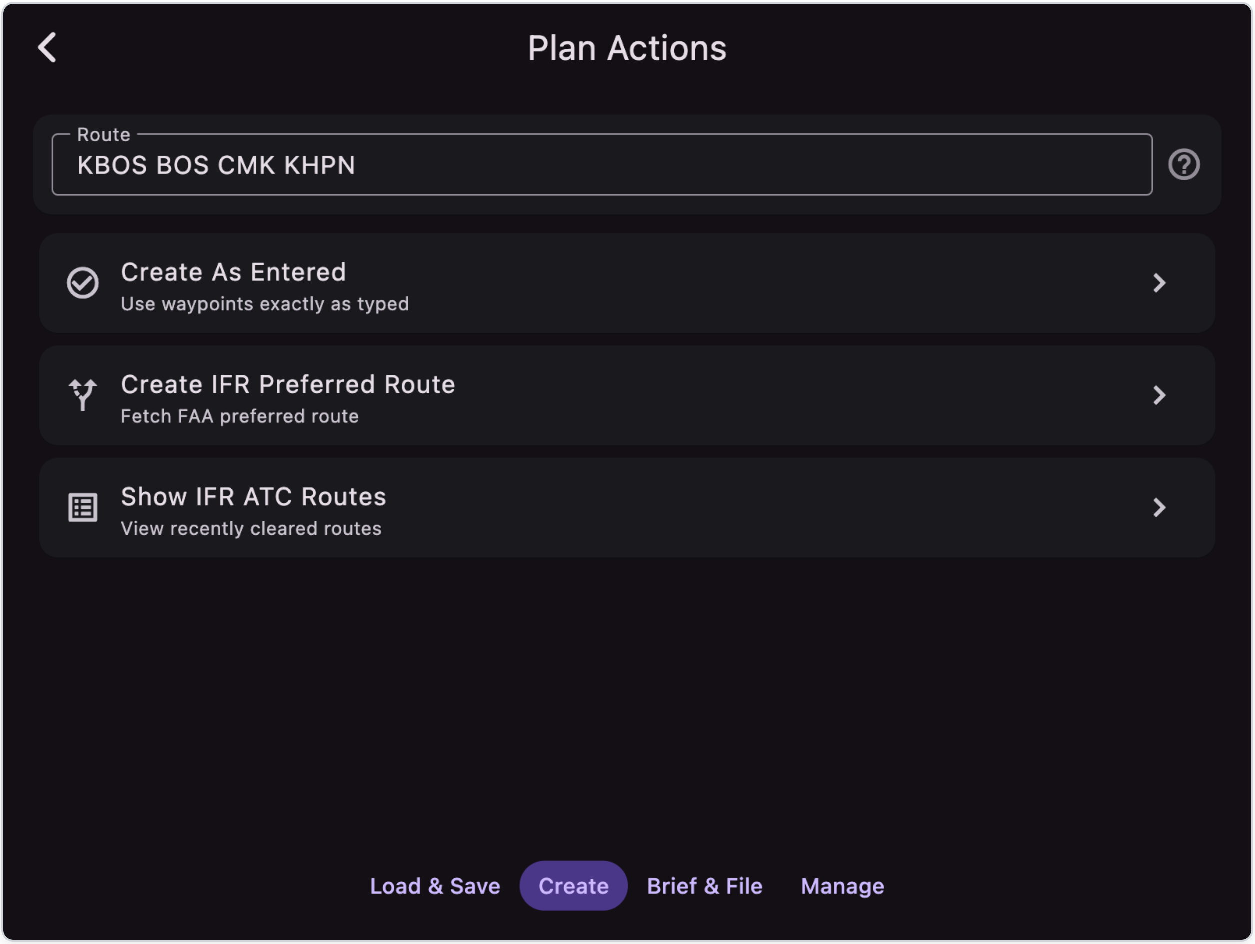
There are three ways to add a waypoint to the plan; the fastest for a brand-new VFR plan is **Plan → Create → Create As Entered**, shown below.



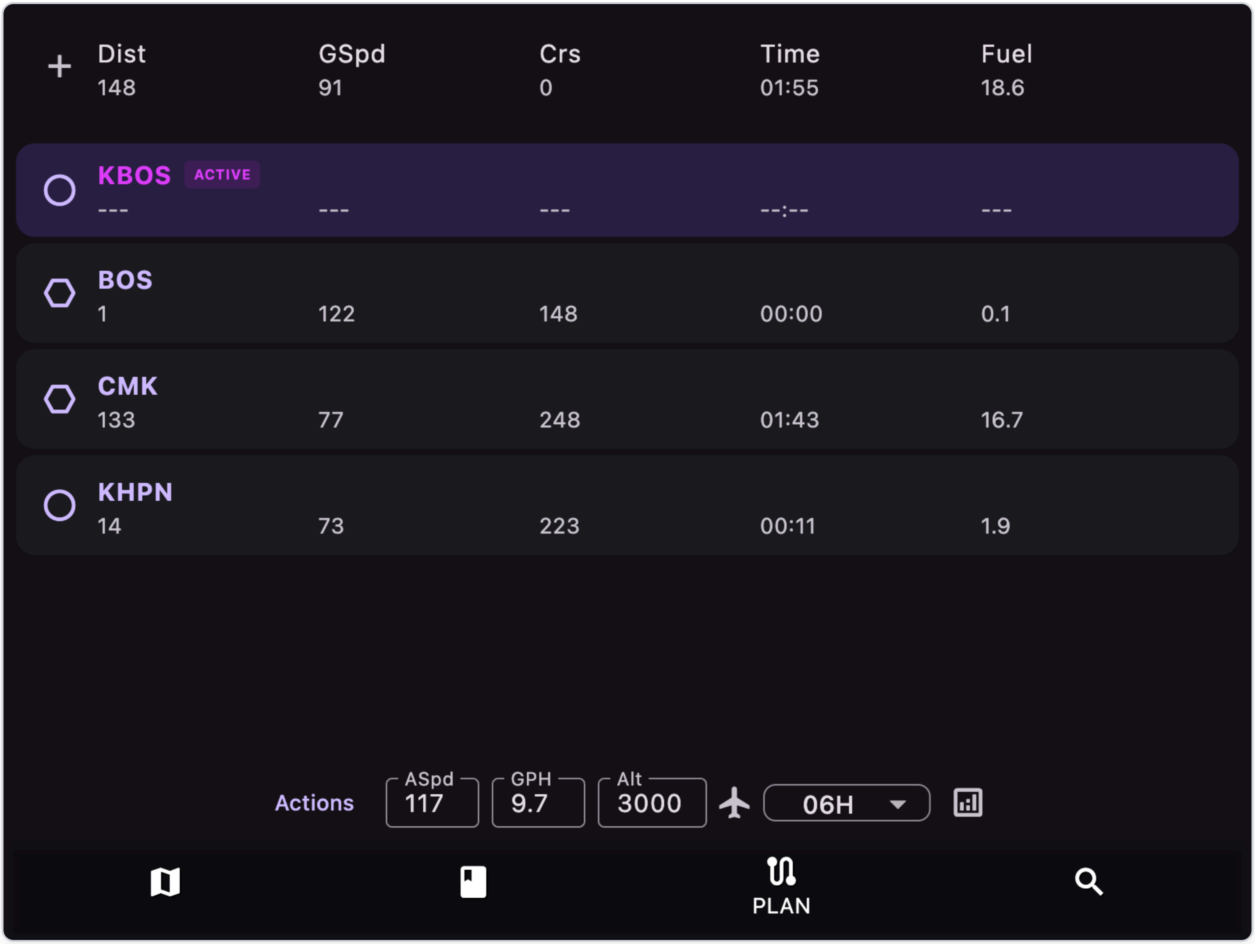
1. Tap the **PLAN** bottom tab. You'll see the empty plan view above (Distance / GSpd / Crs / Time / Fuel show `---`, and `No waypoints in plan` is displayed in the body).
2. Press **Create** at the bottom; the Create sub-page appears inline (the **Create** button is highlighted).



3. If a different sub-page is showing, tap **Create** in the bottom row to switch.



4. Tap into the **Route** text field and type your route as space-separated identifiers (airports, navaids, fixes, airways). Example: `KBOS BOS CMK KHPN`.
5. Tap **Create As Entered**. The app parses each identifier, looks it up in the database, and returns to the main PLAN view.



6. Verify the totals header (Dist 148, GSpd 91, Time 01:55, Fuel 18.6 in the example).
7. Adjust the bottom controls as needed:
   - **ASpd** — true airspeed in knots (or mph if Imperial units)
   - **GPH** — fuel burn (gallons per hour)
   - **Alt** — planned altitude (feet)
   - **Aircraft icon** — auto-fills `ASpd` and `GPH` from your selected aircraft's Performance data at the chosen `Alt`
   - **Forecast horizon** — `06H / 12H / 24H` (winds aloft window for nav-log calculations)
8. To **reorder** legs: long-press and drag rows. To **set a different leg active**, tap its row. To **delete** a leg, swipe right-to-left.
9. To **save** the plan: tap **Load & Save** at the bottom → press **Save** and give it a name.

Alternative add methods (used after the plan exists):

- From **MAP**: long-press near a destination → tap a row in the popup → `+Plan` (insert at current waypoint) or `↓Plan` (append to end).
- From **FIND**: search/select an item → on the destination popup tap `+Plan` or `↓Plan`.
- Long-press a waypoint row in **PLAN** to insert *after* that row (opens FIND; the next selection inserts there).

> See **UC-21** below for using the **navigation log** (analytics icon) to review leg calculations, winds, and terrain along the route.

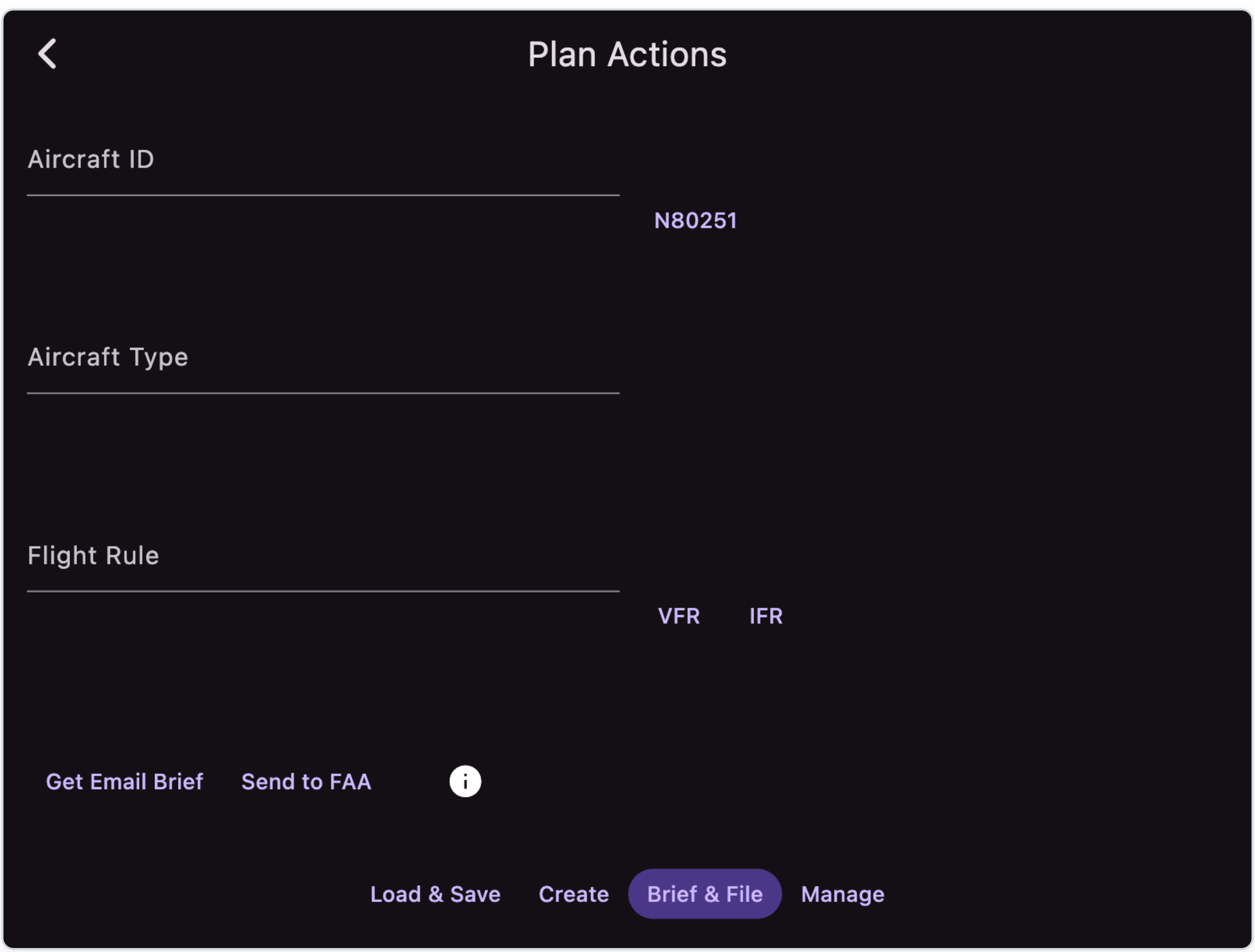
UC-05: Create an IFR route automatically

1. Open `PLAN → Create`.
2. In Route field:
   - Enter route text and use **Create As Entered**, or
   - Enter `DEPART DEST` and use **Create IFR Preferred Route**, or
   - Use **Show IFR ATC Routes** to view recent ATC route options.
3. When route is loaded, return to `PLAN` to review/reorder waypoints.

UC-06: File a flight plan with the FAA

Prereq: set your 1800wxbrief-compatible email in onboarding.

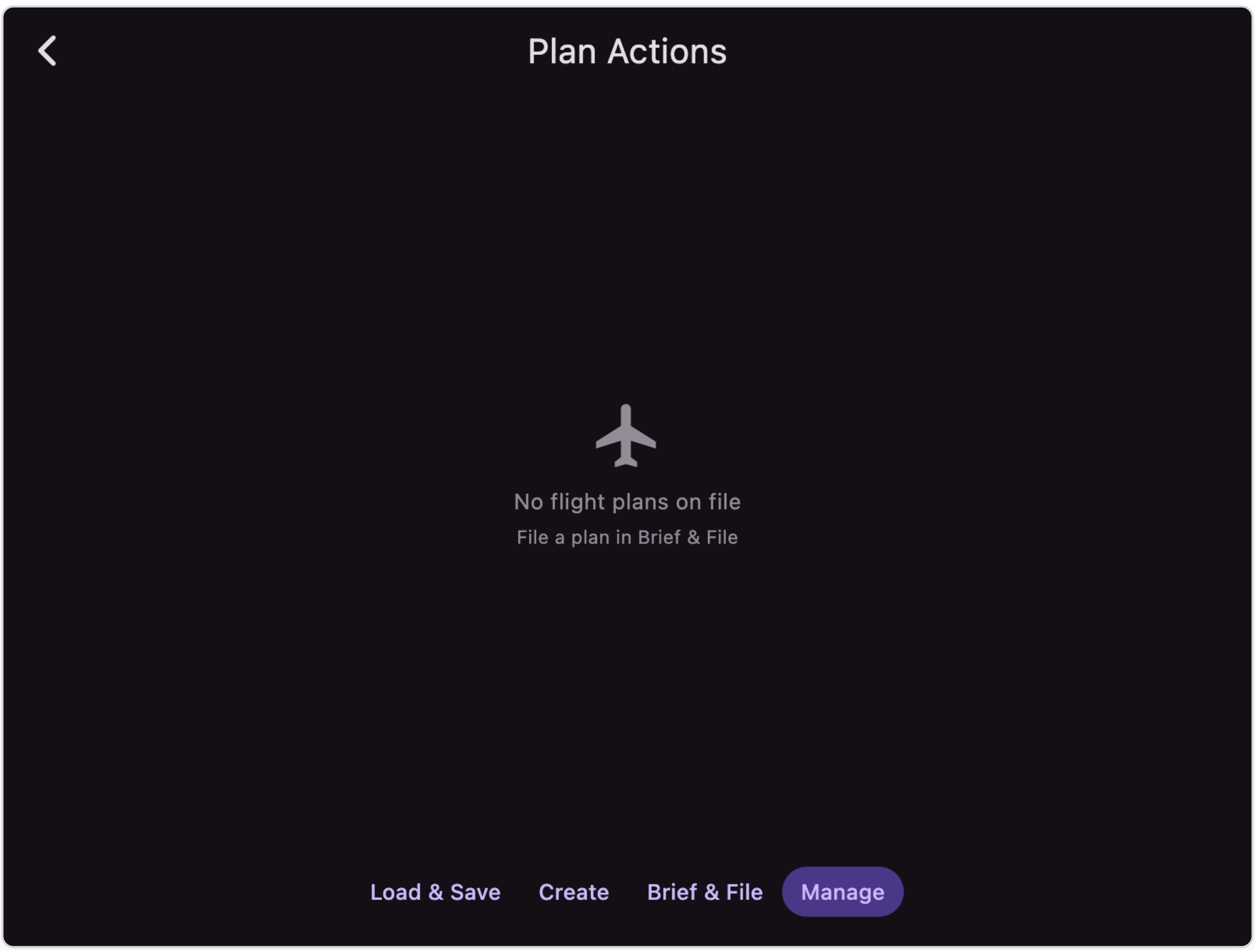
1. Build/verify your route in `PLAN` (see UC-04).
2. Open `PLAN → Brief & File`.



3. Fill required fields (aircraft ID/type, rule, departure, destination, route, times, fuel, POB, etc.).
4. Use quick-fill buttons:
    - "Planned" buttons for departure/destination/route from current plan
    - Aircraft buttons (e.g. `N80251`) to fill from stored aircraft profiles
5. Tap:
    - **Get Email Brief** for briefing email, and/or
    - **Send to FAA** to file.
6. Check status indicator (checkmark=success, info icon=error with tooltip).

UC-07: Activate, close, or cancel an FAA flight plan

1. Open `PLAN → Manage`.



2. Find your plan in the list (color-coded by status).
3. Use action by state:
    - `PROPOSED` (blue): **Depart** (choose time with sunrise/sunset reference) to activate
    - `ACTIVE` (green): **Close** after landing
    - Non-active: **Cancel** if no longer needed

UC-08: Quickly divert to a nearby airport

1. On `MAP`, long-press near destination area.
2. In popup, review nearby list and tap a candidate airport.
3. Tap `→D` to set Direct-To and center map.
4. Optional:
   - Tap `Plates` for immediate airport diagrams/procedures
   - Add to full route with `+Plan`

UC-09: Use plates with procedure profile and add procedure to plan

1. Open `PLATE` tab.
2. Select airport (bottom-right selector).
3. Select desired plate (bottom-left selector, color-coded by type).
4. Use procedure menu (plus/road icon):
   - Choose a procedure to show its profile card, draw the **CIFP route** on georeferenced plates (green→yellow→brown→blue→red→orange from final back, matched on the vertical profile), and append it to the plan.
5. If other IAFs exist for the same chart, tap a **black IAF chip** to switch transition (route and profile update; plan is not duplicated).
6. Fly the approach using the colored route overlay and vertical profile.
7. Keep instruments visible on plate using top-right show/hide toggle.
8. Close profile card with X button when done (clears the route overlay).

UC-10: Save your flown track and retrieve it from Documents

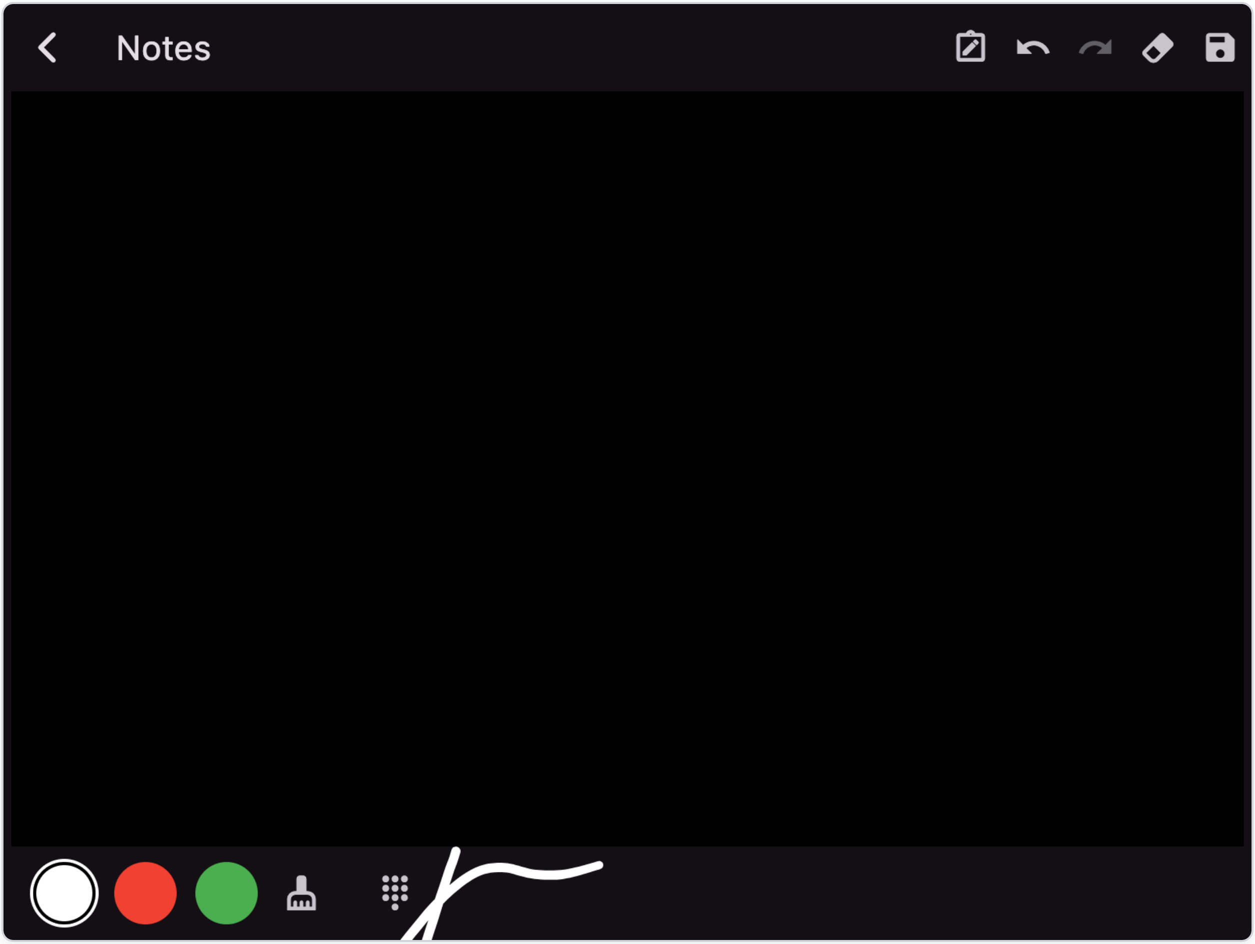
1. In `MAP`, keep `Tracks` layer on during flight (opacity > 0).
2. When done, set `Tracks` layer opacity to `0` in Layers menu.
3. App saves track automatically as KML in Documents/tracks folder.
4. Open `MAP → Menu → Documents → tracks` to access/share file.

UC-10a: View a saved track and create a logbook entry

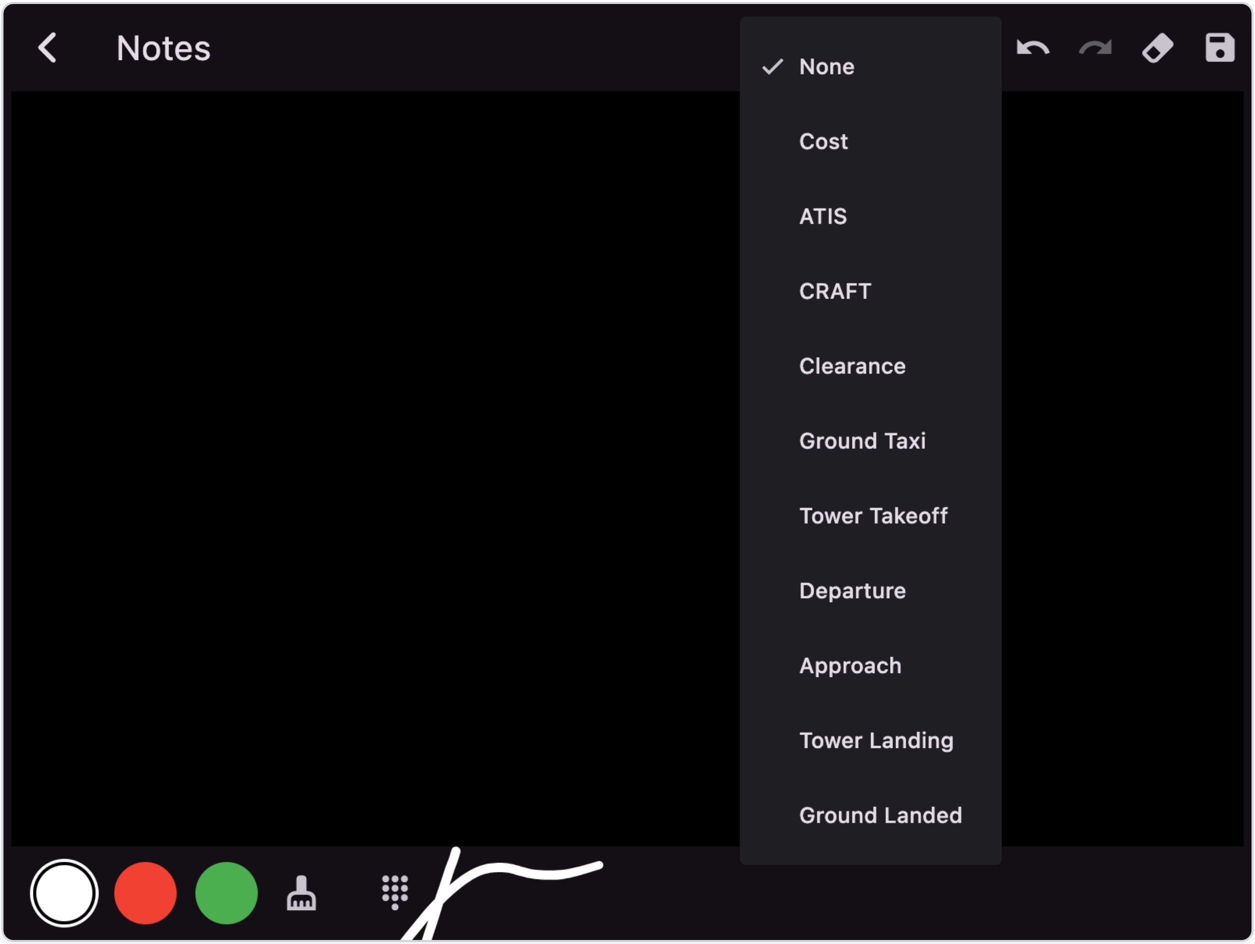
1. Open `MAP → Menu → Documents → tracks`.
2. Tap a KML file to open the Track Viewer.
3. Use view icons to switch between 2D Map, Altitude Profile, and 3D View.
4. In 2D Map view, tap **Log Flight** to create a logbook entry from the track.
5. The entry auto-fills route (nearest airports), time (from timestamps or distance/TAS), and aircraft info from selected profile.

UC-10b: Use aviation sheets for IFR clearance copydown

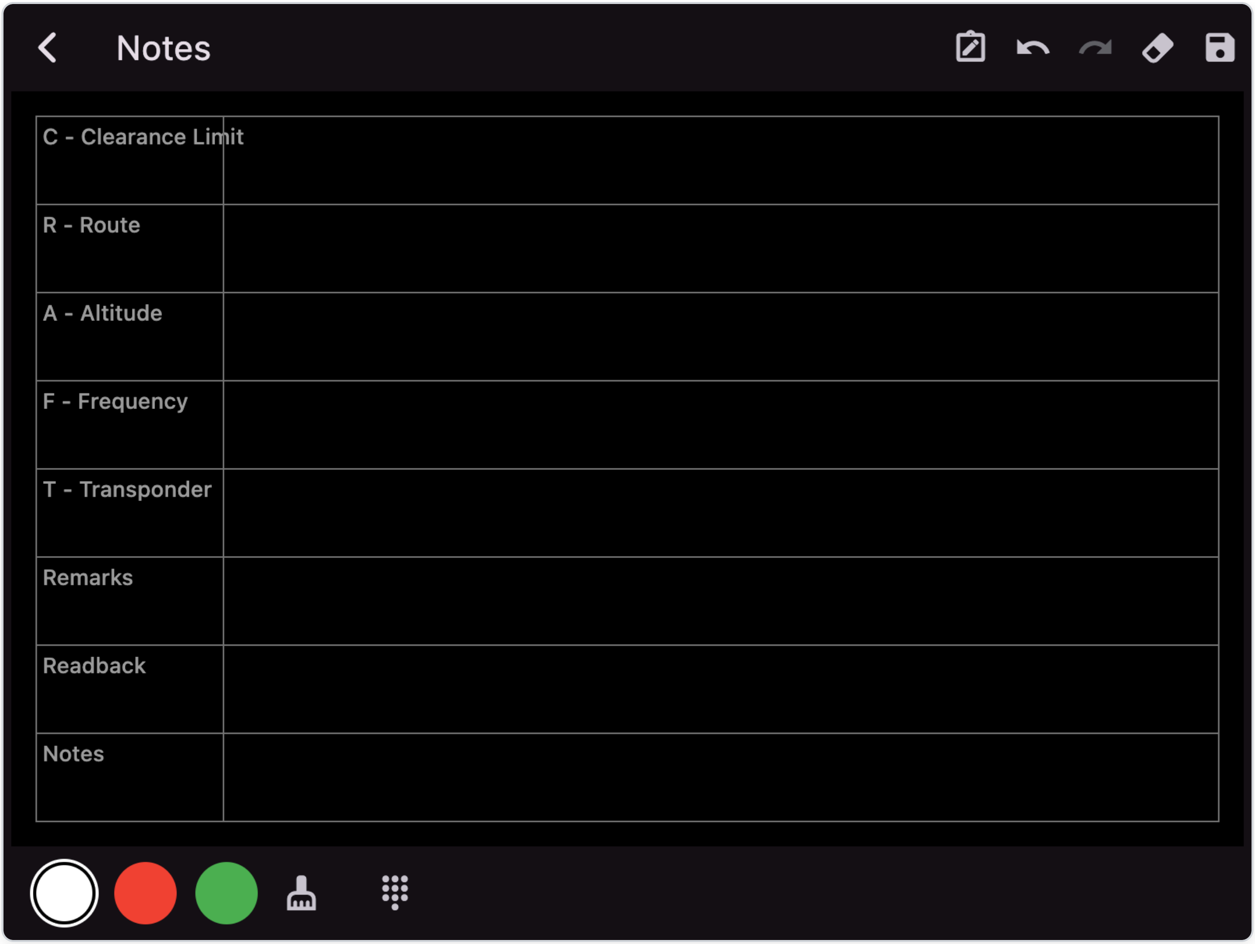
1. On `MAP`, tap the notes/pen icon (in the bottom-right control row).



2. Tap the sheet icon (note_alt) in the app bar.



3. Select **CRAFT** for IFR clearance fields.



4. Use the number keypad (dialpad icon) to type frequencies, altitudes, or squawk codes.
5. **Tap on the canvas** to position the typed text exactly where you want it (e.g., next to the frequency field).
6. Use freehand drawing to fill in route or other notes.
7. The sheet auto-saves when you exit or switch sheets.
8. Return anytime to continue where you left off; each sheet maintains its own state.

UC-11: Import a GeoJSON overlay and display it on map

1. Open `MAP → Menu → Documents`.
2. Tap **Import** and select a `.geojson` file.
3. Open `MAP` and set `GeoJSON` layer opacity > 0.
4. Imported polygons/markers now draw on the map.

UC-12: Send your plan to, or get a plan from, an Avidyne IFD over Wi-Fi

Exchange flight plans with a panel-mounted Avidyne IFD440/540/550 in either direction.

1. Connect this device to the **same Wi-Fi network** as the IFD (configure the network on the IFD's Maintenance-mode page if needed). **iOS**: grant the **Local Network** permission.

2. Open `PLAN → Transfer`.
3. Under **Avidyne IFD (Wi-Fi)**, wait for the IFD to appear in the list.
4. To **send**: build your route with at least 2 waypoints, then tap **Send** next to the IFD. Wait for the success toast, then **review and activate** the uploaded route on the IFD.
5. To **get**: tap **Get** next to the IFD. AvareX downloads the IFD's flight plan and loads it into the active plan (replacing the current one). Matched waypoints keep their database type; unmatched ones become GPS coordinate points.

UC-13: Back up and restore app data (Pro)

1. Open Pro Services from account icon on MAP (top-right).
2. Sign in and pass entitlement/paywall if required.
3. Open `Backup/Sync` (from bottom buttons after login).
4. Use:
    - **Backup** (upload icon) to upload local `user.db`
    - **Restore** (download icon) to overwrite local data from cloud copy
5. Confirm prompts carefully (both operations overwrite existing data).
6. Progress percentage shows during transfer.

UC-14: Get flight help from community resources

1. In onboarding or browser, open Apps4Av forum:
    - `https://groups.google.com/g/apps4av-forum`
2. Search existing threads for similar workflows/devices.
3. Post issue details:
    - Platform/device
    - Receiver type
    - What screen/action failed
    - Any warning drawer messages

UC-15: Use the CAP Grid overlay for search and rescue

1. Go to `MAP → Layers`.
2. Set **CAP Grid** opacity > 0.
3. Zoom to level 9 or higher (grid only renders at sufficient zoom).
4. Grid squares appear with identifiers (e.g., BOS42, SEA123).
5. Use grid coordinates for communication with ground teams or other aircraft.

UC-16: Choose a different aircraft icon on the map

1. Open `MAP → Menu → Aircraft & Performance`.
2. Go to the **Custom** tab.
3. Select or create an aircraft profile.
4. Use the icon dropdown to select from: plane, helicopter, or canard.
5. Save the aircraft profile.
6. The map will display your selected icon for ownship.

UC-17: Load a plan in reverse order for a return flight

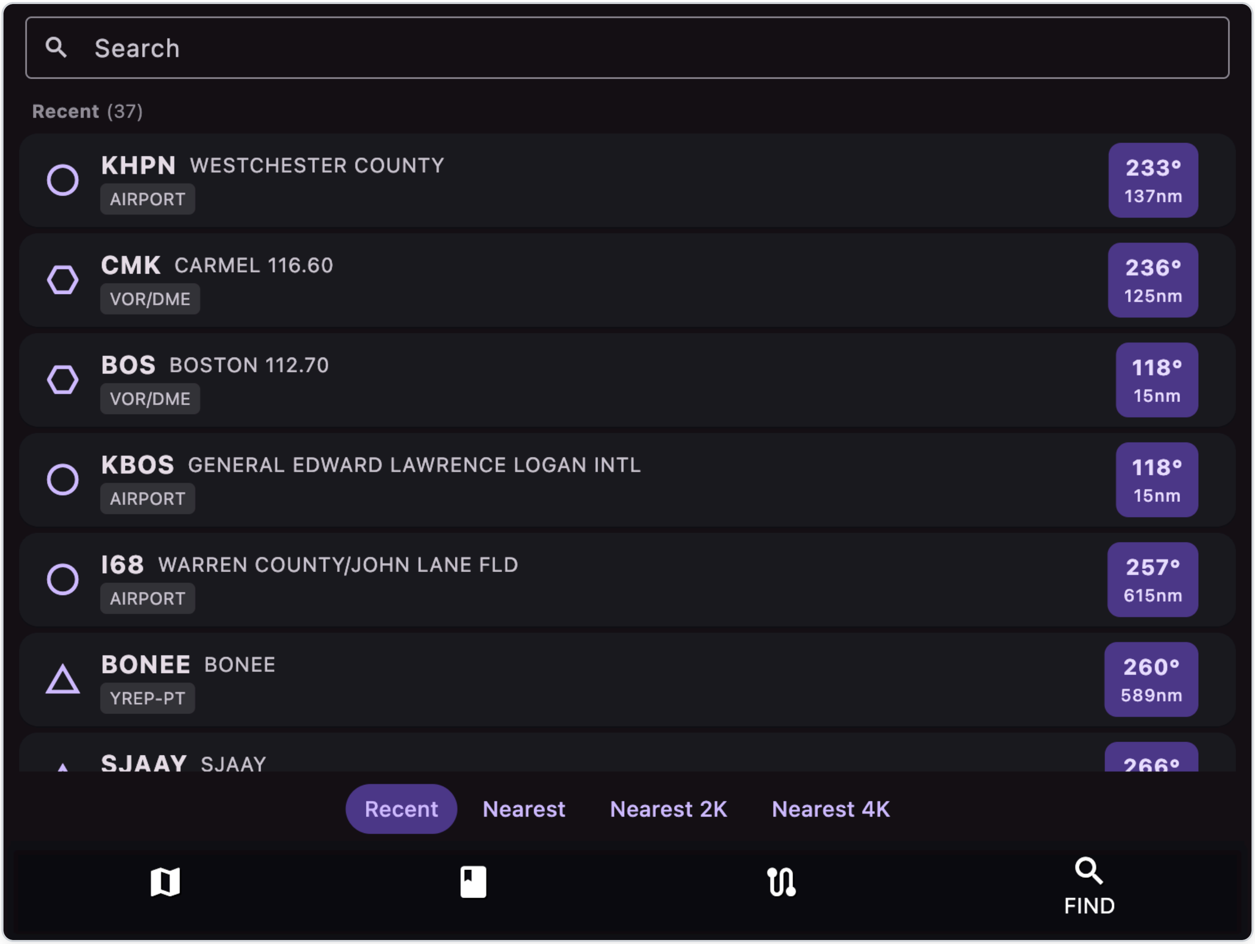
1. Open `PLAN → Load & Save`.
2. Find your saved outbound plan.
3. Tap the 3-dot menu on the plan.
4. Select **Load Reversed**.
5. The plan loads with waypoints in reverse order, ready for your return trip.

UC-18: Check your pilot currency status

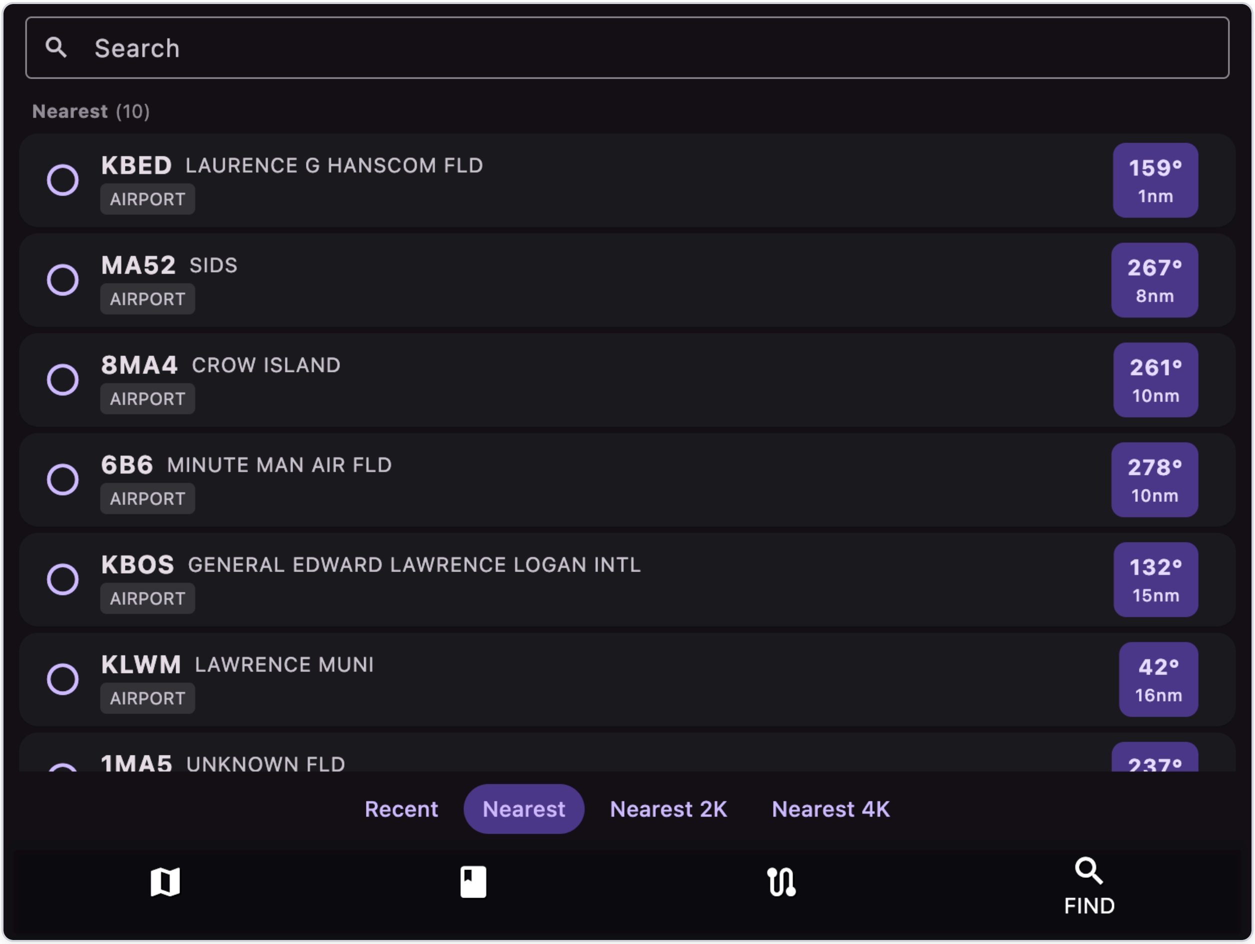
1. Open `MAP → Menu → Log Book`.
2. Tap the **Details** button.
3. View the Currency Status Card at the top showing:
    - Day currency: 3 landings in last 90 days
    - Night currency: 3 night landings in last 90 days
    - IFR currency: 6 approaches in last 6 months
4. Green = current, Red = not current

UC-18a: Search and find a destination

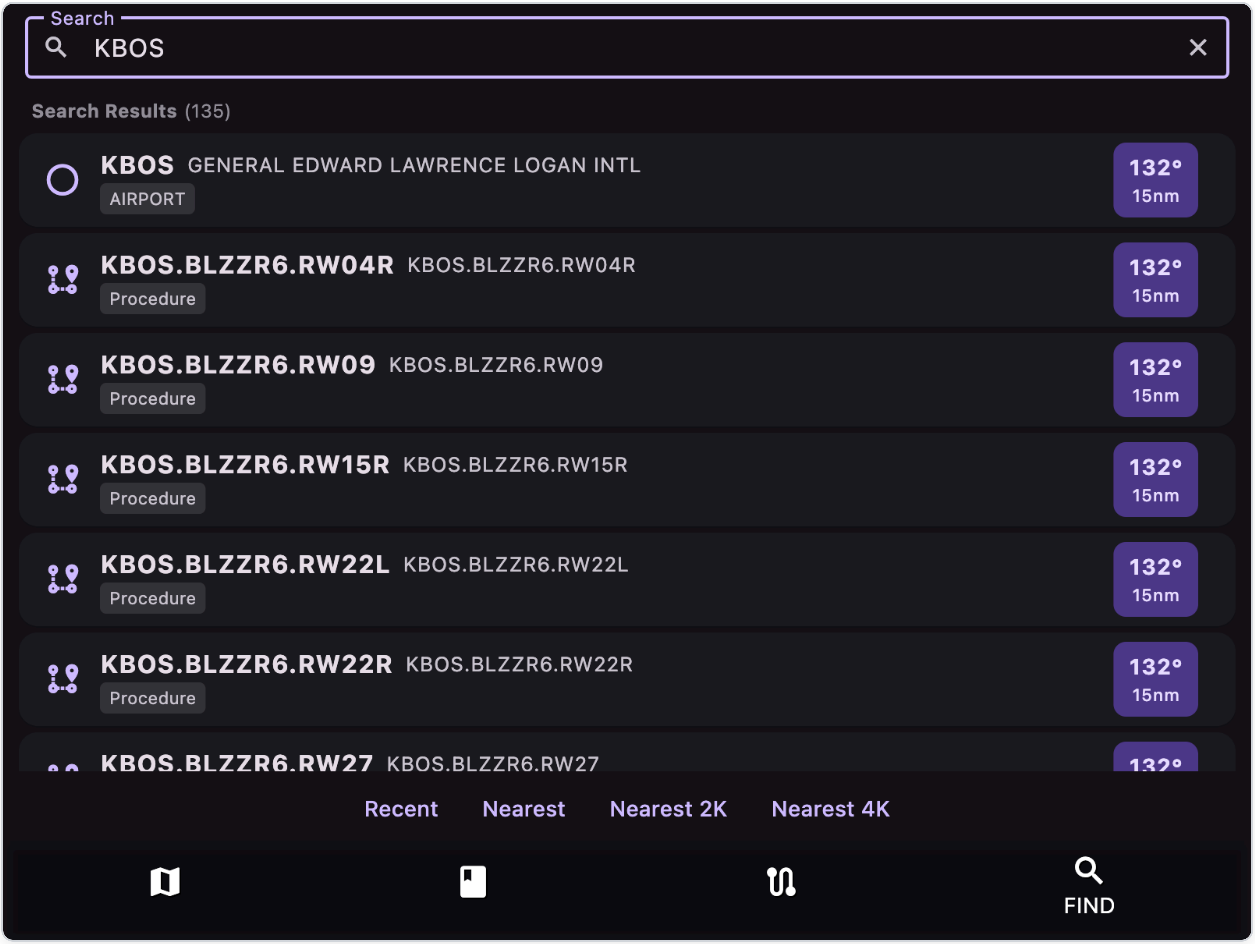
The **FIND** tab is the fastest way to locate any airport, navaid, fix, or procedure and act on it.



1. Tap the **FIND** bottom tab.
2. Use one of the four bottom filter buttons:
    - **Recent** — destinations you have viewed before
    - **Nearest** — every airport within range, sorted by distance from current ownship position
    - **Nearest 2K** — same, but only airports with ≥ 2000 ft of usable runway
    - **Nearest 4K** — same, but ≥ 4000 ft of runway



3. To search by identifier or text, tap into the **Search** field and start typing. Results appear immediately.



4. Each row supports several actions:
    - **Tap row** — open the destination popup with airport info, plates, METAR/TAF, NOTAMs, etc.
    - **Tap the bearing/distance pill on the right** — center the map on that destination and switch to MAP.
    - **Swipe right-to-left** — delete an entry from the Recent list.
    - For GPS-coordinate recent entries, an **edit (pencil) icon** lets you set a custom facility name.
5. From the destination popup you can:
    - **→D** Direct-To this destination
    - **+Plan** insert into the current plan after the active waypoint
    - **↓Plan** append to the end of the plan
    - **Plates** (airports) jump to PLATE tab at this airport

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UC-19: Measure distance and bearing on the map

1. On `MAP`, tap the **Ruler** icon (compass) to enable measure mode (turns red).
2. Long-press on the map to place your first point.

3. Long-press again to add additional points.
4. Distance (NM) and bearing (degrees) are shown between points.
5. Tap the Ruler icon again to disable and clear measurements.

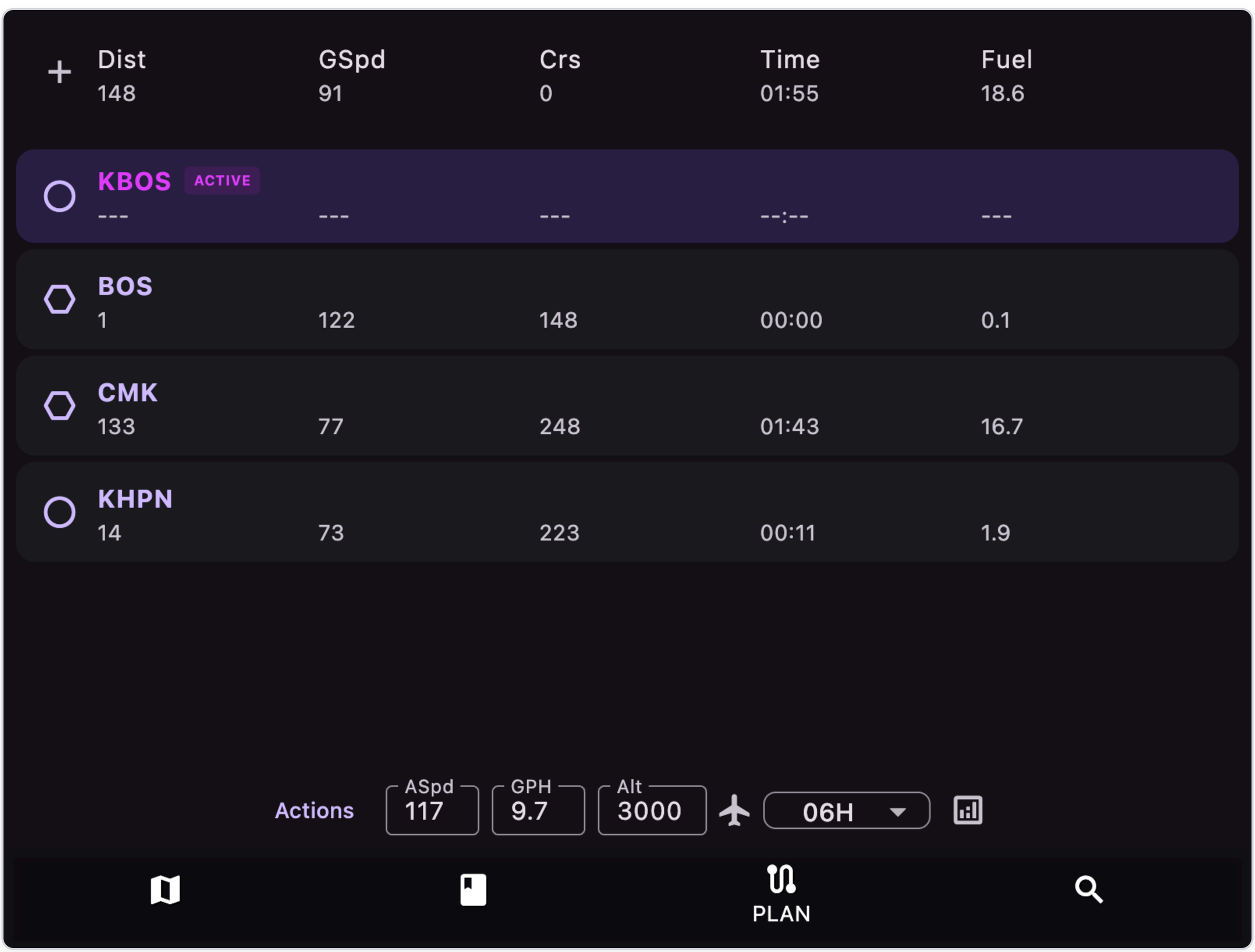
UC-20: Insert a waypoint at a specific position in your plan

1. Open `PLAN` tab with an existing route.
2. Find the waypoint you want to insert after.
3. **Long-press** that waypoint row.
4. The app switches to `FIND` tab.
5. Search for or select the waypoint you want to insert.
6. Tap `+Plan` or `↓Plan` — the new waypoint inserts after the long-pressed position.

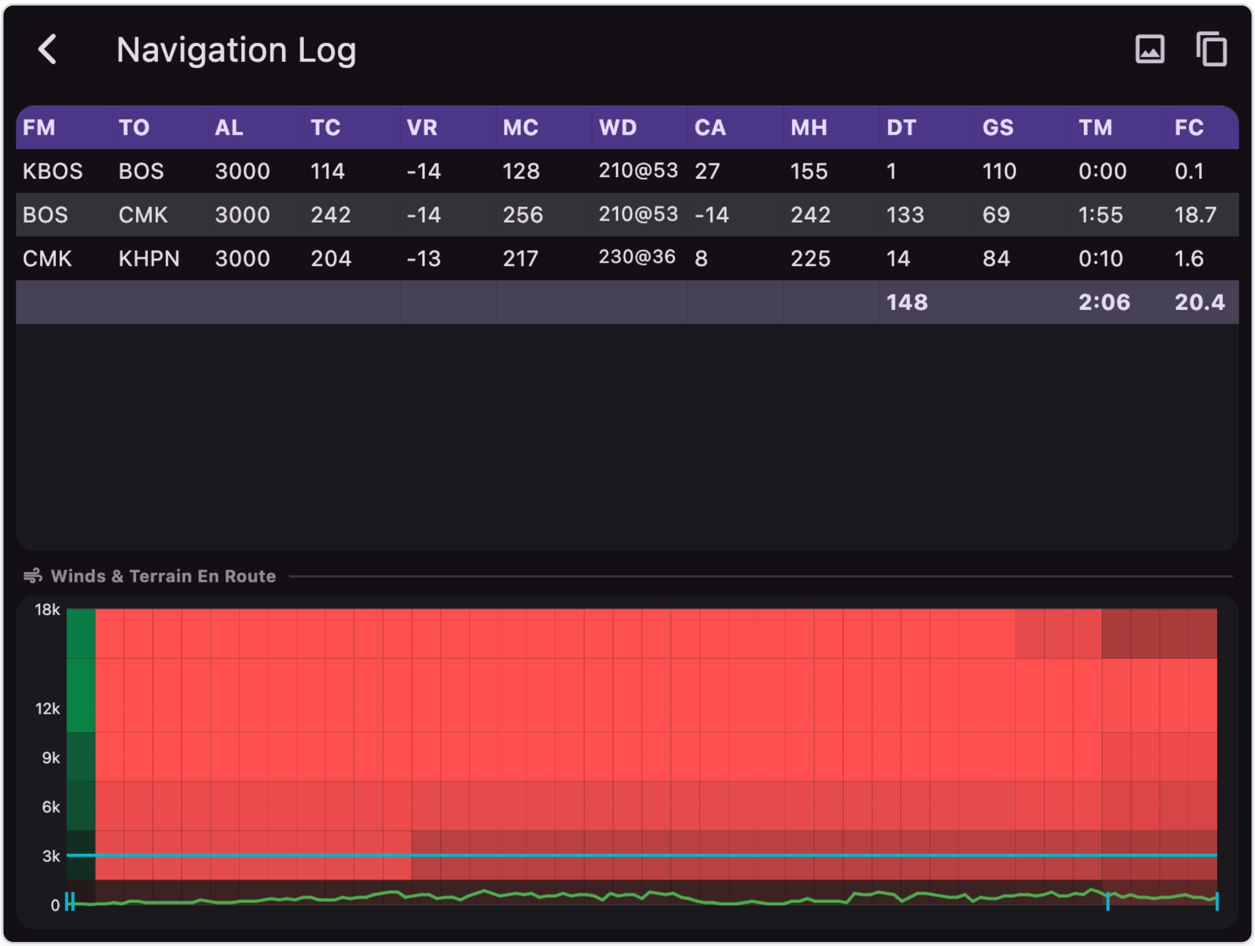
UC-21: Get the navigation log (winds & terrain analysis)

The navigation log gives you per-leg headings, magnetic variation, wind correction, ground speed, time, and fuel; plus a wind-and-terrain heat map at planning altitudes.

1. Open the **PLAN** tab.
2. Ensure your route has at least 2 waypoints (build one with **UC-04** if needed).



3. Set your planned altitude using the **Alt** field at the bottom.
4. Set the **Forecast horizon** dropdown (`06H / 12H / 24H`) so wind data matches your departure window.
5. (Optional) Tap the **Aircraft icon** next to **Alt** to load TAS/GPH from your stored aircraft profile.
6. Tap the **analytics icon** (graph, far right of the bottom row, tooltip *"Navigation log and terrain"*). The **Navigation Log** opens as a full-screen dialog.



7. The **per-leg grid** at the top has 13 columns:

Col	Meaning
FM	From waypoint
TO	To waypoint
AL	Plan altitude (ft)
TC	True course (°)
VR	Magnetic variation (°)

Col	Meaning
MC	Magnetic course (°)
WD	Wind direction@speed (e.g. `210@53`)
CA	Wind correction angle (°)
MH	Magnetic heading to fly (°)
DT	Distance for this leg (NM/SM)
GS	Ground speed for this leg (kt/mph)
TM	Leg time (HH:MM)
FC	Fuel consumed on this leg (gal)

The bottom row of the grid is the **plan total** (sum of DT, TM, FC across all legs).

Double-tap the grid to reset zoom; pinch/scroll to focus on a specific leg.

8. Below the grid, the **Winds & Terrain En Route** diagram is a single combined visualization for the whole route:

    - Y-axis: altitude bands (0, 3000, 6000, 9000, 12000, 18000 ft)
    - X-axis: position along the route from departure (left) to destination (right)
    - **Wind heat map**: each cell colored by tailwind/headwind component — **green = tailwind**, **red = headwind** (darker = lighter winds, brighter = stronger winds, scaled up to 50 kt)
    - **Cyan horizontal line**: your planned altitude (`Alt` from the PLAN tab)
    - **Green/red terrain curve**: terrain elevation; green where it's safely below your plan altitude, red where it's at or above your plan altitude
    - **Cyan tick marks** along the bottom edge: waypoint positions

9. **Tap anywhere on the diagram** to display a tooltip showing:

    - Nearest waypoint (only if your tap is within ~5% of that waypoint's position)
    - Altitude at the tap (snapped to one of the altitude bands)
    - Terrain elevation at that point along the route
    - Course direction
    - Wind direction and speed at that altitude
    - Tailwind/headwind component (kt)

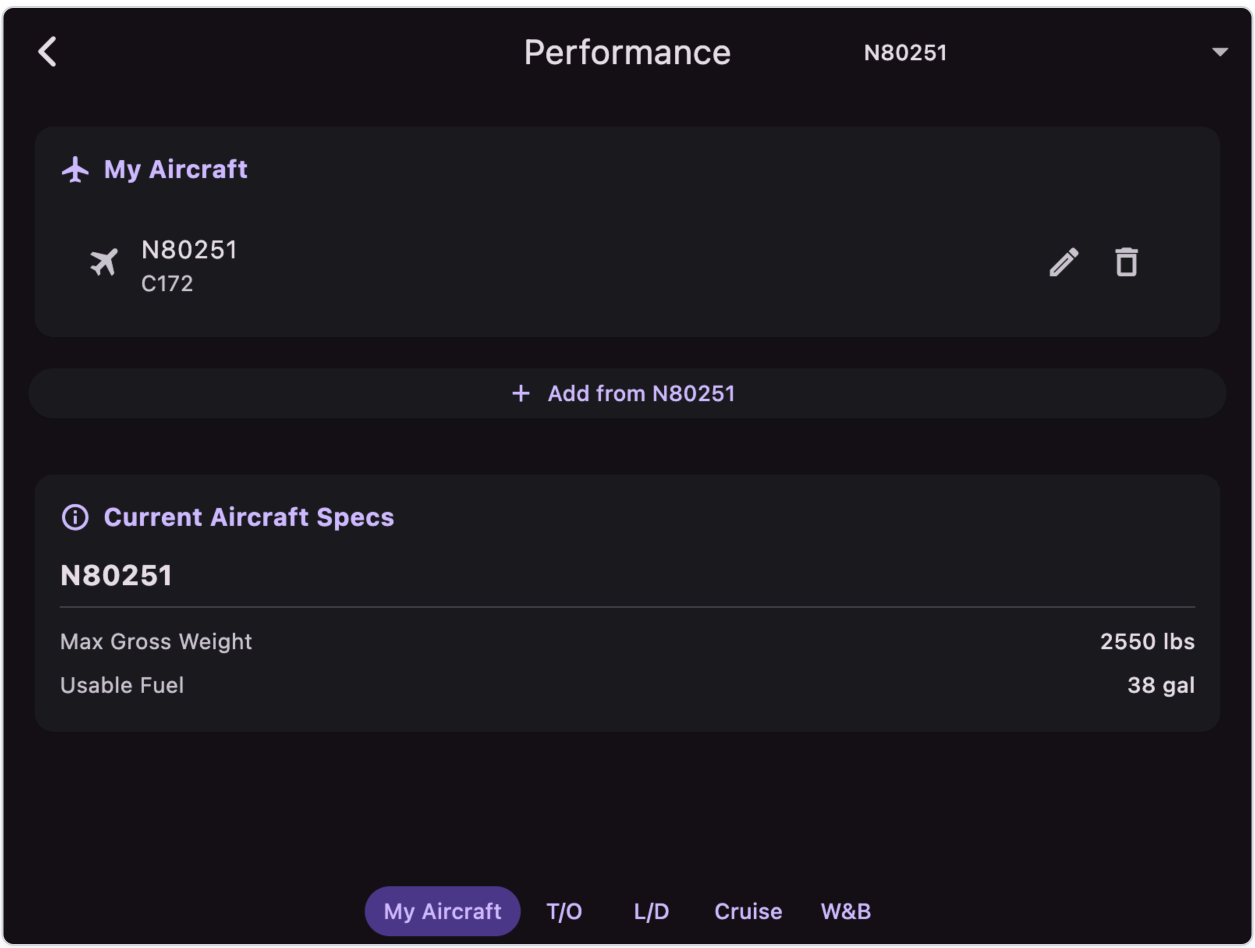
10. Use the **Copy plan to clipboard** icon in the app bar (top-right) to share the per-leg data outside the app.

11. Press **Back** (top-left chevron) to dismiss the dialog and return to the PLAN editor.

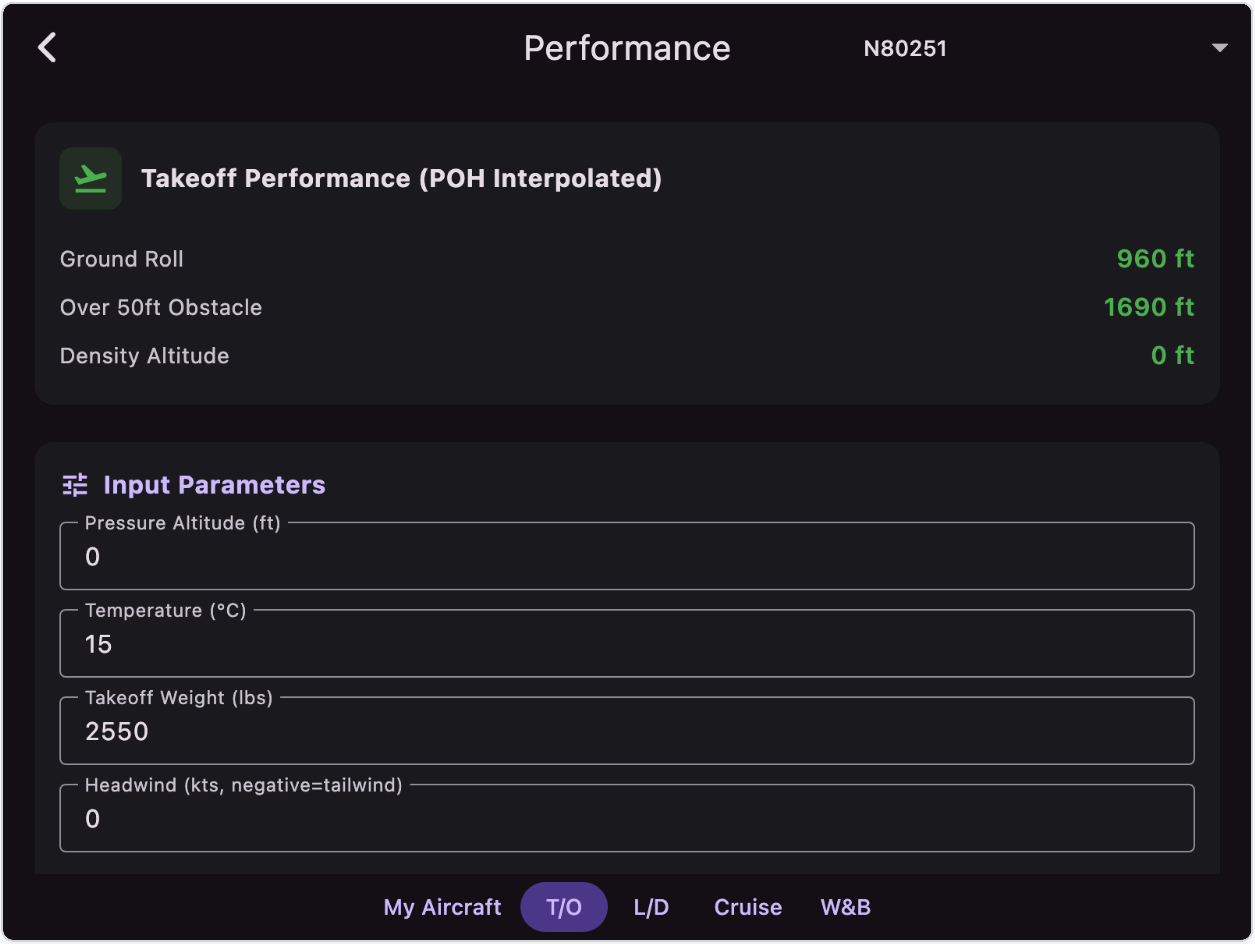
> Tip: The route in the screenshot above (`KBOS → BOS → CMK → KHPN` , plan altitude 3000 ft) shows mostly **red** at the cruise band — this means a headwind across the whole route at the chosen altitude. Use the diagram to see if a higher altitude band gets you a tailwind, or change the **Forecast horizon** dropdown to align with your actual departure window.

UC-22: Calculate takeoff and landing performance

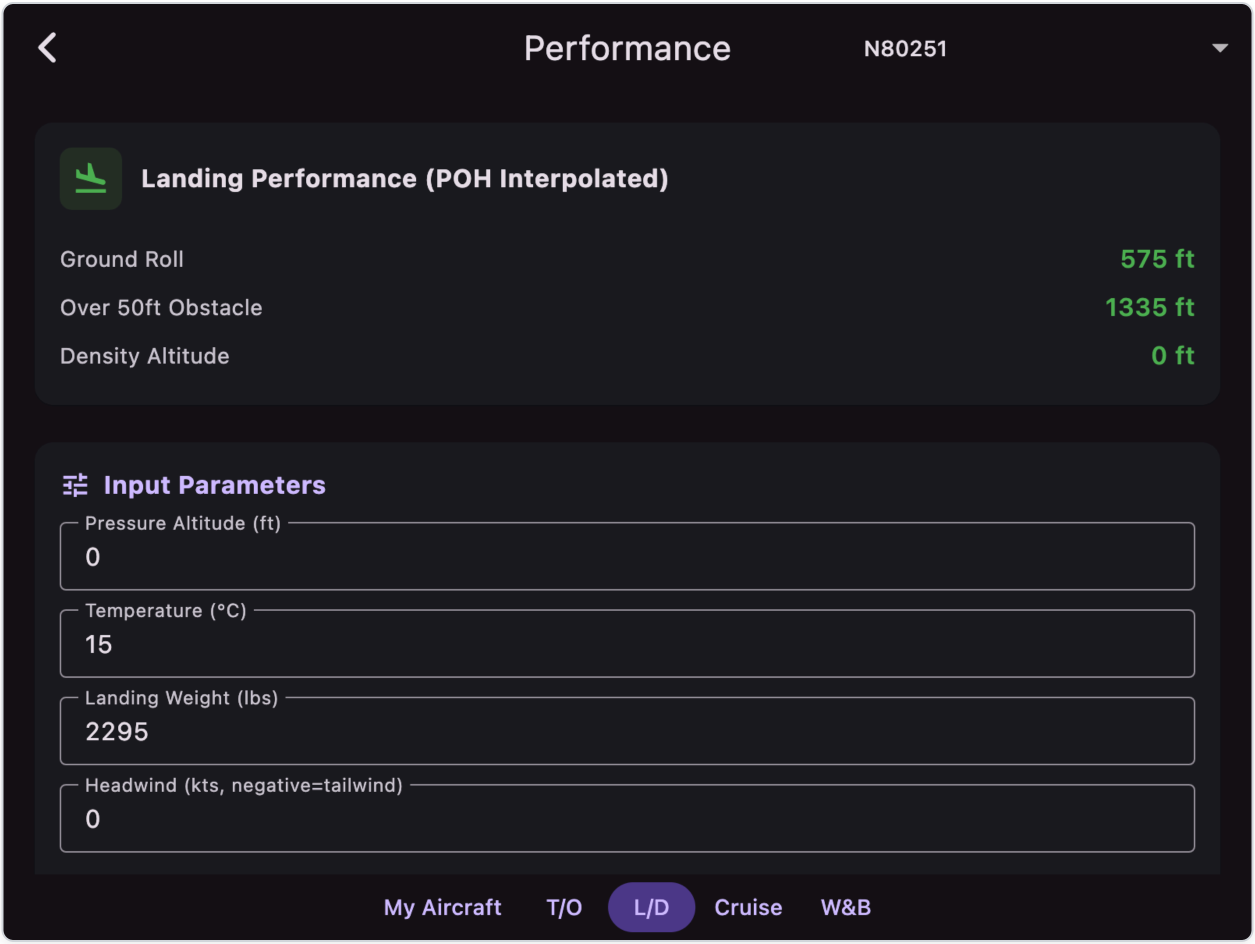
1. Open `MAP → Menu → Aircraft & Performance` .



2. Select your aircraft from the dropdown (built-in or custom). The dropdown is at the top-right of the screen.
3. Go to the **T/O** tab.



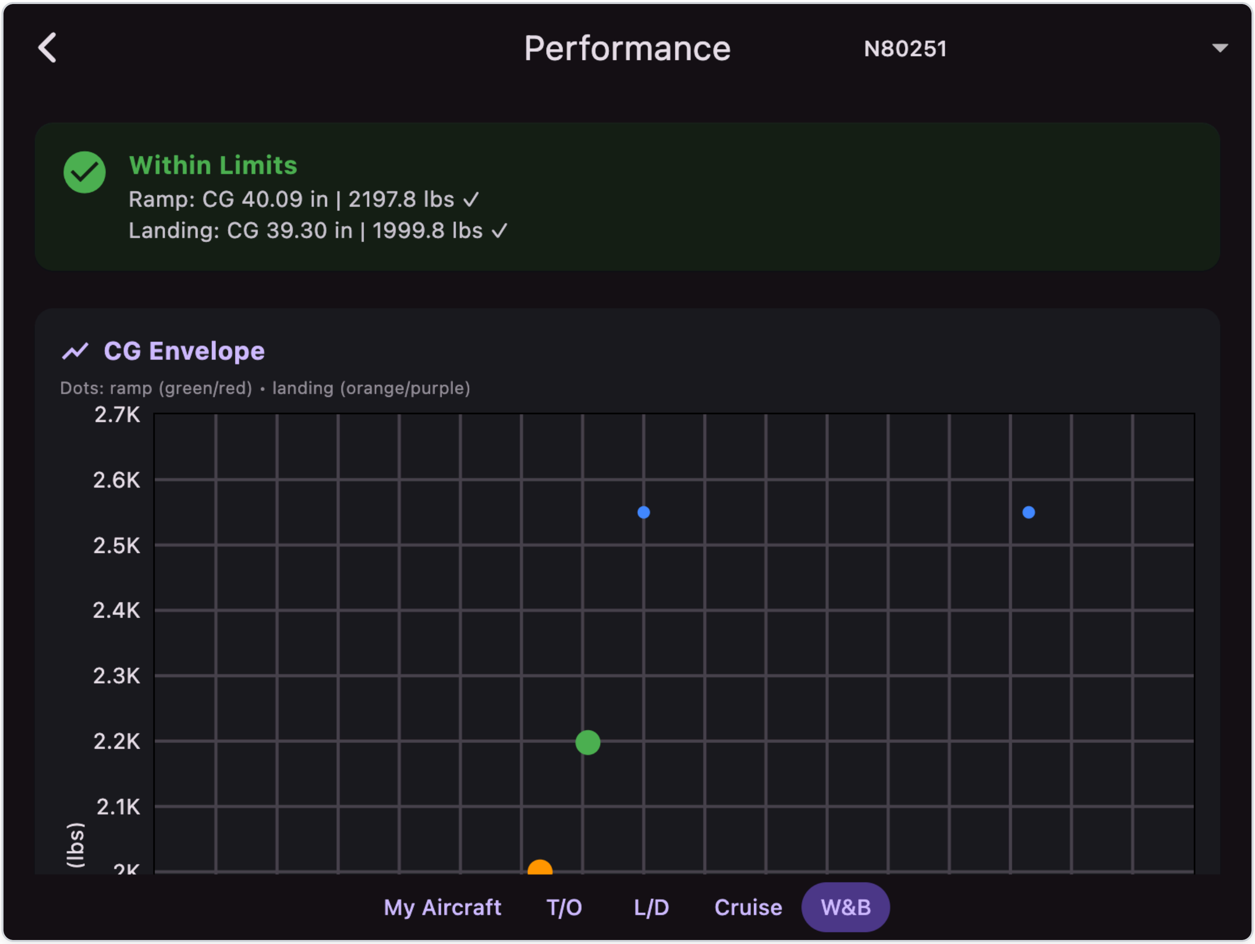
4. Enter current conditions:
    - **Pressure Altitude** (ft)
    - **Temperature** (°C)
    - **Takeoff Weight** (lbs)
    - **Headwind** (use a negative value for a tailwind)
5. View calculated **Ground Roll**, **Over 50ft Obstacle**, and **Density Altitude** distances at the top of the screen.
6. Switch to the **L/D** tab and repeat for landing performance.



7. Results account for aircraft POH data and any configured correction factors (headwind %, tailwind %, soft-field %) on the **My Aircraft** profile.

UC-23: Check weight and balance before flight

1. Open `MAP → Menu → Aircraft & Performance`.
2. Select your aircraft from the dropdown.
3. Go to the **W&B** tab.



4. Enter weights for each station in the table below the chart (passengers, baggage, fuel, plus Landing fuel).
5. The status card shows **Within Limits** (green) or **Outside Limits** (red), with a ✓ or ✕ for ramp and landing rows.
6. Read the CG Envelope chart:
    - Blue dots = envelope boundary
    - Large green/red dot = ramp/takeoff CG
    - Large orange/purple dot = landing CG
7. Adjust loading until both dots are inside the envelope.
8. For custom aircraft, use **Edit** to modify envelope points and station definitions.
9. If your aircraft uses the **helicopter** map icon, check the lateral CG chart and lateral columns as well; all four CG checks must pass for a green status.

UC-24: Create a pilot community group (Pro)

1. Open Pro Services from the account icon on MAP (top-right).
2. Sign in and pass the entitlement/paywall if required.
3. Tap **Community** in the Pro bottom sheet.
4. Tap the **New Group** floating action button.
5. Fill in:

- **Group name** (3–60 characters)
   - **Description** (optional, up to 280 characters)
   - **Home airport** (optional ICAO, e.g. `KBED`)
6. Choose **Visibility**:
   - **Public** — anyone can join immediately.
   - **Private** — discoverable by name, but you approve every new member.
7. Tap **Create Group**. You land on the new group's **Feed** tab as the owner.
8. Tap **Post** to share your first message; optionally attach an airport ICAO to the post.

UC-25: Join a private pilot group (Pro)

1. Open `Community → Discover` and search for the group by name.
2. Tap the group to open it. Private groups show a lock icon and the feed is hidden.
3. Tap **Request to Join**. The banner switches to **Requested** and your status is `pending`.
4. The group owner will see your request in the **Members** tab under **Pending requests**.
5. When approved, the **Feed** unlocks and you can post immediately. You can leave at any time with the **Leave** button.
6. To cancel a pending request, tap **Requested** to withdraw it.

UC-26: Set up an aircraft scheduler and book a flight (Pro)

1. Open Pro Services from the account icon on MAP (top-right).
2. Sign in and pass the entitlement/paywall if required.
3. Tap **Scheduler** in the Pro bottom sheet.
4. Tap the **New Scheduler** floating action button, fill in the name (and optional description/home airport), then tap **Create Scheduler**. Every scheduler is private; you land on the new scheduler as the owner.
5. On the **Schedule** tab, tap **Add resource** to add your aircraft (e.g. `Cessna 172`, tail `N12345`) and any instructors. They appear as rows, with time across the top.
6. (Optional) Tap the **tune** icon in the app bar to open **Booking Rules** and set how many reservations — and how many weekend reservations — each member may hold at once (0 = unlimited). The owner is exempt from these limits.
7. Share the scheduler name with your members; they open `Scheduler → Discover`, search the name, and **Request to Join**. Approve them on the **Members** tab. Use the badge icon to set **Student** / **Instructor** / **Dispatcher** roles and assign students to instructors.
8. On **Dispatch → Fleet**, tap an aircraft and **Update meters / MX** to set hobbs, tach, and inspection due dates for the shared fleet.
9. (Optional) On **Dispatch → Lessons**, create a **lesson pack** of prepaid hours for a student.
10. To book, a member taps a **green** slot on a resource row, picks a **start date/time** and **end date/time** (bookings can span multiple days, up to 14), optionally picks an instructor /

lesson pack, then taps **Book**. The slot turns **blue**. Members can review their bookings on the **Mine** tab.

11. If that time is already taken, the member is added to the **backup** queue. If the main reservation is later cancelled, the first backup is promoted automatically.
12. To cancel, tap the reservation block (or use the **Mine** tab) and choose **Cancel**. Only the member who booked it or the owner can cancel a reservation.
13. For discrepancies, use **Dispatch → Squawks → File squawk**. A **Grounding** squawk blocks booking and shows as GROUNDED on the fleet board. For a hard maintenance hold, turn off **Available for booking** from Schedule or Dispatch (owner/dispatcher).

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16) Resources and Support

- Apps4Av Forum (Google Groups): `https://groups.google.com/g/apps4av-forum`
- Store links and platform download details: see `README.md`.
- For in-app issues, always check the MAP warning drawer first (red icon).

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17) Forum-Derived FAQ and Scenarios (Google Groups)

The following FAQs are derived from recent threads in the Apps4Av forum and mapped to the current AvareX workflows.

FAQ-01: How do I connect an external ADS-B / GPS receiver?

- For Wi-Fi receivers: connect the device to receiver network and AvareX auto-listens on UDP `4000`, `43211`, `49002`.
- **Stratus 3 / 3i**: if traffic/weather never appear, put the unit in Open ADS-B mode from `MAP → ADSB` tile → antenna icon (top-right) (or via Appareo Stratus Horizon Pro).
- For Bluetooth receivers (Android): `Menu → IO`, pair/connect, then return to map.
- **iOS only**: AvareX must have **Local Network** permission enabled (`Settings → Privacy & Security → Local Network → AvareX`). Without it, iOS silently blocks all ADS-B/GDL90 broadcasts and traffic/weather will never show, even though ownship from the internal phone GPS still works. The first time you launch AvareX after install you should see a prompt — tap **Allow**. See FAQ-16.

- Source threads:
  - GPS source discussion:
    `https://groups.google.com/g/apps4av-forum/c/e0ujCJQX1s8`
  - In-flight ADS-B issues:
    `https://groups.google.com/g/apps4av-forum/c/ZP2E9l35Dk8`
  - Android IO functionality:
    `https://groups.google.com/g/apps4av-forum/c/WI-BcDK7rT0`

FAQ-02: How do I transfer a plan to or from an external device?

- Connect this device to the **same Wi-Fi network** as your Avidyne IFD.
- Open `PLAN → Transfer` and wait for the IFD to appear.
- To upload: build a route with at least 2 waypoints, then tap **Send**.
- To download the IFD's plan into AvareX: tap **Get** (this replaces the active plan).

Supported avionics: Panel-mounted Avidyne IFD440/540/550 units with flight-plan input enabled. On iOS, grant the **Local Network** permission for discovery and upload to work.

If the IFD has an ADS-B receiver, AvareX also picks up its **Capstone** ADS-B traffic and weather automatically over the same Wi-Fi network (Capstone is GDL90 data received on UDP `4000`). See FAQ-16 for the iOS Local Network requirement.

FAQ-03: Where do I enable new wind/ceiling map features?

- `MAP → Layers → Weather` (opacity > 0)
- Right-side product menu: choose **Wind** or **Ceil**
- Use the altitude slider to change displayed altitude context
- Source threads:
  - New features v83/v84:
    `https://groups.google.com/g/apps4av-forum/c/qac4-Bb-gVE`
    `https://groups.google.com/g/apps4av-forum/c/Xvhc9yeE42A`

FAQ-04: My track logs are empty. How do I save valid KML logs?

- Keep `Tracks` layer ON (opacity > 0) during flight.
- After flight, turn `Tracks` layer OFF (opacity = 0) — this saves KML to Documents/tracks folder and clears active track buffer.
- Open `Documents → tracks` and verify non-empty KML before sharing.
- Tap the KML file to view it in Track Viewer with 2D map, altitude profile, and 3D terrain views.
- Use **Log Flight** button to create a logbook entry directly from the track.

- Source thread: `https://groups.google.com/g/apps4av-forum/c/goPy8KQpfik`

FAQ-05: Where are FBO/business details?

- In the destination popup, open the **Business** tab (free; sign-in required). The airport's businesses are listed inline; tap one to view and contribute services, fuel, hours, and star reviews per airport (see Section 11.7).
- On the `PLATE` screen, when an airport diagram is active, use the right-side business selector (business icon) to mark a business's location on the diagram. It draws from the same crowd-sourced directory (iOS/Android, sign-in required).
- Source thread: `https://groups.google.com/g/apps4av-forum/c/oMOR-gaIqis`

FAQ-06: What are the colored rings on map?

- In the `Circles` layer:
  - **Black rings**: fixed 2/5/10 NM reference rings; the 10 NM ring is marked with a magnetic compass rose (ticks every 5°, labels at `N` / `NE` / `E` / `SE` / `S` / `SW` / `W` / `NW`)
  - **Blue ring**: speed-based 1-minute travel distance
  - **Purple ring**: glide circle based on the selected aircraft's best glide speed and sink rate, plus winds aloft, terrain, and altitude
- Source thread: `https://groups.google.com/g/apps4av-forum/c/VJ0S3ejWPC8`

FAQ-07: My waypoints appear in wrong order. Can I reorder or insert quickly?

- **Reorder**: In `PLAN`, long-press and drag rows to reorder legs.
- **Insert at specific position**: Long-press a waypoint row to insert after it (opens FIND; next selection inserts after that waypoint).
- **Set current leg**: Tap a waypoint row.
- Source thread: `https://groups.google.com/g/apps4av-forum/c/T-m4BWZynMg`

FAQ-08: How do I enter an ATC reroute quickly?

- Open `PLAN → Create`.
- Use **Create As Entered** and paste/type reroute string (space-separated waypoints/airways).
- Then review/reorder in PLAN list if needed.

- Source thread: `https://groups.google.com/g/apps4av-forum/c/ukaXEEhpvS0`

FAQ-09: Download/cycle issues - what to try first?

- In `Download` screen:
  - Try **This Cycle** vs **Next Cycle**
  - Try **Main Server** vs **Backup Server**
  - Keep download screen open until completion (exiting can abort incomplete downloads)
- Source threads:
  - Cycle download issue: `https://groups.google.com/g/apps4av-forum/c/LS_VaqKEJxw`
  - Background download request: `https://groups.google.com/g/apps4av-forum/c/XcoQzwpCxlw`

FAQ-10: FAA IFR preferred route tools not working - what now?

- Ensure app is updated to latest release and internet is available.
- Confirm your 1800wxbrief-compatible email is configured.
- Retry via `PLAN → Create` and `Brief & File`.
- Source thread: `https://groups.google.com/g/apps4av-forum/c/0wHqcJT-WiY`

FAQ-11: Is Flight Intelligence (AI) available on desktop?

- Current Pro AI workflows are targeted for iOS/Android only.
- Access from map account icon → Pro Services → Flight Intelligence.
- Uses Gemini 2.5 Pro with Google Search integration.
- Source threads:
  - AI feature thread: `https://groups.google.com/g/apps4av-forum/c/wWZUn6TNG1w`
  - Desktop/pro services request: `https://groups.google.com/g/apps4av-forum/c/XcoQzwpCxlw`

FAQ-12: Can I use AvareX with X-Plane/simulator feeds?

- AvareX accepts external NMEA/GDL90-like streams via UDP listener ports (4000, 43211, 49002).
- For simulator setups, configure simulator/network output accordingly.

- Source thread: `https://groups.google.com/g/apps4av-forum/c/QOfnQ-pkT0w`

FAQ-13: What is the CAP Grid layer?

- Civil Air Patrol grid overlay for search and rescue operations.
- Shows grid identifiers (e.g., BOS42, SEA123) at zoom level 9+.
- Enable via `MAP → Layers → CAP Grid` opacity > 0.
- Covers all US sectional chart areas.

FAQ-14: How do I view my flight in 3D?

- Record your flight with `Tracks` layer enabled.
- Turn `Tracks` off to save the KML file.
- Open `Documents → tracks` and tap the KML file.
- Use the 3D View icon (AR) to see interactive 3D visualization with terrain and topo map texture.

FAQ-15: How do I check if I'm current to fly?

- Open `MAP → Menu → Log Book → Details`.
- The Currency Status Card shows your currency for:
  - Day VFR (3 landings in 90 days)
  - Night (3 night landings in 90 days)
  - IFR (6 approaches in 6 months)
- Status is automatically calculated from your logbook entries.

FAQ-16: On iOS, my ownship moves but I never see traffic from Stratux/Stratus/Echo, even though FltPlanGo and ForeFlight see it on the same phone

This is a Local Network permission issue, not a bug in your receiver or in AvareX's traffic engine.

Since iOS 14, Apple silently blocks all incoming UDP broadcasts (including GDL90 traffic on UDP `4000`, ForeFlight on `43211`, and X-Plane on `49002`) until the user grants the **Local Network** permission. The internal phone GPS keeps working, so ownship still moves and the map looks correct — that's the trap. ADS-B traffic, ADS-B weather (NEXRAD/METAR/TAF/AIRMET/SIGMET via FIS-B), and external GPS lock from your receiver will all be missing.

Fix:

1. On the iPhone/iPad, open `Settings → Privacy & Security → Local Network`.
2. Find **AvareX** in the list and turn the toggle **ON**.
3. Force-quit AvareX (swipe up from app switcher) and relaunch it.
4. Make sure the device's Wi-Fi is joined to the receiver's network (e.g. `stratux`, `Stratus-XXXX`, `EchoUAT-XXXX`).
5. Open `MAP` and turn the `Traffic` layer on. Traffic should appear within a few seconds.

If **AvareX** is not listed at all under `Local Network`, you are running an older build that predates the Local Network entitlement — update to the latest AvareX from the App Store.

Notes:

- This setting is per-app and per-device; each iPhone and iPad must be configured separately.
- A factory reset of network settings or "Reset Location & Privacy" will clear this — you'll need to re-enable it.
- This does not affect Android. Android handles Wi-Fi UDP without an extra runtime permission, which is why the same Stratux works fine with AVARE on Android.
- AvareX always listens on UDP `4000` (Stratux/Stratus/Echo GDL90, and Avidyne IFD Capstone), `43211` (ForeFlight), and `49002` (X-Plane). There is no port to configure inside the app — if data is reaching the phone over Wi-Fi and Local Network is allowed, AvareX will pick it up.
